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Distributionally Robust Stochastic Optimization with Wasserstein Distance

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Abstract. Distributionally robust stochastic optimization (DRSO) is an approach to optimization under uncertainty in which, instead of assuming that there is a known true underlying probability distribution, one hedges against a chosen set of distributions. In this paper, we first point out that the set of distributions should be chosen to be appropriate for the application at hand and some of the choices that have been popular until recently are, for many applications, not good choices. We next consider sets of distributions that are within a chosen Wasserstein distance from a nominal distribution. Such a choice of sets has two advantages: (1) The resulting distributions hedged against are more reasonable than those resulting from other popular choices of sets. (2) The problem of determining the worst-case expectation over the resulting set of distributions has desirable tractability properties. We derive a strong duality reformulation of the corresponding DRSO problem and construct approximate worst-case distributions (or an exact worst-case distribution if it exists) explicitly via the first-order optimality conditions of the dual problem. Our contributions are fourfold. (i) We identify necessary and sufficient conditions for the existence of a worst-case distribution, which are naturally related to the growth rate of the objective function. (ii) We show that the worst-case distributions resulting from an appropriate Wasserstein distance have a concise structure and a clear interpretation. (iii) Using this structure, we show that data-driven DRSO problems can be approximated to any accuracy by robust optimization problems, and thereby many DRSO problems become tractable by using tools from robust optimization. (iv) Our strong duality result holds in a very general setting. As examples, we show that it can be applied to infinite dimensional process control and intensity estimation for point processes.

Keywords: distributionally robust optimization • Wasserstein metric • data-driven decision making • ambiguity set • worst-case distribution

1. Introduction

In decision-making problems under uncertainty, a decision maker wants to choose a decision x from a feasible region X. The objective function $\Psi : X \times \Xi \longmapsto \mathbb{R}$ depends on a quantity $\xi \in \Xi$ whose value is not known to the decision maker at the time that the decision has to be made. In some settings, it is reasonable to assume that ξ is a random element with distribution μ supported on Ξ, for example, if multiple realizations of ξ will be encountered. In such settings, the decision-making problems can be formulated as *stochastic optimization* problems as follows:

$$\inf_{x\in X} \mathbb{E}_{\mu}[\Psi(x,\xi)].$$

We refer to Shapiro et al. [48] for a thorough study of stochastic optimization. One major criticism of the preceding formulation for practical applications is the requirement that the underlying distribution μ be known to the decision maker. Even if multiple realizations of ξ are observed, μ still may not be known exactly, and use of a distribution different from μ may sometimes result in bad decisions. Another major criticism is that, in many applications, there are not multiple realizations of ξ that will be encountered, for example, in problems involving events that may either happen once or not happen at all, and thus, the notion of a "true" underlying distribution does not apply. These criticisms motivate the notion of distributionally robust stochastic optimization (DRSO), which does not rely on the notion of a known true underlying distribution. One chooses a set $\mathfrak{M}$ of probability distributions to hedge against and then finds a decision that provides the best hedge against the set $\mathfrak{M}$ of distributions by solving the following minmax problem:

$$\inf_{x\in X}\sup_{\mu\in\mathfrak{M}}\mathbb{E}_\mu[\Psi(x,\xi)]. \tag{DRSO}$$

Such a minmax approach has its roots in Von Neumann's game theory and is used in many fields, such as inventory management (Gallego and Moon [25], Scarf et al. [45]) and statistical decision analysis (Berger [9]) as well as stochastic optimization (Dupačová [20], Shapiro and Kleywegt [47], Žáčková [56]). Recently, it regained attention in the operations research literature and sometimes is called data-driven stochastic optimization or ambiguous stochastic optimization.

A central question is that of how to choose a good set of distributions $\mathfrak{M}$ against which to hedge. A good choice of $\mathfrak{M}$ should take into account the properties of the practical application as well as the tractability of Problem (DRSO). Two typical ways of constructing $\mathfrak{M}$ are moment- and distance-based. The moment-based approach considers distributions whose moments (such as mean and covariance) satisfy certain conditions (Delage and Ye [19], Popescu [43], Scarf et al. [45], Zymler et al. [58]). It is shown that, in many cases, the resulting DRSO problem can be formulated as a conic quadratic or semidefinite program. However, the moment-based approach is based on the curious assumption that certain conditions on the moments are known exactly but that nothing else about the relevant distribution is known. More often in applications, either one has data from repeated observations of the quantity ξ or one has no data, and in both cases, the moment conditions do not describe exactly what is known about ξ. In addition, the resulting worst-case distributions sometimes yield overly conservative decisions (Goh and Sim [29], Wang et al. [52]). For example, Wang et al. [52] show that, for the newsvendor problem, by hedging against all the distributions with fixed mean and variance, Scarf's moment approach yields a two-point worst-case distribution, and the resulting decision does not perform well under other, more likely scenarios.

The distance-based approach considers distributions that are close, in the sense of a chosen statistical distance, to a *nominal distribution* ν, such as an empirical distribution or a Gaussian distribution (Calafiore and El Ghaoui [16], El Ghaoui et al. [21]). Popular choices of the statistical distance are ϕ-divergences (Bayraksan and Love [6], Ben-Tal et al. [8]), which include Kullback–Leibler (KL) divergence (Jiang and Guan [32]), Burg entropy (Wang et al. [52]), and total variation distance (Sun and Xu [49]) as special cases; Prokhorov metric (Erdoğan and Iyengar [22]); and Wasserstein distance (Esfahani and Kuhn [23], Wozabal [53, 54], Zhao and Guan [57]).

1.1. Motivation: Potential Issues with ϕ-Divergence

Despite its widespread use, ϕ-divergence has a number of shortcomings. Here, we highlight some of these shortcomings. In a typical setup using ϕ-divergence, Ξ is partitioned into $\overline{B}+1$ bins represented by points $\xi^0,\xi^1,\ldots,\xi^{\overline{B}}\in\Xi$. The nominal distribution ν associates N_i observations with bin i. That is, the nominal distribution is given by $\nu:=(N_0/N,N_1/N,\ldots,N_{\overline{B}}/N)$, where $N:=\sum_{i=0}^{\overline{B}}N_i$. Let $\Delta_{\overline{B}}:=\{(p_0,p_1,\ldots,p_{\overline{B}})\in\mathbb{R}_+^{\overline{B}+1}:\sum_{j=0}^{\overline{B}}p_j=1\}$ denote the set of probability distributions on the same set of bins. Let $\phi:[0,\infty)\longmapsto\mathbb{R}$ be a chosen convex function such that $\phi(1)=0$ with the conventions that $0\phi(a/0):=a\lim_{t\to\infty}\phi(t)/t$ for all $a>0$, and $0\phi(0/0):=0$. Then, the ϕ-divergence between $\mu=(p_0,\ldots,p_B),\nu=(q_0,\ldots,q_B)\in\Delta_{\overline{B}}$ is defined by

$$I_\phi(\mu,\nu):=\sum_{j=0}^{\overline{B}}q_j\phi\left(\frac{p_j}{q_j}\right).$$

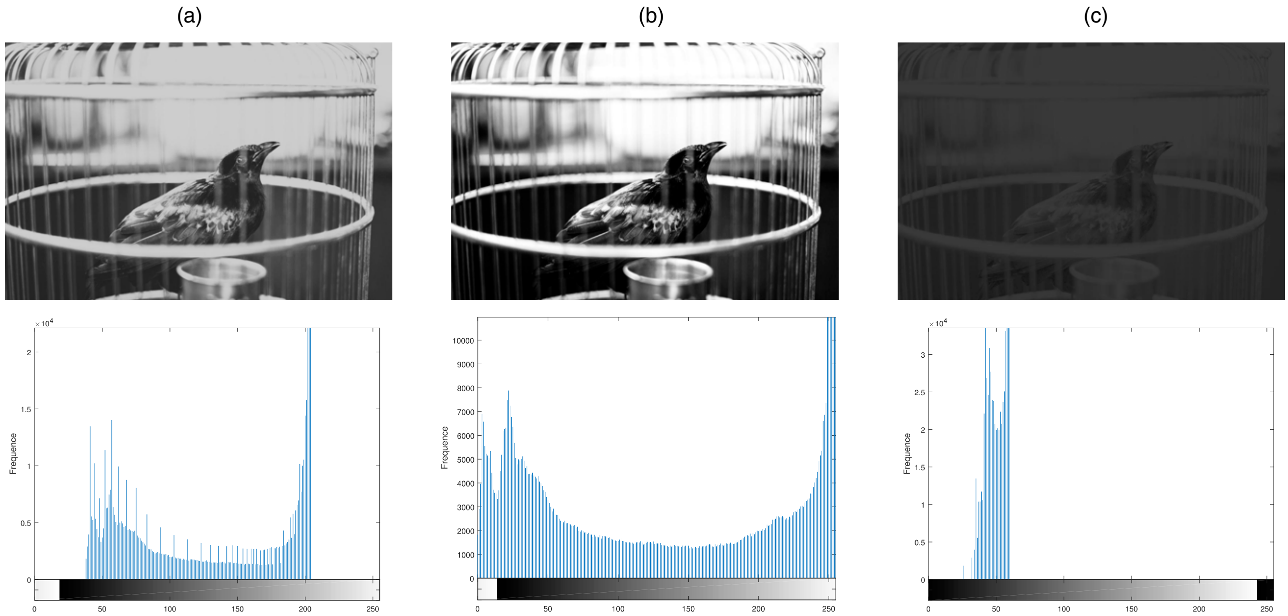
Let $\theta>0$ denote a chosen radius. Then, $\mathfrak{M}_\phi:=\left\{\mu\in\Delta_{\overline{B}}:I_\phi(\mu,\nu)\le\theta\right\}$ denotes the set of probability distributions given by the chosen ϕ-divergence and radius θ. The DRSO problem corresponding to the ϕ-divergence ball $\mathfrak{M}_\phi$ is then given by

$$\inf_{x\in X}\sup_{\mu\in\Delta_{\overline{B}}}\left\{\sum_{j=0}^{\overline{B}}p_j\Psi(x,\xi^j):I_\phi(\mu,\nu)\le\theta\right\}.$$

It is shown in Ben-Tal et al. [8] that the ϕ-divergence ball $\mathfrak{M}_\phi$ can be viewed as a statistical confidence region (Pardo [40]), and for several choices of ϕ, the inner maximization of the preceding problem is tractable.

'One well-known shortcoming of ϕ-divergence balls is that either they are not rich enough to contain distributions that are often relevant or they hedge against many distributions that are too extreme. For example, for some choices of ϕ-divergence, such as Kullback–Leibler divergence, if the nominal $q_i=0$, then $p_i=0$; that is, the

Figure 1. (Color online) Three images and their grayscale histograms.



Notes. For KL divergence, it holds that $I_{\phi_{KL}}(\mu_{true}, \nu) = 5.05 > I_{\phi_{KL}}(\mu_{pathol}, \nu) = 2.33$, whereas in contrast, Wasserstein distance satisfies $W_1(\mu_{true}, \nu) = 30.70 < W_1(\mu_{pathol}, \nu) = 84.03$. (a) Observed image with histogram ν. (b) True image with histogram μ_{true}. (c) Pathological image with histogram μ_{pathol}.

ϕ-divergence ball includes only distributions that are absolutely continuous with respect to the nominal distribution ν and, thus, does not include distributions with support on points at which the nominal distribution ν is not supported. As a result, if $\Xi = \mathbb{R}^K$ and ν is discrete, then there are no continuous distributions in the ϕ-divergence ball $\mathfrak{M}_\phi$. Some other choices of ϕ-divergence exhibit in some sense the opposite behavior. For example, the Burg entropy ball includes distributions with some amount of probability allowed to be shifted from ν to any other bin with the amount of probability allowed to be shifted depending only on θ and not on how extreme the bin is. See Section 5.1 for more details regarding this potential shortcoming.

Next, we illustrate another shortcoming of ϕ-divergence that motivates the use of *Wasserstein distance*.

Example 1. Suppose that there is an underlying true image (Figure 1(b)), and a decision maker possesses, instead of the true image, an approximate image (Figure 1(a)) obtained with a less-than-perfect device that loses some of the contrast. The images are summarized by their grayscale histograms. (In fact, Figure 1(a) was obtained from Figure 1(b) by a low-contrast intensity transformation (Gonzalez and Woods [30]), by which the black pixels become somewhat whiter, and the white pixels become somewhat blacker. This type of transformation changes only the grayscale value of a pixel and not the location of a grayscale value, and therefore, it can also be regarded as a transformation from one grayscale histogram to another grayscale histogram.) As a result, roughly speaking, the observed histogram ν is obtained by shifting the true histogram μ_{true} inward. Also consider the pathological image (Figure 1(c)) that is too dark to see many details with histogram μ_{pathol}. Suppose that the decision maker constructs a KL divergence ball $\mathfrak{M}_{\phi_{KL}} := \{\mu \in \Delta_{\overline{B}} : I_{\phi_{KL}}(\mu, \nu) \le \theta\}$. Note that $I_{\phi_{KL}}(\mu_{true}, \nu) = 5.05 > I_{\phi_{KL}}(\mu_{pathol}, \nu) = 2.33$. Therefore, if θ is chosen small enough (less than 2.33) for $\mathfrak{M}_{\phi_{KL}}$ to exclude the pathological image (Figure 1(c)), then $\mathfrak{M}_{\phi_{KL}}$ also excludes the true image (Figure 1(b)). If θ is chosen large enough (greater than 5.05) for $\mathfrak{M}_{\phi_{KL}}$ to include the true image (Figure 1(b)), then $\mathfrak{M}_{\phi_{KL}}$ also has to include the pathological image (Figure 1(c)), and then, the resulting decision may be overly conservative because of hedging against irrelevant distributions. If an intermediate value is chosen for θ (between 2.33 and 5.05), then $\mathfrak{M}_{\phi_{KL}}$ includes the pathological image (Figure 1(c)) and excludes the true image (Figure 1(b)). In contrast, note that the Wasserstein distance W_1 satisfies $W_1(\mu_{true}, \nu) = 30.7 < W_1(\mu_{pathol}, \nu) = 84.0$, and thus, Wasserstein distance does not exhibit the problem encountered with KL divergence (see also Example 3).

The reason for such behavior is that ϕ-divergence does not incorporate a notion of how close two points $\xi, \xi' \in \Xi$ are to each other, for example, how likely it is that the observation is ξ' given that the true value is ξ.

In Example 1, $\Xi = \{0, 1, \ldots, 255\}$ represents eight-bit grayscale values. In this case, we know that the likelihood that a pixel with grayscale value $\xi \in \Xi$ is observed with grayscale value $\xi' \in \Xi$ is decreasing in the absolute difference between ξ and ξ'. However, in the definition of ϕ-divergence, only the *relative ratio* p_j/q_j for the same grayscale value j is taken into account, whereas the distances between different grayscale values are not taken into account. This phenomenon is observed in studies of image retrieval (Ling and Okada [36], Rubner et al. [44]).

The drawbacks of ϕ-divergence motivates us to consider sets $\mathfrak{M}$ that incorporate a notion of how close two points $\xi, \xi' \in \Xi$ are to each other. One such choice of $\mathfrak{M}$ is based on Wasserstein distance. Specifically, consider any underlying metric d on Ξ that measures the closeness of any two points in Ξ. Let $p \geq 1$, and let $\mathcal{P}(\Xi)$ denote the set of Borel probability measures on Ξ. Then, the Wasserstein distance of order p between two distributions $\mu, \nu \in \mathcal{P}(\Xi)$ is defined as

$$W_p(\mu, \nu) := \min_{\gamma \in \mathcal{P}(\Xi^2)} \left\{ \mathbb{E}^{1/p}_{(\xi,\zeta)\sim\gamma}[d^p(\xi, \zeta)] : \gamma \text{ has marginal distributions } \mu, \nu \right\}.$$

More detailed explanation and discussion on Wasserstein distance are presented in Section 2. Given a radius $\theta > 0$, the Wasserstein ball of probability distributions $\mathfrak{M}$ is defined by

$$\mathfrak{M} := \left\{ \mu \in \mathcal{P}(\Xi) : W_p(\mu, \nu) \leq \theta \right\}.$$

1.2. Related Work

Wasserstein distance and the related field of *optimal transport*, which is a generalization of the transportation problem, are studied in depth. In 1942, together with the linear programming problem (Kantorovich [34]), Kantorovich [33] tackles Monge's problem originally brought up in the study of optimal transport. In the stochastic optimization literature, Wasserstein distance is used for single-stage stochastic optimization (Wozabal [53, 54]) and for multistage stochastic optimization (Pflug and Pichler [42]). The challenge for solving (DRSO) is that the inner maximization involves a supremum over possibly an infinite dimensional space of distributions. To tackle this problem, existing works focus on the setup when ν is the empirical distribution on a finite-dimensional space. Wozabal [53] transformed the inner maximization problem of (DRSO) into a finite-dimensional nonconvex program by using the fact that, if ν is supported on at most N points, then there are extreme distributions of $\mathfrak{M}$ that are supported on at most $N + 3$ points. Recently, using duality theory of conic linear programming (Shapiro [46]), Esfahani and Kuhn [23] and Zhao and Guan [57] show that, under certain conditions, the inner maximization problem of (DRSO) is actually equivalent to a finite-dimensional convex problem.

In this paper, we consider any arbitrary nominal distribution ν on a Polish space and study the tractability of (DRSO) via strong duality. By the time we completed the first version of this paper, we learned that Blanchet and Murthy [10] independently considered a similar problem with more general lower semicontinuous transport cost function and study its strong duality. Our focus and approach to this problem differ from theirs in several important ways. First, we show that the strong duality holds for any measurable function Ψ, whereas Blanchet and Murthy [10] prove the result only when Ψ is upper semicontinuous. The upper semicontinuity is used to ensure the existence of the worst-case distribution but is not needed for the strong duality to hold. Second, we prove the strong duality result for the inner maximization of (DRSO) using a novel, yet simple *constructive* approach in contrast with the nonconstructive approaches in their work and also in Esfahani and Kuhn [23] and Zhao and Guan [57]. This enables us to establish the structural characterization of the worst-case distributions of the data-driven DRSO (Corollary 2(ii)), which improves the result of Wozabal [53] and the more recent result of Owhadi and Scovel [39] on extremal distributions of Wasserstein balls (Remark 6). It also enables us to build a close connection between DRSO and robust optimization (Corollary 2(iii)). Third, we focus on Wasserstein distance of order p ($p \geq 1$), whereas they consider more general transport cost functions in which the distance between two points $\xi, \xi' \in \Xi$ is measured by a lower semicontinuous function rather than a metric $d^p(\xi, \xi')$ as in our case. Nevertheless, our proof remains valid for such transport cost functions and, in fact, for even more general cost functions that are not necessarily lower semicontinuous (Remark 2). In the meantime, focusing on Wasserstein distance enables us to relate the condition for the existence of a worst-case distribution to the important notion of the "growth rate" of the objective function and enables us to provide practical guidance for choosing the ambiguity set and controlling the degree of conservativeness based on the objective function (Remark 1).

1.3. Main Contributions

• General setting: We prove a strong duality result for DRSO problems with Wasserstein distance in a very general setting. We show that

$$\sup_{\mu\in\mathcal{P}(\Xi)}\left\{\mathbb{E}_{\mu}[\Psi(x,\xi)] : W_p(\mu,\nu)\le\theta\right\} = \min_{\lambda\ge 0}\left\{\lambda\theta^p - \int_{\Xi}\inf_{\xi\in\Xi}[\lambda d^p(\xi,\zeta)-\Psi(x,\xi)]\nu(d\zeta)\right\}$$

holds for any Polish space (Ξ,d) and measurable function Ψ (Theorem 1).

1. Both Esfahani and Kuhn [23] and Zhao and Guan [57] assume that Ξ is a convex subset of $\mathbb{R}^K$ with some associated norm. The greater generality of our results enables one to consider interesting problems, such as the process control problems in Sections 4.1 and 4.2, in which Ξ is the set of finite counting measures on $[0,1]$, which is infinite-dimensional and nonconvex.
2. Both Esfahani and Kuhn [23] and Zhao and Guan [57] assume that the nominal distribution ν is an empirical distribution, whereas we allow ν to be any Borel probability measure.
3. Both Esfahani and Kuhn [23] and Zhao and Guan [57] only consider Wasserstein distance of order $p=1$. By considering a bigger family of Wasserstein distances, we establish the importance for DRSO problems of the notion of the growth rate of the objective function, which measures how fast the objective function grows compared with a polynomial of order p. It turns out that the growth rate of the objective function determines the finiteness of the worst-case objective value (Proposition 2), and it plays an important role in the existence conditions for the worst-case distribution (Corollary 1). This is of practical importance because it provides guidance for choosing the proper Wasserstein distance and for controlling the degree of conservativeness based on the structure of the objective function.

• Constructive proof of duality: We prove the strong duality result using a novel, elementary, constructive approach. The results of Esfahani and Kuhn [23] and Zhao and Guan [57] and other strong duality results in the literature are based on the established Hahn–Banach theorem for certain infinite dimensional vector spaces. In contrast, our proof idea is new and relatively elementary and straightforward: we use the weak duality result as well as the first order optimality condition of the dual problem to construct a sequence of primal feasible solutions whose objective values converge to the dual optimal value. Our proof uses relatively elementary tools without resorting to other "big hammers."

• Existence conditions and the structure of worst-case distributions: As a by-product of our constructive proof, we identify necessary and sufficient conditions for the existence of worst-case distributions and a structural characterization of worst-case distributions (Corollary 1). Specifically, for data-driven DRSO problems in which $\nu = 1/N\sum_{i=1}^N \delta_{\hat{\xi}^i}$ (where δ_ξ denotes the unit mass on ξ), whenever a worst-case distribution exists, there is a worst-case distribution μ^* supported on at most $N+1$ points with the following concise structure:

$$\mu^* = \frac{1}{N}\sum_{\substack{i=1\\ i\ne i_0}}^{N}\delta_{\xi^i_*} + \frac{p_0}{N}\delta_{\underline{\xi}^{i_0}_*} + \frac{1-p_0}{N}\delta_{\overline{\xi}^{i_0}_*},$$

for some $i_0\in\{1,\ldots,N\}$, $p_0\in[0,1]$ and

$$\xi^i_*\in\arg\min_{\xi\in\Xi}\left\{\lambda^* d^p(\xi,\hat{\xi}^i)-\Psi(x,\xi)\right\},\quad \forall i\ne i_0,\quad \underline{\xi}^{i_0}_*,\overline{\xi}^{i_0}_*\in\arg\min_{\xi\in\Xi}\left\{\lambda^* d^p(\xi,\hat{\xi}^{i_0})-\Psi(x,\xi)\right\},$$

where λ^* is the dual minimizer (Corollary 2). Thus, μ^* can be viewed as a perturbation of ν, where the mass on $\hat{\xi}^i$ is perturbed to ξ^i_* for all $i\ne i_0$, a fraction p_0 of the mass on $\hat{\xi}^{i_0}$ is perturbed to $\underline{\xi}^{i_0}_*$, and the remaining fraction $1-p_0$ of the mass on $\hat{\xi}^{i_0}$ is perturbed to $\overline{\xi}^{i_0}_*$. In particular, uncertainty quantification problems have a worst-case distribution with this simple structure and can be solved by a greedy procedure (Example 7). Our result regarding the existence of a worst-case distribution with such a structure improves the result of Wozabal [53] and the more recent result of Owhadi and Scovel [39] regarding the extremal distributions of Wasserstein balls.

• Connection with robust optimization: Using the structure of a worst-case distribution, we prove that data-driven DRSO problems can be approximated by robust optimization problems to any accuracy (Corollary 2(iii)). We use this result to show that two-stage linear DRSO problems with linear decision rules have a tractable semidefinite programming approximation (Section 5.2). Moreover, the robust optimization approximation becomes exact when the objective function Ψ is concave in ξ. In addition, if Ψ is convex in x, then the corresponding DRSO problem can be formulated as a convex–concave saddle-point problem.

The rest of this paper is organized as follows. In Section 2, we review some results on the Wasserstein distance. Next, we prove strong duality for general nominal distributions in Section 3.1, and in Section 3.2, we derive additional results for finite-supported nominal distributions. Then, in Sections 4 and 5, we apply our results on strong duality and the structural description of the worst-case distributions to a variety of DRSO problems. We conclude this paper in Section 6. Auxiliary results, as well as proofs of some lemmas, corollaries and propositions, are provided in the appendices.

2. Notation and Preliminaries

In this section, we introduce notation and briefly outline some known results regarding Wasserstein distance. For a more detailed discussion we refer to Villani [50, 51].

Let Ξ be a Polish (separable complete metric) space with metric d. Let $\mathscr{B}(\Xi)$ denote the Borel σ-algebra on Ξ, and let $\mathscr{B}_\nu(\Xi)$ denote the *completion* of $\mathscr{B}(\Xi)$ with respect to a measure ν on $\mathscr{B}(\Xi)$ such that the measure space $(\Xi, \mathscr{B}_\nu(\Xi), \nu)$ is complete (see, e.g., Ambrosio et al. [3, definition 1.11]). Let $\mathcal{B}(\Xi)$ denote the set of Borel measures on Ξ, let $\mathcal{P}(\Xi)$ denote the set of Borel probability measures on Ξ, and let $\mathcal{P}_p(\Xi)$ denote the subset of $\mathcal{P}(\Xi)$ with finite pth moment for $p \in [1, \infty)$:

$$\mathcal{P}_p(\Xi) := \left\{\mu \in \mathcal{P}(\Xi) : \int_\Xi d^p(\xi, \zeta^0)\mu(d\xi) < \infty \text{ for some } \zeta^0 \in \Xi\right\}.$$

It follows from the triangle inequality that the preceding definition does not depend on the choice of ζ^0. A function $\Psi : \Xi \longmapsto \mathbb{R}$ is called ν-measurable if it is $(\mathscr{B}_\nu(\Xi), \mathscr{B}(\mathbb{R}))$-measurable, and a function $T : \Xi \longmapsto \Xi$ is called ν-measurable if it is $(\mathscr{B}_\nu(\Xi), \mathscr{B}(\Xi))$-measurable. To facilitate later discussion, we introduce the push-forward operator on measures.

Definition 1 (Push-Forward Measure). Given measurable spaces $(\Xi, \mathscr{B}(\Xi))$ and $(\Xi', \mathscr{B}(\Xi'))$, a measurable function $T : \Xi \longmapsto \Xi'$, and a measure $\nu \in \mathcal{B}(\Xi)$, let $T_\#\nu \in \mathcal{B}(\Xi')$ denote the push-forward measure of ν through T defined by

$$T_\#\nu(A) := \nu(T^{-1}(A)) = \nu\{\zeta \in \Xi : T(\zeta) \in A\}, \ \forall \text{measurable sets } A \subset \Xi'.$$

That is, $T_\#\nu$ is obtained by *transporting* (pushing forward) ν from Ξ to Ξ' using the function T. For $i \in \{1,2\}$, let $\pi^i : \Xi \times \Xi \longmapsto \Xi$ denote the canonical projections given by $\pi^i(\xi^1, \xi^2) = \xi^i$. Then, for a measure $\gamma \in \mathcal{P}(\Xi \times \Xi)$, $\pi^i_\#\gamma \in \mathcal{P}(\Xi)$ is the ith marginal of γ given by $\pi^1_\#\gamma(A) = \gamma(A \times \Xi)$ and $\pi^2_\#\gamma(A) = \gamma(\Xi \times A)$.

Definition 2 (Wasserstein Distance). The Wasserstein distance $W_p(\mu, \nu)$ between $\mu, \nu \in \mathcal{P}_p(\Xi)$ is defined by

$$W_p^p(\mu, \nu) := \min_{\gamma \in \mathcal{P}(\Xi \times \Xi)} \left\{\int_{\Xi \times \Xi} d^p(\xi, \zeta)\gamma(d\xi, d\zeta) : \pi^1_\#\gamma = \mu, \pi^2_\#\gamma = \nu\right\}. \tag{1}$$

That is, the Wasserstein distance between μ, ν is the minimum cost (in terms of d^p) of redistributing mass from ν to μ, which is why it is also called the "earth mover's distance." Wasserstein distance is a natural way of comparing two distributions when one is obtained from the other by *perturbations*. The minimum on the right side of (1) is attained because d is nonnegative, continuous, and thus lower semicontinuous (Villani [50, theorem 1.3]). The following example is a familiar special case of Problem (1).

Example 2 (Transportation Problem). Consider $\mu = \sum_{i=1}^M p_i \delta_{\xi^i}$ and $\nu = \sum_{j=1}^N q_j \delta_{\hat{\xi}^j}$, where $M, N \geq 1$, $p_i, q_j \geq 0$, $\xi^i, \hat{\xi}^j \in \Xi$ for all i, j, and $\sum_{i=1}^M p_i = \sum_{j=1}^N q_j = 1$. Then, Problem (1) becomes the classic transportation problem in linear programming:

$$\min_{\gamma_{ij} \geq 0} \left\{\sum_{i=1}^M \sum_{j=1}^N d^p(\xi^i, \hat{\xi}^j)\gamma_{ij} : \sum_{j=1}^N \gamma_{ij} = p_i, \ \forall i, \ \sum_{i=1}^M \gamma_{ij} = q_j, \ \forall j\right\}.$$

Example 3 (Revisiting Example 1). Next, we evaluate the Wasserstein distance between the histograms in Example 1. To evaluate $W_1(\mu_{true}, \nu)$, note that the least cost way of transporting mass from ν to μ_{true} is to move the mass outward. In contrast, to evaluate $W_1(\mu_{pathol}, \nu)$, one has to transport mass relatively long distances from right to left (changing the grayscale values of pixels by large amounts), resulting in a larger cost than $W_1(\mu_{true}, \nu)$. Therefore, $W_1(\mu_{pathol}, \nu) > W_1(\mu_{true}, \nu)$.

Wasserstein distance has a dual representation due to Kantorovich's duality (Villani [51, theorem 5.10]):

$$W_p^p(\mu, \nu) = \sup_{u \in L^1(\mu), v \in L^1(\nu)} \left\{\int_\Xi u(\xi)\mu(d\xi) + \int_\Xi v(\zeta)\nu(d\zeta) : u(\xi) + v(\zeta) \leq d^p(\xi, \zeta), \ \forall \xi, \zeta \in \Xi\right\}, \tag{2}$$

where $L^1(\nu)$ represents the L^1 space of ν-measurable functions. In addition, $u \in L^1(\mu), v \in L^1(\nu)$ under the preceding supremum can be replaced by $u, v \in C_b(\Xi)$, where $C_b(\Xi)$ denotes the set of continuous and bounded real-valued functions on Ξ. Particularly, when $p = 1$, by the Kantorovich–Rubinstein theorem, (2) can be simplified to (see, e.g., Villani [51, equation (5.11)])

$$W_1(\mu,\nu) = \sup_{u \in L^1(\mu)} \left\{ \int_\Xi u(\xi) d(\mu - \nu)(\xi) : u \text{ is 1-Lipschitz} \right\}.$$

So, for an L-Lipschitz function $\Psi : \Xi \mapsto \mathbb{R}$, it holds that $|\mathbb{E}_\mu[\Psi(\xi)] - \mathbb{E}_\nu[\Psi(\xi)]| \le L W_1(\mu,\nu) \le L\theta$ for all $\mu \in \mathfrak{M}$.

Definition 2 and the preceding results can be extended to finite Borel measures. Moreover, we have the following result.

Lemma 1. *For any finite Borel measures $\mu, \nu \in \mathcal{B}(\Xi)$ with $\mu(\Xi) \neq \nu(\Xi)$, it holds that $W_p(\mu,\nu) = \infty$.*

Another important feature of Wasserstein distance is that W_p metrizes weak convergence in $\mathcal{P}_p(\Xi)$ (cf. Villani [51, theorem 6.9]). That is, for any sequence $\{\mu_k\}_{k=1}^\infty$ of measures in $\mathcal{P}_p(\Xi)$ and $\mu \in \mathcal{P}_p(\Xi)$, it holds that $\lim_{k\to\infty} W_p(\mu_k, \mu) = 0$ if and only if μ_k converges weakly to μ and $\int_\Xi d^p(\xi,\zeta^0)\mu_k(d\xi) \to \int_\Xi d^p(\xi,\zeta^0)\mu(d\xi)$ as $k \to \infty$. Therefore, convergence in the Wasserstein distance of order p implies convergence up to the pth moment. Villani [51, chapter 6] discusses the advantages of Wasserstein distance relative to other distances, such as the Prokhorov metric, that metrize weak convergence.

3. Tractable Reformulation via Duality

In this section, we develop a tractable reformulation by deriving its strong dual. We suppress the variable x of Ψ in this section, and results are interpreted pointwise for each x. Given $\nu \in \mathcal{P}(\Xi)$ and $\Psi \in L^1(\nu)$, for any $\theta > 0$ and $p \in [1,\infty)$, the inner maximization problem of (DRSO) is written as

$$v_P := \sup_{\mu \in \mathfrak{M}} \int_\Xi \Psi(\xi)\mu(d\xi) = \sup_{\mu \in \mathcal{P}(\Xi)} \left\{ \int_\Xi \Psi(\xi)\mu(d\xi) : W_p(\mu,\nu) \le \theta \right\}. \tag{Primal}$$

Our main goal is to derive its strong dual:

$$v_D := \inf_{\lambda \ge 0} \left\{ \lambda\theta^p - \int_\Xi \inf_{\xi\in\Xi} [\lambda d^p(\xi,\zeta) - \Psi(\xi)]\nu(d\zeta) \right\}. \tag{Dual}$$

The dual problem is a one-dimensional convex minimization problem with respect to λ, the Lagrangian multiplier of the Wasserstein constraint in the primal problem. The term $\inf_{\xi\in\Xi}[\lambda d^p(\xi,\zeta) - \Psi(\xi)]$ is called the *Moreau–Yosida regularization* of $-\Psi$ with parameter $1/\lambda$ in the literature (cf. Parikh and Boyd [41]). Its measurability with respect to ν is established in Lemma 3(i) in Section 3.1.

3.1. General Nominal Distribution

In this section, we prove the strong duality result for a general nominal distribution ν on a Polish space Ξ. Such generality broadens the applicability of the result for (DRSO). For example, the result is useful when the nominal distribution is some distribution such as a Gaussian distribution on $\mathbb{R}^K$ (Section 4.3) or even some stochastic process (Sections 4.1 and 4.2). We begin with a weak duality result, which is an application of Lagrangian weak duality.

Proposition 1 (Weak Duality). *Consider any $\nu \in \mathcal{P}(\Xi)$ and $\Psi \in L^1(\nu)$. Then, for any $p \in [1,\infty)$ and $\theta > 0$, it holds that $v_P \le v_D$.*

To prove the strong duality result, we consider two separate cases: $v_D = \infty$ and $v_D < \infty$. As can be seen from (Dual), if the term $-\int_\Xi \inf_{\xi\in\Xi}[\lambda d^p(\xi,\zeta) - \Psi(\xi)]\nu(d\zeta)$ is infinite for all $\lambda \ge 0$, then $v_D = \infty$. Thus, to facilitate our analysis, we introduce the following definitions.

Definition 3 (Regularization Operator Φ). Let $\Phi : \mathbb{R} \times \Xi \mapsto \mathbb{R} \cup \{-\infty\}$ be given by

$$\Phi(\lambda,\zeta) := \inf_{\xi\in\Xi} \{\lambda d^p(\xi,\zeta) - \Psi(\xi)\}.$$

For any $\lambda \geq 0$ and any $\zeta \in \Xi$ such that $\Phi(\lambda, \zeta) > -\infty$, let

$$\overline{D}(\lambda, \zeta) := \limsup_{\varepsilon \downarrow 0} \left\{ \sup_{\xi \in \Xi} \{ d^p(\xi, \zeta) : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon \} \right\},$$

$$\underline{D}(\lambda, \zeta) := \liminf_{\varepsilon \downarrow 0} \left\{ \inf_{\xi \in \Xi} \{ d^p(\xi, \zeta) : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon \} \right\}. \tag{3}$$

For any $\lambda \geq 0$ and any $\zeta \in \Xi$ such that $\arg\min_{\xi \in \Xi} \{ \lambda d^p(\xi, \zeta) - \Psi(\xi) \}$ is nonempty, let

$$\overline{D}_0(\lambda, \zeta) := \sup_{\xi \in \Xi} \{ d^p(\xi, \zeta) : \lambda d^p(\xi, \zeta) - \Psi(\xi) = \Phi(\lambda, \zeta) \}$$

$$\underline{D}_0(\lambda, \zeta) := \inf_{\xi \in \Xi} \{ d^p(\xi, \zeta) : \lambda d^p(\xi, \zeta) - \Psi(\xi) = \Phi(\lambda, \zeta) \}. \tag{4}$$

Then, $\underline{D}_0(\lambda, \zeta)$ and $\overline{D}_0(\lambda, \zeta)$ represent, respectively, the closest and furthest distances between ζ and any point in $\arg\min_{\xi \in \Xi} \{ \lambda d^p(\xi, \zeta) - \Psi(\xi) \}$. Note that $\overline{D}_0(\lambda, \zeta)$ (respectively, $\underline{D}_0(\lambda, \zeta)$) may not be equal to $\overline{D}(\lambda, \zeta)$ ($\underline{D}(\lambda, \zeta)$).

Definition 4 (Growth Rate). Define the growth rate κ of Ψ as

$$\kappa := \inf \left\{ \lambda \geq 0 : \int_{\Xi} \Phi(\lambda, \zeta) \nu(d\zeta) > -\infty \right\}.$$

Particularly, if $\int_{\Xi} \Phi(\lambda, \zeta) \nu(d\zeta) = -\infty$ for all $\lambda \geq 0$, then $\kappa = \infty$.

If Ξ is bounded and Ψ is bounded above, then $\kappa = 0$, and if Ξ is bounded and Ψ is not bounded above, then $\kappa = \infty$. The possibilities when Ξ is not bounded are more interesting. Next, Lemma 2 establishes some additional properties of κ, including the property that, if Ξ is not bounded and $\kappa < \infty$, then

$$\kappa = \limsup_{\xi \in \Xi : d^p(\xi, \zeta) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta)\}}{d^p(\xi, \zeta)}$$

for any $\zeta \in \Xi$, which motivates why we call κ the growth rate of Ψ.

Lemma 2 (Properties of the Growth Rate κ).

i. *Suppose that Ξ is unbounded. Then, the quantity*

$$\limsup_{\xi \in \Xi : d^p(\xi, \zeta) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta)\}}{d^p(\xi, \zeta)}$$

is independent of ζ.

ii. *Suppose that $\nu \in \mathcal{P}_p(\Xi)$. Then, the growth rate κ is finite if and only if there exists $\zeta^0 \in \Xi$ and $L, M > 0$ such that*

$$\Psi(\xi) - \Psi(\zeta^0) \leq L d^p(\xi, \zeta^0) + M \quad \forall \xi \in \Xi. \tag{5}$$

iii. *Suppose that $\nu \in \mathcal{P}_p(\Xi)$. If Ξ is unbounded and $\kappa < \infty$, then*

$$\kappa = \limsup_{\xi \in \Xi : d^p(\xi, \zeta) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta)\}}{d^p(\xi, \zeta)}$$

for any $\zeta \in \Xi$.

Lemma 3 (Measurability). *For any $p \in [1, \infty)$, $\nu \in \mathcal{P}(\Xi)$, and $\Psi \in L^1(\nu)$, the following hold:*

i. *$\Phi(\lambda, \cdot)$, $\overline{D}(\lambda, \cdot)$, $\underline{D}(\lambda, \cdot)$, $\overline{D}_0(\lambda, \cdot)$, and $\underline{D}_0(\lambda, \cdot)$ are ν-measurable.*

ii. *Suppose that $\kappa < \infty$. Then, for any $\lambda, \delta, \varepsilon \geq 0$ such that the sets*

$$\overline{F}_\delta^\varepsilon(\lambda, \zeta) := \{ \xi \in \Xi : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon, \ d^p(\xi, \zeta) \geq \overline{D}(\lambda, \zeta) - \delta \},$$

$$\underline{F}_\delta^\varepsilon(\lambda, \zeta) := \{ \xi \in \Xi : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon, \ d^p(\xi, \zeta) \leq \underline{D}(\lambda, \zeta) + \delta \}$$

are nonempty for ν-almost all $\zeta \in \Xi$, there exists ν-measurable mappings $\overline{T}^{\varepsilon}_{\delta}(\lambda,\cdot), \underline{T}^{\varepsilon}_{\delta}(\lambda,\cdot) : \Xi \longmapsto \Xi$ such that $\overline{T}^{\varepsilon}_{\delta}(\lambda,\zeta) \in \overline{F}^{\varepsilon}_{\delta}(\lambda,\zeta)$ and $\underline{T}^{\varepsilon}_{\delta}(\lambda,\zeta) \in \underline{F}^{\varepsilon}_{\delta}(\lambda,\zeta)$ for ν-almost all $\zeta \in \Xi$.

iii. *Suppose that $\kappa < \infty$. Then, for any $\lambda, \delta \geq 0$ such that the sets*

$$\overline{F}(\lambda,\zeta) := \{\xi \in \Xi : \lambda d^p(\xi,\zeta) - \Psi(\xi) = \Phi(\lambda,\zeta),\, d^p(\xi,\zeta) \geq \overline{D}_0(\lambda,\zeta) - \delta\},$$

$$\underline{F}(\lambda,\zeta) := \{\xi \in \Xi : \lambda d^p(\xi,\zeta) - \Psi(\xi) = \Phi(\lambda,\zeta),\, d^p(\xi,\zeta) \leq \underline{D}_0(\lambda,\zeta) + \delta\}$$

are nonempty for ν-almost all $\zeta \in \Xi$, there exists ν-measurable mappings $\overline{T}(\lambda,\cdot), \underline{T}(\lambda,\cdot) : \Xi \longmapsto \Xi$ such that $\overline{T}(\lambda,\zeta) \in \overline{F}(\lambda,\zeta)$ and $\underline{T}(\lambda,\zeta) \in \underline{F}(\lambda,\zeta)$ for ν-almost all $\zeta \in \Xi$.

iv. *For any $E \in \mathscr{B}_\nu(\Xi)$ and $a, b > 0$ such that*

$$F(\zeta) := \{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) > a d^p(\xi,\zeta) + b\}$$

is nonempty for ν-almost all $\zeta \in E$, there exists a ν-measurable mapping $T : E \longmapsto \Xi$ such that $T(\zeta) \in F(\zeta)$ for ν-almost all $\zeta \in E$.

v. *Suppose that $\kappa < \infty$. For any $\kappa' \in (0,\kappa)$ and any ν-measurable function $M : \Xi \longmapsto \mathbb{R}$ such that the set*

$$F(\zeta) := \{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi,\zeta),\, d^p(\xi,\zeta) \geq M(\zeta)\}$$

is nonempty for ν-almost all $\zeta \in \Xi$, there exists a ν-measurable mapping $T : \Xi \longmapsto \Xi$ such that $T(\zeta) \in F(\zeta)$ for ν-almost all $\zeta \in \Xi$.

Proposition 2 (Strong Duality with Infinite Optimal Value). *Consider any $p \in [1,\infty)$, $\nu \in \mathcal{P}(\Xi)$, and $\Psi \in L^1(\nu)$. Suppose that $\theta > 0$ and $\kappa = \infty$. Then, $v_P = v_D = \infty$.*

Remark 1 (Choosing Wasserstein Order *p*). Let

$$\underline{p} := \inf\left\{p \geq 1 : \limsup_{d(\zeta,\zeta^0)\to\infty} \frac{\Psi(\zeta) - \Psi(\zeta^0)}{d^p(\zeta,\zeta^0)} < \infty\right\}.$$

Proposition 2 suggests that a meaningful formulation of (DRSO) should be such that the Wasserstein order p is greater than or equal to $\underline{p}$. In both Esfahani and Kuhn [23] and Zhao and Guan [57], only $p = 1$ is considered. By considering higher orders p in our analysis, we can accommodate a greater set of functions Ψ, and we also have more flexibility to choose the ambiguity set and to control the degree of conservativeness.

Lemma 4 (Properties of the Regularization Operator Φ]). *Let (Ξ, d) be a Polish space. Consider any $p \in [1,\infty)$, $\nu \in \mathcal{P}(\Xi)$, and $\Psi \in L^1(\nu)$ such that $\kappa < \infty$. Then, there is a set $B \in \mathscr{B}_\nu(\Xi)$ such that $\nu(B) = 1$, and the following holds.*

i. *(Monotonicity) For all $\zeta \in \Xi$ $\Phi(\cdot,\zeta)$ is nondecreasing and upper semicontinuous.* $\Phi(\lambda,\zeta) > -\infty$ for all $\lambda > \kappa$ and all $\zeta \in B$. *$\Phi(\cdot,\zeta)$ is concave for all $\zeta \in B$. For any $\varepsilon \geq 0$, any $\lambda_2 > \lambda_1$, and any $\zeta \in \Xi$ such that $\Phi(\lambda_1,\zeta) > -\infty$, it holds that*

$$\sup_{\xi\in\Xi} \{d^p(\xi,\zeta) : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \leq \Phi(\lambda_2,\zeta) + \varepsilon\} \leq \sup_{\xi\in\Xi} \{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \leq \Phi(\lambda_1,\zeta) + \varepsilon\}.$$

For any $\lambda_2 > \lambda_1$ and any $\zeta \in \Xi$ such that $\Phi(\lambda_1,\zeta) > -\infty$, it holds that $\overline{D}(\lambda_2,\zeta) \leq \underline{D}(\lambda_1,\zeta) \leq \overline{D}(\lambda_1,\zeta)$.

ii. *(Bounds) For any $\lambda_2 > \lambda_1$ such that $\Phi(\lambda_1,\zeta) > -\infty$, it holds that*

$$(\lambda_2 - \lambda_1)\overline{D}(\lambda_2,\zeta) \leq -\Psi(\zeta) - \Phi(\lambda_1,\zeta).$$

iii. *(Derivative) For all $\lambda > \kappa$ and all $\zeta \in B$, the left partial derivative $\partial\Phi(\lambda,\zeta)/\partial\lambda-$ exists and satisfies*

$$\overline{D}(\lambda,\zeta) \leq \frac{\partial\Phi(\lambda,\zeta)}{\partial\lambda-} \leq \lim_{\lambda_1\uparrow\lambda} \underline{D}(\lambda_1,\zeta).$$

For all $\lambda \geq 0$ and $\zeta \in \Xi$ such that $\Phi(\lambda,\zeta) > -\infty$, the right partial derivative $\partial\Phi(\lambda,\zeta)/\partial\lambda+$ exists and satisfies

$$\lim_{\lambda_2\downarrow\lambda} \overline{D}(\lambda_2,\zeta) \leq \frac{\partial\Phi(\lambda,\zeta)}{\partial\lambda+} \leq \underline{D}(\lambda,\zeta).$$

Let $h: \mathbb{R}_+ \longmapsto \mathbb{R} \cup \{\infty\}$ denote the dual objective function given by

$$h(\lambda) := \lambda\theta^p - \int_\Xi \Phi(\lambda, \zeta)\nu(d\zeta).$$

Lemma 5 (Dual Objective Function). *The dual objective function h has the following properties:*

i. $h(\lambda) = \infty$ *for all* $\lambda \in [0, \kappa)$ *and* $h(\lambda) < \infty$ *for all* $\lambda \in (\kappa, \infty)$.
ii. h *is a convex function.*
iii. h *is a lower semicontinuous function.*
iv. $h(\lambda) \longrightarrow \infty$ *as* $\lambda \longrightarrow \infty$.
v. h *has a minimizer* $\lambda^* \in [\kappa, \infty)$.

Lemma 6 (Structure of ε-Optimal Primal Solution with $\lambda^* > \kappa$). *Consider any* $p \in [1, \infty)$, *any* $\nu \in \mathcal{P}(\Xi)$, *any* $\theta > 0$, *and any* $\Psi \in L^1(\nu)$ *such that* $\kappa < \infty$. *Suppose that h has a minimizer* $\lambda^* > \kappa$. *Then, for any* $\varepsilon > 0$, *there are a* $\lambda_1^\varepsilon \in (\max\{\kappa, \lambda^* - \varepsilon\}, \lambda^*)$, *a* $\lambda_2^\varepsilon \in (\lambda^*, \lambda^* + \varepsilon)$, *a* $\delta^\varepsilon \in (0, \varepsilon)$, ν*-measurable mappings* $T_1^\varepsilon, T_2^\varepsilon : \Xi \longmapsto \Xi$, *and* $p_1^\varepsilon, p_2^\varepsilon, p_3^\varepsilon \geq 0$ *such that* $p_1^\varepsilon + p_2^\varepsilon + p_3^\varepsilon = 1$,

$$T_1^\varepsilon(\zeta) \in \left\{\xi \in \Xi : \lambda_1^\varepsilon d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda_1^\varepsilon, \zeta) + \varepsilon,\ d^p(\xi, \zeta) \geq \overline{D}(\lambda_1^\varepsilon, \zeta) - \delta^\varepsilon\right\},$$
$$T_2^\varepsilon(\zeta) \in \left\{\xi \in \Xi : \lambda_2^\varepsilon d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda_2^\varepsilon, \zeta) + \varepsilon,\ d^p(\xi, \zeta) \leq \underline{D}(\lambda_2^\varepsilon, \zeta) + \delta^\varepsilon\right\}$$

for ν*-almost all* $\zeta \in \Xi$, *and*

$$\mu^\varepsilon := p_1^\varepsilon T_{1\#}^\varepsilon \nu + p_2^\varepsilon T_{2\#}^\varepsilon \nu + p_3^\varepsilon \nu$$

satisfies

$$\int_\Xi \Psi(\xi)\mu^\varepsilon(d\xi) \geq \lambda^*\theta^p - \int_\Xi \Phi(\lambda^*, \zeta)\nu(d\zeta) - \varepsilon, \tag{6}$$

and

$$W_p^p(\mu^\varepsilon, \nu) \leq p_1 \int_\Xi d^p(T_1^\varepsilon(\zeta), \zeta)\nu(d\zeta) + p_2 \int_\Xi d^p(T_2^\varepsilon(\zeta), \zeta)\nu(d\zeta) \leq \theta^p. \tag{7}$$

Proof of Lemma 6. Note that, for any $\lambda > \kappa$ and $\delta, \varepsilon > 0$, the sets $\overline{F}(\lambda, \zeta), \underline{F}(\lambda, \zeta)$ in Lemma 3(ii) are nonempty for ν-almost all $\zeta \in \Xi$. Hence, there exist ν-measurable mappings $\overline{T}_\delta^\varepsilon(\lambda, \cdot), \underline{T}_\delta^\varepsilon(\lambda, \cdot) : \Xi \longmapsto \Xi$ such that

$$\overline{T}_\delta^\varepsilon(\lambda, \zeta) \in \{\xi \in \Xi : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon,\ d^p(\xi, \zeta) \geq \overline{D}(\lambda, \zeta) - \delta\},$$
$$\underline{T}_\delta^\varepsilon(\lambda, \zeta) \in \{\xi \in \Xi : \lambda d^p(\xi, \zeta) - \Psi(\xi) \leq \Phi(\lambda, \zeta) + \varepsilon,\ d^p(\xi, \zeta) \leq \underline{D}(\lambda, \zeta) + \delta\}$$

for ν-almost all $\zeta \in \Xi$.

The first order optimality conditions $\frac{\partial}{\partial\lambda-} h(\lambda^*) \leq 0$ and $\frac{\partial}{\partial\lambda+} h(\lambda^*) \geq 0$ imply that

$$\frac{\partial}{\partial\lambda+}\left(\int_\Xi \Phi(\lambda^*, \zeta)\nu(d\zeta)\right) \leq \theta^p \leq \frac{\partial}{\partial\lambda-}\left(\int_\Xi \Phi(\lambda^*, \zeta)\nu(d\zeta)\right). \tag{8}$$

Next, we verify that we can interchange the partial derivative and integration in (8). Recall from Lemma 4(i) that there is a set $B \in \mathscr{B}_\nu(\Xi)$ such that $\nu(B) = 1$, and $\Phi(\lambda, \zeta) > -\infty$ for all $\lambda > \kappa$ and all $\zeta \in B$, and $\Phi(\cdot, \zeta)$ is concave for all $\zeta \in B$. To show it for the right derivative in (8), consider any $\zeta \in B$ and any decreasing sequence $\lambda^n \downarrow \lambda^*$. Let

$$f_n(\zeta) := \frac{\Phi(\lambda^n, \zeta) - \Phi(\lambda^*, \zeta)}{\lambda^n - \lambda^*}.$$

Because $\Phi(\cdot, \zeta)$ is nondecreasing for all ζ, $f_n(\zeta) \geq 0$ for all ζ. In addition, because $\Phi(\cdot, \zeta)$ is concave, $f_n \leq f_{n+1}$ for all n. Note that $\lim_{n\to\infty} f_n(\zeta) = \frac{\partial}{\partial\lambda+}\Phi(\lambda^*, \zeta)$. Thus, it follows from the monotone convergence theorem that

$$\frac{\partial}{\partial\lambda+}\left(\int_\Xi \Phi(\lambda^*, \zeta)\nu(d\zeta)\right) = \int_\Xi \frac{\partial}{\partial\lambda+}\Phi(\lambda^*, \zeta)\nu(d\zeta). \tag{9}$$

To show it for the left derivative in (8), consider any $\zeta \in B$ and any increasing sequence $\lambda^n \uparrow \lambda^*$ with $\lambda^1 > \kappa$. Let f_n be defined as before, and thus, $f_n(\zeta) \geq 0$ for all ζ. In addition, because $\Phi(\cdot,\zeta)$ is concave, $f_n \geq f_{n+1}$ for all n. That is, for all n, it holds that

$$\left|f_n(\zeta)\right| = f_n(\zeta) \leq f_1(\zeta) \leq \frac{\left|\Phi(\lambda^1,\zeta)\right| + |\Phi(\lambda^*,\zeta)|}{\lambda^* - \lambda^1}.$$

It follows from $\lambda^1 > \kappa$ that

$$\int_\Xi f_1(\zeta)\nu(d\zeta) \leq \frac{\int_\Xi \left|\Phi(\lambda^1,\zeta)\right|\nu(d\zeta) + \int_\Xi |\Phi(\lambda^*,\zeta)|\nu(d\zeta)}{\lambda^* - \lambda^1} < \infty.$$

Also, $\lim_{n\to\infty} f_n(\zeta) = \frac{\partial}{\partial\lambda-}\Phi(\lambda^*,\zeta)$. Thus, it follows from the dominated convergence theorem that

$$\frac{\partial}{\partial\lambda-}\left(\int_\Xi \Phi(\lambda^*,\zeta)\nu(d\zeta)\right) = \int_\Xi \frac{\partial}{\partial\lambda-}\Phi(\lambda^*,\zeta)\nu(d\zeta). \tag{10}$$

Therefore, it follows from (8)–(10) and Lemma 4(iii) that

$$\begin{aligned}
\theta^p &\geq \frac{\partial}{\partial\lambda+}\left(\int_\Xi \Phi(\lambda^*,\zeta)\nu(d\zeta)\right) = \int_\Xi \frac{\partial}{\partial\lambda+}\Phi(\lambda^*,\zeta)\nu(d\zeta) \geq \int_\Xi \lim_{\lambda\downarrow\lambda^*} \overline{D}(\lambda,\zeta)\nu(d\zeta),\\
\theta^p &\leq \frac{\partial}{\partial\lambda-}\left(\int_\Xi \Phi(\lambda^*,\zeta)\nu(d\zeta)\right) = \int_\Xi \frac{\partial}{\partial\lambda-}\Phi(\lambda^*,\zeta)\nu(d\zeta) \leq \int_\Xi \lim_{\lambda\uparrow\lambda^*} \underline{D}(\lambda,\zeta)\nu(d\zeta).
\end{aligned} \tag{11}$$

In particular, for any λ_1, λ_2 with $\kappa < \lambda_1 < \lambda^* < \lambda_2$, it follows from (11) and Lemma 4(i) that

$$\begin{aligned}
\theta^p &\geq \int_\Xi \lim_{\lambda\downarrow\lambda^*} \overline{D}(\lambda,\zeta)\nu(d\zeta) \geq \int_\Xi \overline{D}(\lambda_2,\zeta)\nu(d\zeta) \geq \int_\Xi \underline{D}(\lambda_2,\zeta)\nu(d\zeta) \geq \int_\Xi d^p(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta),\zeta)\nu(d\zeta) - \delta,\\
\theta^p &\leq \int_\Xi \lim_{\lambda\uparrow\lambda^*} \underline{D}(\lambda,\zeta)\nu(d\zeta) \leq \int_\Xi \underline{D}(\lambda_1,\zeta)\nu(d\zeta) \leq \int_\Xi \overline{D}(\lambda_1,\zeta)\nu(d\zeta) \leq \int_\Xi d^p(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta),\zeta)\nu(d\zeta) + \delta.
\end{aligned} \tag{12}$$

Based on (12), we now construct a feasible primal solution. Note that there is a $q^\varepsilon_\delta(\lambda_1,\lambda_2) \in [0,1]$ such that

$$q^\varepsilon_\delta(\lambda_1,\lambda_2)\left[\int_\Xi d^p(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta),\zeta)\nu(d\zeta) + \delta\right] + (1 - q^\varepsilon_\delta(\lambda_1,\lambda_2))\left[\int_\Xi d^p(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta),\zeta)\nu(d\zeta) - \delta\right] = \theta^p. \tag{13}$$

Let $q^\delta := \theta^p/(\theta^p + \delta)$. Define a distribution $\mu^\varepsilon_\delta(\lambda_1,\lambda_2)$ by

$$\mu^\varepsilon_\delta(\lambda_1,\lambda_2) := q^\delta q^\varepsilon_\delta(\lambda_1,\lambda_2)\overline{T}^\varepsilon_\delta(\lambda_1,\cdot)_\#\nu + q^\delta(1 - q^\varepsilon_\delta(\lambda_1,\lambda_2))\underline{T}^\varepsilon_\delta(\lambda_2,\cdot)_\#\nu + (1 - q^\delta)\nu. \tag{14}$$

Then, $\mu^\varepsilon_\delta(\lambda_1,\lambda_2)$ is primal feasible because

$$\begin{aligned}
W^p_p(\mu^\varepsilon_\delta(\lambda_1,\lambda_2),\nu) &\leq q^\delta q^\varepsilon_\delta(\lambda_1,\lambda_2)\int_\Xi d^p(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta),\zeta)\nu(d\zeta) + q^\delta(1 - q^\varepsilon_\delta(\lambda_1,\lambda_2))\int_\Xi d^p(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta),\zeta)\nu(d\zeta)\\
&= q^\delta(\theta^p + [1 - 2q^\varepsilon_\delta(\lambda_1,\lambda_2)]\delta) \leq \theta^p.
\end{aligned} \tag{15}$$

Furthermore, recall that

$$\begin{aligned}
\lambda_1 d^p(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta),\zeta) - \Phi(\lambda_1,\zeta) - \varepsilon &\leq \Psi(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta)) \leq \lambda_1 d^p(\overline{T}^\varepsilon_\delta(\lambda_1,\zeta),\zeta) - \Phi(\lambda_1,\zeta),\\
\lambda_2 d^p(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta),\zeta) - \Phi(\lambda_2,\zeta) - \varepsilon &\leq \Psi(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta)) \leq \lambda_2 d^p(\underline{T}^\varepsilon_\delta(\lambda_2,\zeta),\zeta) - \Phi(\lambda_2,\zeta).
\end{aligned}$$

This is for ν-almost all $\zeta \in \Xi$. This, together with (13), implies that

$$\begin{aligned}
&\int_{\Xi} \Psi(\xi)\mu_{\delta}^{\varepsilon}(\lambda_1,\lambda_2)(d\xi) \\
&= q^{\delta} q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2)\int_{\Xi}\Psi(\overline{T}_{\delta}^{\varepsilon}(\lambda_1,\zeta))\nu(d\zeta) + q^{\delta}(1-q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2))\int_{\Xi}\Psi(\underline{T}_{\delta}^{\varepsilon}(\lambda_2,\zeta))\nu(d\zeta) + (1-q^{\delta})\int_{\Xi}\Psi(\zeta)\nu(d\zeta) \\
&\geq q^{\delta} q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2)\int_{\Xi}[\lambda_1 d^p(\overline{T}_{\delta}^{\varepsilon}(\lambda_1,\zeta),\zeta) - \Phi(\lambda_1,\zeta) - \varepsilon]\nu(d\zeta) + q^{\delta}(1-q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2))\int_{\Xi}[\lambda_2 d^p(\underline{T}_{\delta}^{\varepsilon}(\lambda_2,\zeta),\zeta) - \Phi(\lambda_2,\zeta) - \varepsilon]\nu(d\zeta) \\
&\quad + (1-q^{\delta})\int_{\Xi}\Psi(\zeta)\nu(d\zeta) \\
&\geq q^{\delta}\lambda_1[\theta^p + (1-2q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2))\delta] - q^{\delta}q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2)\int_{\Xi}\Phi(\lambda_1,\zeta)\nu(d\zeta) \\
&\quad - q^{\delta}(1-q_{\delta}^{\varepsilon}(\lambda_1,\lambda_2))\int_{\Xi}\Phi(\lambda_2,\zeta)\nu(d\zeta) - q^{\delta}\varepsilon + (1-q^{\delta})\int_{\Xi}\Psi(\zeta)\nu(d\zeta).
\end{aligned} \tag{16}$$

Recall that $\Phi(\lambda,\zeta) \leq -\Psi(\zeta)$ for all $\zeta \in \Xi$. Also, consider any $\lambda_0 \in (\kappa,\lambda_1)$. Recall that $\Phi(\cdot,\zeta)$ is nondecreasing, and thus, $|\Phi(\lambda,\zeta)| \leq |\Psi(\zeta)| + |\Phi(\lambda_0,\zeta)|$ for all $\lambda \geq \lambda_0$ and all $\zeta \in \Xi$. Also, it follows from $\lambda_0 > \kappa$ that $\int_{\Xi}\Phi(\lambda_0,\zeta)\nu(d\zeta) > -\infty$. Hence, it follows from the dominated convergence theorem that $\lim_{\lambda_1\uparrow\lambda^*}\int_{\Xi}\Phi(\lambda_1,\zeta)\nu(d\zeta) = \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)$ and $\lim_{\lambda_2\downarrow\lambda^*}\int_{\Xi}\Phi(\lambda_2,\zeta)\nu(d\zeta) = \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)$. Thus, given any $\varepsilon > 0$, choose $\lambda_1^{\varepsilon} \in (\max\{\kappa, \lambda^* - \varepsilon\}, \lambda^*)$ such that $\int_{\Xi}\Phi(\lambda_1^{\varepsilon},\zeta)\nu(d\zeta) \leq \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) + \varepsilon$ and $(\lambda^* - \lambda_1^{\varepsilon})\theta^p \leq \varepsilon$, choose $\lambda_2^{\varepsilon} \in (\lambda^*, \lambda^* + \varepsilon)$ such that $\int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta) \leq \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) + \varepsilon$, and choose $\delta^{\varepsilon} \in (0,\varepsilon)$ such that $2\lambda_1^{\varepsilon}\delta^{\varepsilon} \leq \varepsilon$, $-\delta^{\varepsilon}\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) \leq \theta^p\varepsilon$ and $-\delta^{\varepsilon}\int_{\Xi}\Psi(\zeta)\nu(d\zeta) \leq \theta^p\varepsilon$. Set

$$T_1^{\varepsilon} := \overline{T}_{\delta^{\varepsilon}}^{\varepsilon}(\lambda_1^{\varepsilon},\cdot),\ T_2^{\varepsilon} := \underline{T}_{\delta^{\varepsilon}}^{\varepsilon}(\lambda_2^{\varepsilon},\cdot),\ p_1^{\varepsilon} := q^{\delta^{\varepsilon}} q_{\delta^{\varepsilon}}^{\varepsilon}(\lambda_1^{\varepsilon},\lambda_2^{\varepsilon}),\ p_2^{\varepsilon} := q^{\delta^{\varepsilon}}(1 - q_{\delta^{\varepsilon}}^{\varepsilon}(\lambda_1^{\varepsilon},\lambda_2^{\varepsilon})),\ p_3^{\varepsilon} := 1 - q^{\delta^{\varepsilon}}.$$

Thus, (7) follows from (15). Also, it follows from (16) that

$$\begin{aligned}
&\int_{\Xi}\Psi(\xi)\mu^{\varepsilon}(d\xi) \\
&\geq \frac{\theta^p}{(\theta^p+\delta^{\varepsilon})}\lambda_1^{\varepsilon}(\theta^p - \delta^{\varepsilon}) - \frac{\theta^p}{(\theta^p+\delta^{\varepsilon})}\left(\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) + \varepsilon\right) - \frac{\theta^p}{(\theta^p+\delta^{\varepsilon})}\varepsilon + \frac{\delta^{\varepsilon}}{(\theta^p+\delta^{\varepsilon})}\int_{\Xi}\Psi(\zeta)\nu(d\zeta) \\
&= \lambda^*\theta^p - (\lambda^* - \lambda_1^{\varepsilon})\theta^p - 2\frac{\delta^{\varepsilon}}{(\theta^p+\delta^{\varepsilon})}\lambda_1^{\varepsilon}\theta^p - \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) + \frac{\delta^{\varepsilon}}{(\theta^p+\delta^{\varepsilon})}\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) \\
&\quad - 2\frac{\theta^p}{(\theta^p+\delta^{\varepsilon})}\varepsilon + \frac{\delta^{\varepsilon}}{(\theta^p+\delta^{\varepsilon})}\int_{\Xi}\Psi(\zeta)\nu(d\zeta) \\
&\geq \lambda^*\theta^p - \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) - 6\varepsilon. \quad \square
\end{aligned}$$

Lemma 7 (Structure of ε-Optimal Primal Solution with $\lambda^* = \kappa$). *Consider any $p \in [1,\infty)$, any $\nu \in \mathcal{P}(\Xi)$, any $\theta > 0$, and any $\Psi \in L^1(\nu)$ such that $\kappa < \infty$. Suppose that κ is the unique minimizer of h. If $\kappa = 0$, then, for any $\varepsilon > 0$, there are a $\lambda^{\varepsilon} \in (0,\varepsilon)$ and a ν-measurable mapping $T^{\varepsilon} : \Xi \longmapsto \Xi$ such that $\lambda^{\varepsilon} d^p(T^{\varepsilon}(\zeta),\zeta) - \Psi(T^{\varepsilon}(\zeta)) \leq \Phi(\lambda^{\varepsilon},\zeta) + \varepsilon$ for ν-almost all $\zeta \in \Xi$ and*

$$\mu^{\varepsilon} := T_{\#}^{\varepsilon}\nu$$

satisfies

$$\int_{\Xi}\Psi(\xi)\mu^{\varepsilon}(d\xi) \geq \kappa\theta^p - \int_{\Xi}\Phi(\kappa,\zeta)\nu(d\zeta) - \varepsilon, \tag{17}$$

and

$$W_p^p(\mu^{\varepsilon},\nu) \leq \int_{\Xi} d^p(T^{\varepsilon}(\zeta),\zeta)\nu(d\zeta) \leq \theta^p. \tag{18}$$

If $\kappa > 0$, then for any $\varepsilon > 0$, there is a $\lambda_1^\varepsilon \in (\kappa - \varepsilon, \kappa)$, a $\lambda_2^\varepsilon \in (\kappa, \kappa + \varepsilon)$, ν-measurable mappings $T_1^\varepsilon, T_2^\varepsilon : \Xi \longmapsto \Xi$, and $p^\varepsilon \in (0,1)$ such that $\Psi(T_1^\varepsilon(\zeta)) - \Psi(\zeta) \geq \lambda_1^\varepsilon d^p(T_1^\varepsilon(\zeta), \zeta)$ and $\lambda_2^\varepsilon d^p(T_2^\varepsilon(\zeta), \zeta) - \Psi(T_2^\varepsilon(\zeta)) \leq \Phi(\lambda_2^\varepsilon, \zeta) + \varepsilon$ for ν-almost all $\zeta \in \Xi$, $\int_\Xi d^p(T_1^\varepsilon(\zeta), \zeta)\nu(d\zeta) > \theta^p + \theta^p/\varepsilon$ and

$$\mu^\varepsilon := p^\varepsilon T_{1\#}^\varepsilon \nu + (1 - p^\varepsilon) T_{2\#}^\varepsilon \nu$$

satisfies (17) *and*

$$W_p^p(\mu^\varepsilon, \nu) \leq p^\varepsilon \int_\Xi d^p(T_1^\varepsilon(\zeta), \zeta)\nu(d\zeta) + (1 - p^\varepsilon) \int_\Xi d^p(T_2^\varepsilon(\zeta), \zeta)\nu(d\zeta) = \theta^p. \tag{19}$$

Proof of Lemma 7. If κ is the unique minimizer of h, then h is increasing and convex on $[\kappa, \infty)$. For any $\lambda > \kappa$, it follows from h being increasing that

$$\int_\Xi [\Phi(\lambda, \zeta) - \Phi(\kappa, \zeta)]\nu(d\zeta) < (\lambda - \kappa)\theta^p. \tag{20}$$

Consider any $\varepsilon \in \left(0, (\lambda - \kappa)\theta^p - \int_\Xi [\Phi(\lambda, \zeta) - \Phi(\kappa, \zeta)]\nu(d\zeta)\right)$. It follows from Lemma 3 that there exists a ν-measurable map $\underline{T}_\varepsilon(\lambda, \cdot) : \Xi \longmapsto \Xi$ such that $\lambda d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta) - \Psi(\underline{T}_\varepsilon(\lambda, \zeta)) \leq \Phi(\lambda, \zeta) + \varepsilon$ for ν-almost all $\zeta \in \Xi$. Also, note that $\Phi(\kappa, \zeta) \leq \kappa d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta) - \Psi(\underline{T}_\varepsilon(\lambda, \zeta))$. Thus,

$$\Phi(\lambda, \zeta) - \Phi(\kappa, \zeta) \geq (\lambda - \kappa) d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta) - \varepsilon$$

for ν-almost all $\zeta \in \Xi$. This, together with (20), yield that

$$(\lambda - \kappa) \int_\Xi d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta)\nu(d\zeta) \leq \int_\Xi [\Phi(\lambda, \zeta) - \Phi(\kappa, \zeta)]\nu(d\zeta) + \varepsilon < (\lambda - \kappa)\theta^p \Rightarrow \int_\Xi d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta)\nu(d\zeta) < \theta^p. \tag{21}$$

Hence, the distribution $\underline{T}_\varepsilon(\lambda, \cdot)_\# \nu$ is primal feasible.

Next, we separately consider the cases $\kappa = 0$ and $\kappa > 0$. If $\kappa = 0$, then for the distribution

$$\mu_\varepsilon(\lambda) = \underline{T}_\varepsilon(\lambda, \cdot)_\# \nu, \tag{22}$$

it holds that

$$\int_\Xi \Psi(\xi)\mu_\varepsilon(\lambda)(d\xi) = \int_\Xi \Psi(\underline{T}_\varepsilon(\lambda, \zeta))\nu(d\zeta) \geq \int_\Xi [\lambda d^p(\underline{T}_\varepsilon(\lambda, \zeta), \zeta) - \Phi(\lambda, \zeta) - \varepsilon]\nu(d\zeta) \geq -\int_\Xi \Phi(\lambda, \zeta)\nu(d\zeta) - \varepsilon. \tag{23}$$

Thus, given any $\varepsilon > 0$, choose $\lambda^\varepsilon \in (0, \varepsilon)$ such that $\lambda^\varepsilon \theta^p \leq \varepsilon$ and choose $\varepsilon' \in \left(0, \lambda^\varepsilon \theta^p - \int_\Xi [\Phi(\lambda^\varepsilon, \zeta) - \Phi(0, \zeta)]\nu(d\zeta)\right)$ (note that $\varepsilon' \leq \varepsilon$). Set $T^\varepsilon := \underline{T}_{\varepsilon'}(\lambda^\varepsilon, \cdot)$. Thus, (18) follows from (21). Also, it follows from (23) that

$$\begin{aligned}
\int_\Xi \Psi(\xi)\mu^\varepsilon(d\xi) &\geq -\int_\Xi \Phi(\lambda^\varepsilon, \zeta)\nu(d\zeta) - \varepsilon' \geq \lambda^\varepsilon \theta^p - \int_\Xi \Phi(\lambda^\varepsilon, \zeta)\nu(d\zeta) - 2\varepsilon \\
&\geq \kappa\theta^p - \int_\Xi \Phi(\kappa, \zeta)\nu(d\zeta) - 2\varepsilon.
\end{aligned}$$

Otherwise, if $\kappa > 0$, then consider any $\kappa' \in (0, \kappa)$. First, note that $\zeta \in \{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\}$, and hence, $\{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\} \neq \emptyset$ for all $\zeta \in \Xi$. Let

$$\hat{D}(\kappa', \zeta) := \sup_{\xi \in \Xi} \{d^p(\xi, \zeta) : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\}.$$

Next, we show that $\int_\Xi \hat{D}(\kappa', \zeta)\nu(d\zeta) = \infty$. Note that $\Psi(\xi) \leq \lambda d^p(\xi, \zeta) - \Phi(\lambda, \zeta)$ for all $\xi \in \Xi$. Thus,

$$\begin{aligned}
\int_\Xi \Phi(\kappa', \zeta)\nu(d\zeta) &= \int_\Xi \inf_{\xi \in \Xi} \{\kappa' d^p(\xi, \zeta) - \Psi(\xi) : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\}\nu(d\zeta) \\
&\geq \int_\Xi \inf_{\xi \in \Xi} \{-\Psi(\xi) : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\}\nu(d\zeta) \\
&\geq \int_\Xi \inf_{\xi \in \Xi} \{-\lambda d^p(\xi, \zeta) + \Phi(\lambda, \zeta) : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi, \zeta)\}\nu(d\zeta) \\
&= -\lambda \int_\Xi \hat{D}(\kappa', \zeta)\nu(d\zeta) + \int_\Xi \Phi(\lambda, \zeta)\nu(d\zeta).
\end{aligned}$$

It follows from the definition of κ that $\int_\Xi \Phi(\kappa',\zeta)\nu(d\zeta) = -\infty$ from $\lambda > \kappa$ that $\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) > -\infty$, and thus, $\int_\Xi \hat{D}(\kappa',\zeta)\nu(d\zeta) = \infty$. Consider any $R > \theta^p$. It follows that there exists $M \in L^1(\nu)$ such that $\int_\Xi M(\zeta)\nu(d\zeta) > R$ and

$$\{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) \geq \kappa' d^p(\xi,\zeta),\ d^p(\xi,\zeta) \geq M(\zeta)\} \neq \varnothing$$

for ν-almost all $\zeta \in \Xi$. Thus, it follows from Lemma 3(v) that, for any $\kappa' \in (0,\kappa)$ and $R > \theta^p$, there exists a ν-measurable mapping $T^R(\kappa',\cdot) : \Xi \longmapsto \Xi$ such that

$$\Psi(T^R(\kappa',\zeta)) - \Psi(\zeta) \geq \kappa' d^p(T^R(\kappa',\zeta),\zeta)$$

for ν-almost all $\zeta \in \Xi$ and

$$\int_\Xi d^p(T^R(\kappa',\zeta),\zeta)\nu(d\zeta) \geq \int_\Xi M(\zeta)\nu(d\zeta) > R.$$

If $\int_\Xi d^p(T^R(\kappa',\zeta),\zeta)\nu(d\zeta) = \infty$, let $\Xi_r := \{\zeta \in \Xi : d^p(T^R(\kappa',\zeta),\zeta) \leq r\}$. Note that $\lim_{r\to\infty}\Xi_r = \Xi$, and thus, there exists a $\bar{r} > R$ such that $R < \int_{\Xi_{\bar{r}}} d^p(T^R(\kappa',\zeta),\zeta)\nu(d\zeta) \leq \bar{r} < \infty$. Then, let $\overline{T}^R(\kappa',\cdot) : \Xi \longmapsto \Xi$ be given by $\overline{T}^R(\kappa',\zeta) := T^R(\kappa',\zeta)$ for all $\zeta \in \Xi_{\bar{r}}$ and $\overline{T}^R(\kappa',\zeta) := \zeta$ for all $\zeta \in \Xi\backslash\Xi_{\bar{r}}$. Note that $\overline{T}^R(\kappa',\cdot)$ is ν-measurable and $\int_\Xi d^p(\overline{T}^R(\kappa',\zeta),\zeta)\ \nu(d\zeta) = \int_{\Xi_{\bar{r}}} d^p(T^R(\kappa',\zeta),\zeta)$ $\nu(d\zeta) \in (R,\infty)$. Otherwise, if $\int_\Xi d^p(T^R(\kappa',\zeta),\zeta)\nu(d\zeta) < \infty$, then let $\Xi_{\bar{r}} = \Xi$ and $\overline{T}^R(\kappa',\cdot) := T^R(\kappa',\cdot)$. Also, in this case, $\overline{T}^R(\kappa',\cdot)$ is ν-measurable and $\int_\Xi d^p(\overline{T}^R(\kappa',\zeta),\zeta)\nu(d\zeta) = \int_{\Xi_{\bar{r}}} d^p(T^R(\kappa',\zeta),\zeta)\ \nu(d\zeta) \in (R,\infty)$. Let $q_\varepsilon^R(\kappa',\lambda) \in (0,1)$ be such that

$$q_\varepsilon^R(\kappa',\lambda)\int_\Xi d^p(\underline{T}_\varepsilon(\lambda,\zeta),\zeta)\nu(d\zeta) + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_\Xi d^p(\overline{T}^R(\kappa',\zeta),\zeta)\nu(d\zeta) = \theta^p. \tag{24}$$

Define a distribution $\mu_\varepsilon^R(\kappa',\lambda)$ by

$$\mu_\varepsilon^R(\kappa',\lambda) := q_\varepsilon^R(\kappa',\lambda)\underline{T}_\varepsilon(\lambda,\cdot)_\#\nu + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\overline{T}^R(\kappa',\cdot)_\#\nu. \tag{25}$$

By construction, $\mu_\varepsilon^R(\kappa',\lambda)$ is primal feasible. Also,

$$\begin{aligned}
&\int_\Xi \Psi(\xi)\mu_\varepsilon^R(\kappa',\lambda)(d\xi)\\
&= q_\varepsilon^R(\kappa',\lambda)\int_\Xi \Psi(\underline{T}_\varepsilon(\lambda,\zeta))\nu(d\zeta) + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_\Xi \Psi\left(\overline{T}^R(\kappa',\zeta)\right)\nu(d\zeta)\\
&\geq q_\varepsilon^R(\kappa',\lambda)\int_\Xi [\lambda d^p(\underline{T}_\varepsilon(\lambda,\zeta),\zeta) - \Phi(\lambda,\zeta) - \varepsilon]\nu(d\zeta)\\
&\quad + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_{\Xi_{\bar{r}}} [\kappa' d^p(T^R(\kappa',\zeta),\zeta) + \Psi(\zeta)]\nu(d\zeta) + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_{\Xi\backslash\Xi_{\bar{r}}} \Psi(\zeta)\nu(d\zeta)\\
&\geq \kappa'\left(q_\varepsilon^R(\kappa',\lambda)\int_\Xi d^p(\underline{T}_\varepsilon(\lambda,\zeta),\zeta)\nu(d\zeta) + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_{\Xi_{\bar{r}}} d^p(T^R(\kappa',\zeta),\zeta)\nu(d\zeta)\right)\\
&\quad - q_\varepsilon^R(\kappa',\lambda)\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) - q_\varepsilon^R(\kappa',\lambda)\varepsilon + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_\Xi \Psi(\zeta)\nu(d\zeta)\\
&= \kappa'\theta^p - q_\varepsilon^R(\kappa',\lambda)\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) - q_\varepsilon^R(\kappa',\lambda)\varepsilon + \left(1 - q_\varepsilon^R(\kappa',\lambda)\right)\int_\Xi \Psi(\zeta)\nu(d\zeta).
\end{aligned} \tag{26}$$

Thus, given any $\varepsilon > 0$, choose $\lambda_1^\varepsilon \in (\kappa-\varepsilon,\kappa)$ such that $(\kappa - \lambda_1^\varepsilon)\theta^p \leq \varepsilon$, choose $\lambda_2^\varepsilon \in (\kappa,\kappa+\varepsilon)$ such that $(\lambda_2^\varepsilon - \kappa)\theta^p \leq \varepsilon$, choose $\varepsilon' \in \left(0, (\lambda_2^\varepsilon - \kappa)\theta^p - \int_\Xi [\Phi(\lambda_2^\varepsilon,\zeta) - \Phi(\kappa,\zeta)]\nu(d\zeta)\right)$ such that $\varepsilon'\left|\int_\Xi \Phi(\lambda_2^\varepsilon,\zeta)\nu(d\zeta)\right| \leq \varepsilon$ and $\varepsilon'\left|\int_\Xi \Psi(\zeta)\right.$ $\left.\nu(d\zeta)\right| \leq \varepsilon$ (note that $\varepsilon' \leq \varepsilon$), and choose $R \geq \theta^p + \theta^p/\varepsilon'$. Set

$$T_1^\varepsilon := \overline{T}^R(\lambda_1^\varepsilon,\cdot),\quad T_2^\varepsilon := \underline{T}_{\varepsilon'}(\lambda_2^\varepsilon,\cdot),\quad p^\varepsilon := 1 - q_{\varepsilon'}^R(\lambda_1^\varepsilon,\lambda_2^\varepsilon).$$

Thus, (19) follows from (24). It also follows from (24) that

$$p^{\varepsilon} = \frac{\theta^p - \int_{\Xi} d^p(\underline{T}_{\varepsilon'}(\lambda_2^{\varepsilon},\zeta),\zeta)\nu(d\zeta)}{\int_{\Xi} d^p(\overline{T}^R(\lambda_1^{\varepsilon},\zeta),\zeta)\nu(d\zeta) - \int_{\Xi} d^p(\underline{T}_{\varepsilon'}(\lambda_2^{\varepsilon},\zeta),\zeta)\nu(d\zeta)} \le \frac{\theta^p}{\theta^p + \frac{\theta^p}{\varepsilon'} - \theta^p} = \varepsilon'.$$

Thus, it follows from (26) that

$$\begin{aligned}
&\int_{\Xi} \Psi(\xi)\mu^{\varepsilon}(d\xi)\\
&\ge \lambda_1^{\varepsilon}\theta^p - (1-p^{\varepsilon})\int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta) - (1-p^{\varepsilon})\varepsilon + p^{\varepsilon}\int_{\Xi}\Psi(\zeta)\nu(d\zeta)\\
&= \lambda_2^{\varepsilon}\theta^p - (\lambda_2^{\varepsilon}-\kappa)\theta^p - (\kappa-\lambda_1^{\varepsilon})\theta^p - \int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta) + p^{\varepsilon}\int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta) - (1-p^{\varepsilon})\varepsilon + p^{\varepsilon}\int_{\Xi}\Psi(\zeta)\nu(d\zeta)\\
&\ge \lambda_2^{\varepsilon}\theta^p - \int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta) - \varepsilon'\left|\int_{\Xi}\Phi(\lambda_2^{\varepsilon},\zeta)\nu(d\zeta)\right| - \varepsilon'\left|\int_{\Xi}\Psi(\zeta)\nu(d\zeta)\right| - 3\varepsilon\\
&\ge \kappa\theta^p - \int_{\Xi}\Phi(\kappa,\zeta)\nu(d\zeta) - 5\varepsilon. \quad \square
\end{aligned}$$

The next theorem establishes a strong duality result when the growth rate κ is finite. The result follows from Lemmas 6 and 7 and by noting that

$$v_P \ge \int_{\Xi}\Psi(\xi)\mu^{\varepsilon}(d\xi) \ge \lambda^*\theta^p - \int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta) - \varepsilon = v_D - \varepsilon$$

for any $\varepsilon > 0$.

Theorem 1 (Strong Duality with Finite Optimal Value). *Consider any $p \in [1,\infty)$, any $\nu \in \mathcal{P}(\Xi)$, any $\theta > 0$, and any $\Psi \in L^1(\nu)$ such that $\kappa < \infty$. Then, $v_P = v_D < \infty$.*

Remark 2. All these results and proofs in Section 3.1 (except for Lemma 2) continue to hold if we replace the transportation cost $d^p(\cdot,\cdot)$ with any measurable, nonnegative cost function $c(\cdot,\cdot)$ that satisfies $c(\xi,\zeta) = 0$ if $\xi = \zeta$.

Next, we investigate existence conditions for worst-case distributions and their structure. In the remainder of this section, we assume that Ψ is upper semicontinuous, and every bounded subset in (Ξ, d) is totally bounded (see, for example, Munkres [37, section 45]), which is satisfied by, for example, any finite-dimensional normed space. Under this assumption, if $\lambda > \kappa$ and $\zeta \in B$, then Lemma 4(ii) and the upper semicontinuity of Ψ imply that the set $\arg\min_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta) - \Psi(\xi)\}$ is nonempty and that $\min/\max_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda d^p(\xi,\zeta) - \Psi(\xi) = \Phi(\lambda,\zeta)\}$ can be attained. If $\lambda = \kappa$ and $\nu(\{\zeta\in\Xi : \arg\min_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi(\xi)\} = \emptyset\}) = 0$, then the upper semicontinuity of Ψ imply that $\min_{\xi\in\Xi}\{d^p(\xi,\zeta) : \kappa d^p(\xi,\zeta) - \Psi(\xi) = \Phi(\kappa,\zeta)\}$ can be attained for ν-almost all $\zeta\in\Xi$, but $\sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \kappa d^p(\xi,\zeta) - \Psi(\xi) = \Phi(\kappa,\zeta)\}$ can be infinite. Thus, if (i) $\lambda > \kappa$ or (ii) $\lambda = \kappa$ and $\nu(\{\zeta\in\Xi : \arg\min_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi(\xi)\} = \emptyset\}) = 0$, then the quantities $\underline{D}_0(\lambda,\zeta)$ and $\overline{D}_0(\lambda,\zeta)$ in (4) are well-defined for ν-almost all $\zeta\in\Xi$ (in which $\overline{D}_0(\lambda,\zeta)$ can be infinite if $\lambda = \kappa$).

Corollary 1 (Worst-Case Distribution). *Consider any $p\in[1,\infty)$, $\nu\in\mathcal{P}(\Xi)$, $\theta > 0$, and $\Psi\in L^1(\nu)$ such that $\kappa < \infty$. Assume that Ψ is upper semicontinuous and that bounded subsets of (Ξ, d) are totally bounded. Then, the following holds:*

(i) *(Existence condition) A worst-case distribution exists if and only if any of the following conditions hold:*

(a) *There exists a dual minimizer $\lambda^* > \kappa$.*

(b) *$\lambda^* = \kappa > 0$ is the unique dual minimizer $\nu(\{\zeta\in\Xi : \arg\min_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi(\xi)\} = \emptyset\}) = 0$, and*

$$\int_{\Xi}\underline{D}_0(\kappa,\zeta)\nu(d\zeta) \le \theta^p \le \int_{\Xi}\overline{D}_0(\kappa,\zeta)\nu(d\zeta).$$

(c) *$\lambda^* = \kappa = 0$ is the unique dual minimizer,* $\arg\max_{\xi\in\Xi}\{\Psi(\xi)\}$ *is nonempty, and*

$$\int_{\Xi}\underline{D}_0(0,\zeta)\nu(d\zeta) \le \theta^p.$$

In addition, if $\nu(\{\zeta\in\Xi : -\Psi(\zeta) > \inf_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi(\xi)\}\}) = 0$, then $\lambda^* = \kappa$ for any $\theta > 0$. Otherwise, there is $\theta_0 > 0$ such that $\lambda^* > \kappa$ for any $\theta < \theta_0$.

(ii) *(Structure) Whenever a worst-case distribution exists, there exists a worst-case distribution μ^*, which can be represented as a convex combination of two distributions $\overline{T}^*_{\#}\nu$ and $\underline{T}^*_{\#}\nu$, each of which is a perturbation of ν as follows:*

$$\mu^* = p^*\overline{T}^*_{\#}\nu + (1-p^*)\underline{T}^*_{\#}\nu,$$

where $p^* \in [0,1]$ and $\overline{T}^*, \underline{T}^* : \Xi \longmapsto \Xi$ satisfy

$$\nu\left(\left\{\zeta \in \Xi : \overline{T}^*(\zeta), \underline{T}^*(\zeta) \notin \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\zeta) - \Psi(\xi)\}\right\}\right) = 0. \tag{27}$$

Let $\gamma^T \in \mathcal{P}(\Xi\times\Xi)$ be the joint distribution given by $\overline{T}^*, T^*$ and p^*. For any measurable set $A \subset \Xi\times\Xi$, let

$$\gamma^T(A) := p^*\nu(\{\zeta : (\zeta, \overline{T}^*(\zeta)) \in A\}) + (1-p^*)\nu(\{\zeta : (\zeta, T^*(\zeta)) \in A\}),$$

and if $\lambda^* > 0$, then γ^T is an optimal joint distribution in the definition (1) of $W_p^p(\mu^*,\nu)$.

(iii) *Suppose Ξ is convex, Ψ is a concave function, and $d^p(\cdot,\zeta)$ is a convex function for ν-almost all $\zeta\in\Xi$, then*

$$v_P = v_D = \sup_{T:\Xi\longmapsto\Xi}\left\{\mathbb{E}_{T_{\#}\nu}[\Psi(\xi)] : W_p(T_{\#}\nu,\nu) \le \theta,\ T \text{ is } \nu\text{-measurable}\right\}.$$

Moreover, whenever a worst-case distribution exists, there exists $T^* : \Xi \longmapsto \Xi$ such that $T^*_{\#}\nu$ is primal optimal and

$$\nu\left(\left\{\zeta\in\Xi : T^*(\zeta) \notin \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\zeta) - \Psi(\xi)\}\right\}\right) = 0.$$

Remark 3. Compared with Esfahani and Kuhn [23, corollary 4.7], Corollary 1(i) provides a complete description of the necessary and sufficient conditions for the existence of a worst-case distribution. Note that Esfahani and Kuhn [23, example 1] corresponds to $\lambda^* = \kappa = 1$ and $p = 1$. We also remark that Yue et al. [55] derives a sufficient condition on the existence of the worst-case distribution that does not require the knowledge of λ^*.

Example 4. In Figure 2, we present several examples that correspond to different cases in Corollary 1(i). In all these examples, $\Xi = [0,\infty)$, $d(\xi,\zeta) = |\xi-\zeta|$ for all $\xi,\zeta\in\Xi$, $p = 1$, $\theta > 0$, and $\nu = \delta_0$.

1. $\Psi_a(\xi) := \max\{0, \xi - a\}$ for some $a \in \mathbb{R}$. It follows that $\lambda^* = \kappa = 1$.

   - If $a \le 0$, then $\arg\min_{\xi\in\Xi}\{d^p(\xi,0) - \Psi_a(\xi)\} = [0,\infty)$; hence, $\underline{D}_0(\kappa,\zeta) = 0$ and $\overline{D}_0(\kappa,\zeta) = \infty$. Thus, the condition in Corollary 1(i)(b) is satisfied. One of the worst-case distributions is $\mu^* = \delta_\theta$ with $v_P = v_D = \theta - a$.

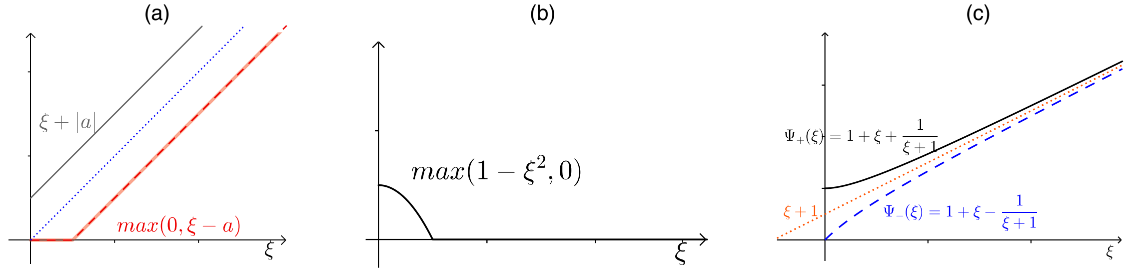
   - If $a > 0$, then $\arg\min_{\xi\in\Xi}\{d^p(\xi,0) - \Psi_a(\xi)\} = \{0\}$; hence, $\underline{D}_0(\kappa,\zeta) = \overline{D}_0(\kappa,\zeta) = 0 < \theta$. Thus, the condition in Corollary 1(i)(b) is violated. There is no worst-case distribution, but the objective value of $\mu_\varepsilon = (1-\varepsilon)\delta_0 + \varepsilon\delta_{\theta/\varepsilon}$ converges to $v_P = v_D = \theta - a$ as $\varepsilon \to 0$.

2. $\Psi(\xi) = \max\{0, 1-\xi^2\}$. It follows that $\lambda^* = \kappa = 0$ and $\arg\max_{\xi\in\Xi}\Psi(\xi) = \{0\}$. Thus, the condition in Corollary 1(i)(c) is satisfied, and the worst-case distribution is $\mu^* = \delta_0 = \nu$.

3. $\Psi_\pm(\xi) = 1 + \xi \pm \frac{1}{\xi+1}$. It follows that $\kappa = 1$. Note that $\Psi'_\pm(\xi) = 1 \mp \frac{1}{(\xi+1)^2}$.

   - Note that $\Psi'_+(\xi) < \kappa = 1$ on Ξ. Also, $-\Psi_+(0) = -2 = \inf_{\xi\ge 0}\{\xi - \Psi_+(\xi)\}$ and thus Ψ_+ satisfies $\nu(\{\zeta\in\Xi : -\Psi_+(\zeta) > \inf_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi_+(\xi)\}\}) = 0$ thus, for all $\theta > 0$, it holds that $\lambda^*_+ = \kappa = 1$ and $\arg\min_{\xi\in\Xi}\{\lambda^*_+ d^p(\xi,0) - \Psi_+(\xi)\} = \{0\}$. There is no worst-case distribution, but the objective value of $\mu_\varepsilon = (1-\varepsilon)\delta_0 + \varepsilon\delta_{\theta/\varepsilon}$ converges to $v_P = v_D = 2 + \theta$ as $\varepsilon \to 0$.

Figure 2. (Color online) Examples for existence and nonexistence of the worst-case distribution.



Notes. (a) $\Psi_a(\xi) = \max\{0, \xi - a\}$. (b) $\Psi(\xi) = \max\{0, 1-\xi^2\}$. (c) $\Psi_\pm(\xi) = 1 + \xi \pm \frac{1}{\xi+1}$.

- Note that $\Psi'_-(\xi) > \kappa = 1$ on Ξ. Also, $\arg\min_{\lambda\ge 0}\left\{\lambda\theta - \inf_{\xi\in\Xi}\left\{\lambda\xi - \left(1+\xi-\frac{1}{\xi+1}\right)\right\}\right\} = \arg\min_{\lambda\ge 1}\{\lambda(\theta+1) - 2\sqrt{\lambda-1}\} = \left\{1+\frac{1}{(\theta+1)^2}\right\}$. Thus, $\lambda^*_- > 1 = \kappa$.

3.2. Finite-Supported Nominal Distribution

In this section, we consider the setting in which the nominal distribution has finite support. Let $\nu = 1/N\sum_{i=1}^N \delta_{\hat{\xi}^i}$ for some $\hat{\xi}^i \in \Xi$, $i = 1,\dots,N$. This occurs, for example, in data-driven settings in which the nominal distribution is given by an empirical distribution constructed with N observations.

Corollary 2 (Data-Driven DRSO). *Consider any $\nu = 1/N\sum_{i=1}^N \delta_{\hat{\xi}^i}$, $p \in [1,\infty)$, and $\theta > 0$. The following hold:*

i. *(Strong duality) The primal problem (Primal) has a strong dual problem*

$$v_P = v_D = \inf_{\lambda\ge 0}\left\{\lambda\theta^p - \frac{1}{N}\sum_{i=1}^N \inf_{\xi\in\Xi}\left[\lambda d^p\left(\xi,\hat{\xi}^i\right) - \Psi(\xi)\right]\right\}. \tag{28}$$

ii. *(Structure of the worst-case distribution) Whenever a worst-case distribution exists, there exists one that is supported on at most $N+1$ points and has the form*

$$\mu^* = \frac{1}{N}\sum_{i\ne i_0}\delta_{\xi^i_*} + \frac{p_0}{N}\delta_{\underline{\xi}^{i_0}_*} + \frac{1-p_0}{N}\delta_{\overline{\xi}^{i_0}_*}, \tag{29}$$

where $i_0 \in \{1,\dots,N\}$, $p_0 \in [0,1]$, $\underline{\xi}^{i_0}_*, \overline{\xi}^{i_0}_* \in \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\hat{\xi}^{i_0}) - \Psi(\xi)\}$, and $\xi^i_* \in \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\hat{\xi}^i) - \Psi(\xi)\}$ for all $i \ne i_0$.

iii. *(Approximation by robust optimization) Suppose that there exists $\zeta^0 \in \Xi$, $L, M \ge 0$ such that $|\Psi(\xi) - \Psi(\zeta^0)| < Ld^p(\xi,\zeta^0) + M$ for all $\xi \in \Xi$. For any positive integer K, consider the robust optimization problem*

$$v_K := \sup_{(\xi^{ik})_{i,k}\in\mathfrak{M}_K} \frac{1}{NK}\sum_{i=1}^N\sum_{k=1}^K \Psi(\xi^{ik}),$$

with uncertainty set

$$\mathfrak{M}_K := \left\{\left(\xi^{ik}\right)_{i,k} : \frac{1}{NK}\sum_{i=1}^N\sum_{k=1}^K d^p\left(\xi^{ik},\hat{\xi}^i\right) \le \theta^p,\ \xi^{ik}\in\Xi\ \forall i,k\right\}.$$

If $\lambda^* > \kappa$, then there exists a constant D independent of K such that

$$v_K \le \sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[\Psi(\xi)] \le v_K + \frac{LD+M}{NK},$$

and in addition, if Ξ is convex and Ψ is concave, then $v_1 = v_P = v_D$.

Statement (ii) shows that the worst-case distribution μ^* is a perturbation of $\nu = 1/N\sum_{i=1}^N \delta_{\hat{\xi}^i}$, where $N-1$ out of the N points, $\{\hat{\xi}^i\}_{i\ne i_0}$, are perturbed with all their probability mass to an associated maximizer ξ^i_*, whereas at most one point $\hat{\xi}^{i_0}$ is split and perturbed to two maximizers $\underline{\xi}^{i_0}_*$ and $\overline{\xi}^{i_0}_*$. If, for each $\hat{\xi}^i$, the set of maximizers is a singleton, then there is no need to split, and the worst-case distribution μ^* has support on N points as well. Using this structure, we obtain result (iii), which states that the primal problem can be approximated by a robust optimization problem with uncertainty set $\mathfrak{M}_K$, which is a subset of $\mathfrak{M}$ that contains all distributions supported on NK points (not necessarily distinct) with probability $1/(NK)$ each. Particularly, when Ψ is concave, such approximation is exact, and when Ψ is Lipschitz and $p = 1$, then v_1 is an $O(1/N)$-approximation of $v_P = v_D$.

Remark 4. As can easily be seen in the proof, the result of statement (ii) can be generalized as follows. Suppose that $\nu = \sum_{i=1}^N \nu_i\delta_{\hat{\xi}^i}$; then, whenever a worst-case distribution exists, there exists one of the form

$$\mu^* = \sum_{i\ne i_0}\nu_i\delta_{\xi^i_*} + p_0\nu_{i_0}\delta_{\underline{\xi}^{i_0}_*} + (1-p_0)\nu_{i_0}\delta_{\overline{\xi}^{i_0}_*}.$$

Remark 5. The results in Corollary 2 hold for arbitrary metric spaces (Ξ, d). The Polish space assumption on (Ξ, d) is used only for the measurability results in Lemma 3, and in the finite-supported setting, such an assumption is not needed.

Remark 6. Under a compactness assumption on (Ξ, d), Wozabal [53] points out that to solve (Primal), it suffices to consider the extreme points of the Wasserstein ball $\mathfrak{M}$, which are distributions that are supported on at most $N+3$ points. Later, in Owhadi and Scovel [39], this result is improved for Polish spaces or Borel subsets of Polish spaces to distributions that are supported on at most $N+2$ points. Statement 2 further strengthens these results; for arbitrary metric spaces (see Remark 5), it suffices to consider distributions that are supported on at most $N+1$ points, and this bound is tight as shown by Example 7. Moreover, $N-1$ out of the at most $N+1$ points in the support of the extreme distribution has the same probability masses as the associated points in the support of the nominal distribution. We also remark that, after this work, Yue et al. [55] show that the worst-case distribution is supported on at most $N+1$ points by employing the Richter–Rogosinski theorem.

Remark 7 (Total Variation Metric). By choosing the discrete metric $d(\xi,\zeta) = \mathbb{1}_{\{\xi\neq\zeta\}}$ on Ξ, the Wasserstein distance is equal to the total variation distance (Gibbs and Su [28]), which can be used for settings in which it matters whether two points are the same or different, but no other notion of difference is relevant. In this case, if θ is chosen such that $N\theta$ is an integer, then there is no point indexed by i_0 in (29) such that fractions of its probability are transported to different points to obtain worst-case distribution μ^*, and the primal problem (Primal) is reduced to the robust optimization problem with uncertainty set $\mathfrak{M}_1$ whether Ξ (Ψ) is convex (concave) or not.

Remark 8 (Choosing Radius θ). There are multiple ways to choose the radius θ of the Wasserstein ball. Practically, we find that the cross-validation method seems to work well for many problems. We use this method for the numerical experiments on intensity estimation in Section 4.2 and obtain good performance. It is also used in Esfahani and Kuhn [23] for various numerical examples, including portfolio optimization and uncertainty quantification. On the other hand, one can use concentration inequalities to determine an upper bound on the radius with a statistical guarantee. For example, in the context of a model with an underlying true distribution, one can estimate a theoretical upper bound on the distance between the empirical distribution and the true distribution. The classical concentration inequalities (see, e.g., Esfahani and Kuhn [23], Fournier and Guillin [24]) provide a radius θ of the order $O(N^{-1/s})$, where s is the dimension of the random variable. We use this technique for a one-dimensional newsvendor problem in Section 5.1. However, note that such a bound is pessimistic for problems with high-dimensional random variables because it suggests that the number of data points needed grows exponentially (in the dimension s) to make the radius θ smaller while maintaining the statistical guarantee. Fortunately, for many decision problems it is *unnecessary* to choose a radius θ such that the ball $\mathfrak{M}$ contains a true distribution with guaranteed probability; our goal is not to approximate the true distribution, but only the objective value. This also means that, even if we know the exact distance between the empirical distribution and the true distribution, we should not choose the distance to be the radius. Instead, we can choose the radius such that, with high probability, for every x the worst-case expectation in the DRSO problem dominates the true expectation. This idea has been exploited recently in Gao [26], which shows that the radius can be in the order of $N^{-1/2}$ in many settings, independent of the dimension of the random variable. Another principle is to choose the radius θ such that the uncertainty set yields a confidence region for the true minimizer, and it is shown in Blanchet et al. [11, 12] that such a choice would lead to a $N^{-1/2}$-bound asymptotically under proper conditions.

Remark 9 (Algorithmic Complexity). According to the dual reformulation, fixing the dual variable λ, the dual objective function is the sum of the optimal objective values of N separate maximization problems. Suppose that we have an oracle for solving the inner maximization problem $\sup_{\xi\in\Xi}\{\Psi(x,\xi) - \lambda\|\xi - \hat{\xi}^i\|\}$ for any $\hat{\xi}^i$. Thereby the number of oracle calls to evaluate the worst-case expectation scales linearly with the sample size N. To optimize the finite sum problem over x, there are optimal algorithms whose complexity match the theoretical lower bound (see, e.g., Agarwal and Bottou [1], Lan and Zhou [35]).

For an important class of problems, the inner maximization problem is easy. Examples include the following problem types:

1. Problems in which the inner maximization problem has a closed-form solution, for example, when $\Psi(x,\xi)$ is linear in ξ or, more generally, $\Psi(x,\xi) = \max_{1\le k\le K} a_k(x)^\top \xi + b_k(x)$ and $\Xi = \mathbb{R}^s$ (Esfahani and Kuhn [23, remark 6.6], Gao et al. [27]). In this case, the minimax DRSO problem is equivalent to a finite-sum optimization problem with regularization, so the complexity is linear in the sample size N and can be (nearly) independent of the dimension of Ξ under suitable conditions.

2. Problems in which $\Psi(x,\xi)$ is (piecewise) concave in ξ, for which the overall DRSO is equivalent to a saddle-point problem (Examples 5 and 6). In this case, the complexity is more involved because it may also depend on the size θ of the Wasserstein ball and the geometry of the sample space. For example, consider use of the Mirror–Prox algorithm (Nemirovski [38]) to solve the saddle-point problem in which $\Psi(x,\xi)$ is convex in x. The complexity is proportional to the square of the radius $N\theta$ of the Wasserstein ball and is nearly independent of the dimension of Ξ for uncertainty sets with nice geometry. As indicated in Remark 8, a good way to choose θ is roughly $O(N^{-1/2})$, and similar to the first case, this leads to a complexity that is linear in N and nearly independent of the dimension of the random variable.

Proof of Corollary 2.

i. The result follows directly from Theorem 1 and Proposition 2.

ii. By Corollary 1(ii), whenever a worst-case distribution exists, there is one supported on at most $2N$ points with the form

$$\hat{\mu} = \hat{p}\frac{1}{N}\sum_{i=1}^{N}\delta_{\underline{\xi}_*^i} + (1-\hat{p})\frac{1}{N}\sum_{i=1}^{N}\delta_{\overline{\xi}_*^i}, \tag{30}$$

where $\hat{p}\in[0,1]$ and $\underline{\xi}_*^i, \overline{\xi}_*^i \in \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\hat{\xi}^i) - \Psi(\xi)\}$. Given $\underline{\xi}_*^i, \overline{\xi}_*^i$ for all i, note that

$$\mathbb{E}_{\mu^*}[\Psi(\xi)] \le \max_{0\le p_i\le 1}\left\{\frac{1}{N}\sum_{i=1}^{N}\Big[p_i\Psi(\underline{\xi}_*^i) + (1-p_i)\Psi(\overline{\xi}_*^i)\Big] : \frac{1}{N}\sum_{i=1}^{N}\Big[p_i d^p(\underline{\xi}_*^i,\hat{\xi}^i) + (1-p_i)d^p(\overline{\xi}_*^i,\hat{\xi}^i)\Big] \le \theta^p\right\}.$$

This problem is a linear program with N variables, one perturbation distance constraint, and $2N$ constraints $0\le p_i\le 1$, $i=1,\dots,N$. At an extreme point optimal solution p^* of the linear program, if the perturbation distance constraint holds as an equality, then at least $N-1$ of the constraints $0\le p_i\le 1$ hold as equalities, or equivalently, for at most one index i_0, it holds that p_{i_0} is fractional. Otherwise, if the perturbation distance constraint holds as a strict inequality, then exactly N of the constraints $0\le p_i\le 1$ hold as equalities, or equivalently, none of the p_i is fractional. Such an extreme point optimal solution p^* gives a worst-case distribution

$$\mu^* = \frac{1}{N}\sum_{i=1}^{N}\Big[p_i^*\delta_{\underline{\xi}_*^i} + (1-p_i^*)\delta_{\overline{\xi}_*^i}\Big] = \frac{1}{N}\sum_{i\ne i_0}\delta_{\xi_*^i} + \frac{p_0}{N}\delta_{\underline{\xi}_*^{i_0}} + \frac{1-p_0}{N}\delta_{\overline{\xi}_*^{i_0}},$$

where $\xi_*^i = \underline{\xi}_*^i$ if $p_i^*=1$, $\xi_*^i = \overline{\xi}_*^i$ if $p_i^*=0$ and $p_0 = p_{i_0}^*$.

iii. Note that $\nu\in\mathcal{P}_p(\Xi)$. If Ξ is bounded, then $\kappa=0$, and if Ξ is unbounded, then it follows from the assumption and Lemma 2(iii) that $\kappa\le L<\infty$.

It follows from Lemma 4(ii) that $\overline{D}(\lambda,\hat{\xi}^i)<\infty$ for all $\lambda>\kappa$ and all i. Let $D := \max\{\overline{D}([\lambda^*+\kappa]/2,\hat{\xi}^i) : i\in\{1,\dots,N\}\}<\infty$. Consider any $\varepsilon'>0$. For each i, let $\varepsilon^i>0$ be such that $\sup_{\xi\in\Xi}\{d^p(\xi,\hat{\xi}^i) : [\lambda^*/2+\kappa/2]d^p(\xi,\hat{\xi}^i) - \Psi(\xi) \le \Phi([\lambda^*+\kappa]/2,\hat{\xi}^i)+\varepsilon\} \le \overline{D}([\lambda^*+\kappa]/2,\hat{\xi}^i)+\varepsilon'$ for all $\varepsilon\in(0,\varepsilon^i)$. Consider any $\varepsilon\in(0,[\lambda^*-\kappa]/2)$ such that $\varepsilon<\varepsilon^i$ for all i. By Lemma 6, there are a $\lambda_1^\varepsilon\in(\lambda^*-\varepsilon,\lambda^*)$, a $\lambda_2^\varepsilon\in(\lambda^*,\lambda^*+\varepsilon)$, $\hat{p}_1^\varepsilon,\hat{p}_2^\varepsilon,\hat{p}_3^\varepsilon\ge 0$, and $\hat{\xi}_{i1}^\varepsilon,\hat{\xi}_{i2}^\varepsilon,\hat{\xi}_{i3}^\varepsilon\in\Xi$ for $i=1,\dots,N$ such that $\hat{p}_1^\varepsilon+\hat{p}_2^\varepsilon+\hat{p}_3^\varepsilon=1$,

$$\begin{aligned}\lambda_1^\varepsilon d^p(\hat{\xi}_{i1}^\varepsilon,\hat{\xi}^i) - \Psi(\hat{\xi}_{i1}^\varepsilon) &\le \Phi(\lambda_1^\varepsilon,\hat{\xi}^i)+\varepsilon\\ \lambda_2^\varepsilon d^p(\hat{\xi}_{i2}^\varepsilon,\hat{\xi}^i) - \Psi(\hat{\xi}_{i2}^\varepsilon) &\le \Phi(\lambda_2^\varepsilon,\hat{\xi}^i)+\varepsilon\\ \hat{\xi}_{i3}^\varepsilon &= \hat{\xi}^i\end{aligned}$$

for all $i=1,\dots,N$, $1/N\sum_{i=1}^N\sum_{j=1}^3 \hat{p}_j^\varepsilon d^p(\hat{\xi}_{ij}^\varepsilon,\hat{\xi}^i)\le\theta^p$, and

$$\hat{\mu}^\varepsilon := \hat{p}_1^\varepsilon\frac{1}{N}\sum_{i=1}^{N}\delta_{\hat{\xi}_{i1}^\varepsilon} + \hat{p}_2^\varepsilon\frac{1}{N}\sum_{i=1}^{N}\delta_{\hat{\xi}_{i2}^\varepsilon} + \hat{p}_3^\varepsilon\frac{1}{N}\sum_{i=1}^{N}\delta_{\hat{\xi}_{i3}^\varepsilon}$$

satisfies $\sup_{\mu\in\mathfrak{M}}\mathbb{E}_\mu[\Psi(\xi)] \le \mathbb{E}_{\hat{\mu}^\varepsilon}[\Psi(\xi)]+\varepsilon$. If $\Psi(\hat{\xi}_{ij}^\varepsilon)<\Psi(\hat{\xi}^i)$ for any (i,j), then $\hat{\xi}_{ij}^\varepsilon$ can be replaced with $\hat{\xi}^i$, and the preceding statements continue to hold. Therefore, without loss of generality, we assume that $\Psi(\hat{\xi}_{ij}^\varepsilon)\ge\Psi(\hat{\xi}^i)$.

Given $\{\hat{\xi}^{\varepsilon}_{ij} : 1 \le i \le N, 1 \le j \le 3\}$, note that

$$\mathbb{E}_{\hat{\mu}^{\varepsilon}}[\Psi(\xi)] \le \max_{p_{ij}\ge 0}\left\{\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{3} p_{ij}\Psi(\hat{\xi}^{\varepsilon}_{ij}) : \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{3} p_{ij} d^p(\hat{\xi}^{\varepsilon}_{ij}, \hat{\xi}^{i}) \le \theta^p,\ \sum_{j=1}^{3} p_{ij} = 1\ \forall i = 1,\ldots,N\right\}.$$

This problem is a linear program with $3N$ variables and $N+1$ constraints in addition to the variable bounds $p_{ij} \ge 0$. At an extreme point optimal solution of the linear program, at most $N+1$ of the variables are nonzero. In addition, for each $i = 1,\ldots,N$, the constraint $p_{i1} + p_{i2} + p_{i3} = 1$ requires at least one of the variables p_{i1}, p_{i2}, p_{i3} to be nonzero. Thus, for at least $N-1$ indices i, exactly one of p_{i1}, p_{i2}, p_{i3} is equal to one, and for at most one index i_0, two of $p_{i_0,1}, p_{i_0,2}, p_{i_0,3}$ are positive. Hence, an extreme point optimal solution of the linear program gives a solution of the form

$$\mu^{\varepsilon} = \frac{1}{N}\sum_{i\ne i_0}\delta_{\xi^i_{\varepsilon}} + \frac{p_{\varepsilon}}{N}\delta_{\underline{\xi}^{i_0}_{\varepsilon}} + \frac{1-p_{\varepsilon}}{N}\delta_{\overline{\xi}^{i_0}_{\varepsilon}},$$

where $\xi^i_{\varepsilon} \in \left\{\hat{\xi}^{\varepsilon}_{i1}, \hat{\xi}^{\varepsilon}_{i2}, \hat{\xi}^{\varepsilon}_{i3}\right\}$ for all $i \in \{1,\ldots,N\}\setminus\{i_0\}$, $\underline{\xi}^{i_0}_{\varepsilon}, \overline{\xi}^{i_0}_{\varepsilon} \in \left\{\hat{\xi}^{\varepsilon}_{i_0,1}, \hat{\xi}^{\varepsilon}_{i_0,2}, \hat{\xi}^{\varepsilon}_{i_0,3}\right\}$ such that $d^p\left(\underline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) \le d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right)$,

$$\frac{1}{N}\sum_{i\ne i_0} d^p\left(\xi^i_{\varepsilon}, \hat{\xi}^i\right) + \frac{p_{\varepsilon}}{N} d^p\left(\underline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) + \frac{1-p_{\varepsilon}}{N} d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) \le \theta^p,$$

and $\mathbb{E}_{\hat{\mu}^{\varepsilon}}[\Psi(\xi)] \le \mathbb{E}_{\mu^{\varepsilon}}[\Psi(\xi)]$. Consider

$$\xi^{ik}_{\varepsilon} := \begin{cases} \xi^i_{\varepsilon} \text{ for } i \ne i_0, k = 1,\ldots,K, \\ \underline{\xi}^{i_0}_{\varepsilon} \text{ for } k = 1,\ldots,\lceil Kp_{\varepsilon}\rceil, \\ \overline{\xi}^{i_0}_{\varepsilon} \text{ for } k = \lceil Kp_{\varepsilon}\rceil + 1,\ldots,K. \end{cases}$$

Then, because $d^p\left(\underline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) \le d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right)$, it follows that

$$\begin{aligned}\frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K} d^p\left(\xi^{ik}_{\varepsilon}, \hat{\xi}^i\right) &= \frac{1}{N}\sum_{i\ne i_0} d^p\left(\xi^i_{\varepsilon}, \hat{\xi}^i\right) + \frac{\lceil Kp_{\varepsilon}\rceil/K}{N} d^p\left(\underline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) + \frac{1-\lceil Kp_{\varepsilon}\rceil/K}{N} d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) \\ &\le \frac{1}{N}\sum_{i\ne i_0} d^p\left(\xi^i_{\varepsilon}, \hat{\xi}^i\right) + \frac{p_{\varepsilon}}{N} d^p\left(\underline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) + \frac{1-p_{\varepsilon}}{N} d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) \le \theta^p,\end{aligned}$$

and thus, $\{\xi^{ik}_{\varepsilon}\}_{i,k} \in \mathfrak{M}_K$. Because $0 \le \lceil Kp_{\varepsilon}\rceil/K - p_{\varepsilon} < 1/K$, it follows that

$$\begin{aligned}\mathbb{E}_{\mu^{\varepsilon}}[\Psi(\xi)] &= \frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K}\Psi(\xi^{ik}_{\varepsilon}) + \frac{1}{N}\left(\frac{\lceil Kp_{\varepsilon}\rceil}{K} - p_{\varepsilon}\right)[\Psi(\overline{\xi}^{i_0}_{\varepsilon}) - \Psi(\underline{\xi}^{i_0}_{\varepsilon})] \\ &\le \frac{1}{NK}\sum_{i=1}^{N}\sum_{k=1}^{K}\Psi(\xi^{ik}_{\varepsilon}) + \frac{1}{N}\left(\frac{\lceil Kp_{\varepsilon}\rceil}{K} - p_{\varepsilon}\right)[\Psi(\overline{\xi}^{i_0}_{\varepsilon}) - \Psi(\hat{\xi}^{i_0})] \\ &\le v_K + \frac{1}{NK}\left[L d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) + M\right].\end{aligned}$$

Because $\overline{\xi}^{i_0}_{\varepsilon} \in \left\{\hat{\xi}^{\varepsilon}_{i_0,1}, \hat{\xi}^{\varepsilon}_{i_0,2}, \hat{\xi}^{\varepsilon}_{i_0,3}\right\}$, it follows that

$$\begin{aligned}d^p\left(\overline{\xi}^{i_0}_{\varepsilon}, \hat{\xi}^{i_0}\right) &\le \sup_{\xi\in\Xi}\left\{d^p(\xi, \hat{\xi}^{i_0}) : \lambda^{\varepsilon}_1 d^p(\xi, \hat{\xi}^{i_0}) - \Psi(\xi) \le \Phi(\lambda^{\varepsilon}_1, \hat{\xi}^{i_0}) + \varepsilon\right\} \\ &\le \sup_{\xi\in\Xi}\left\{d^p(\xi, \hat{\xi}^{i_0}) : \frac{\lambda^* + \kappa}{2} d^p(\xi, \hat{\xi}^{i_0}) - \Psi(\xi) \le \Phi([\lambda^* + \kappa]/2, \hat{\xi}^{i_0}) + \varepsilon\right\} \\ &\le \overline{D}\left([\lambda^* + \kappa]/2, \hat{\xi}^{i_0}\right) + \varepsilon',\end{aligned}$$

where the second inequality follows from Lemma 4(i) and $\lambda_1^\varepsilon > [\lambda^* + \kappa]/2$. Therefore,

$$\sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[\Psi(\xi)] \le \mathbb{E}_{\mu^\varepsilon}[\Psi(\xi)] + \varepsilon \le v_K + \frac{1}{NK}\Big[L\Big(\overline{D}\Big([\lambda^*+\kappa]/2, \hat{\xi}^{i_0}\Big) + \varepsilon'\Big) + M\Big] + \varepsilon$$

$$\Rightarrow \sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[\Psi(\xi)] \le v_K + \frac{1}{NK}[LD + M]. \quad \square$$

Example 5 (Saddle-Point Problem). When $\Psi(x,\xi)$ is convex in x and concave in ξ, $p=1$, and $d(\xi,\zeta) = \|\xi-\zeta\|_2$, then Corollary 2(iii) shows that (DRSO) is equivalent to a convex-concave saddle-point problem

$$\min_{x\in X} \max_{(\xi^1,\dots,\xi^N)\in Y} \frac{1}{N}\sum_{i=1}^N \Psi(x,\xi^i),$$

with ℓ_1/ℓ_2-norm uncertainty set

$$Y = \left\{(\xi^1,\dots,\xi^N)\in\Xi^N : \sum_{i=1}^N \left\|\xi^i - \hat{\xi}^i\right\|_2 \le N\theta\right\}.$$

Example 6 (Piecewise Concave Objective). Esfahani and Kuhn [23] show that, if Ξ is a closed convex subset of $\mathbb{R}^K$ with norm $\|\cdot\|$, ν has finite support, $p=1$, and $\Psi(\xi) = \max_{1\le j\le J}\Psi^j(\xi)$, where each Ψ^j is concave, then the DRSO can be formulated as a convex optimization problem. Here, we show that the result can be obtained as a corollary from the structure of a worst-case distribution. If $\lambda^* > \kappa$, then by Corollary 1(i)(a), a worst-case distribution exists and, by Corollary 2, has the form $1/N\sum_{i=1}^N\sum_{k=1}^2 p_{ik}\delta_{\xi^{ik}}$, where for each i it holds that $\sum_{k=1}^2 p_{ik} = 1$ and $\xi^{ik}\in \arg\min_{\xi\in\Xi}\{\lambda^*\|\xi-\hat{\xi}^i\| - \Psi(\xi)\}$. Moreover, because of the concavity of Ψ^j, if $\Psi(\xi^{i1}) = \Psi^j(\xi^{i1})$ and $\Psi(\xi^{i2}) = \Psi^j(\xi^{i2})$ for some j, then any convex combination $\xi_\alpha = \alpha\xi^{i1} + (1-\alpha)\xi^{i2}$ satisfies $\lambda^*\|\xi_\alpha - \hat{\xi}^i\| - \Psi(\xi_\alpha) \le \lambda^*\ \|\xi_\alpha - \hat{\xi}^i\| - \Psi^j(\xi_\alpha) \le \alpha\Big[\lambda^*\|\xi^{i1}-\hat{\xi}^i\| - \Psi^j(\xi^{i1})\Big] + (1-\alpha)\Big[\lambda^*\|\xi^{i2}-\hat{\xi}^i\| - \Psi^j(\xi^{i2})\Big] = \min_{\xi\in\Xi}\{\lambda^*\|\xi-\hat{\xi}^i\| - \Psi(\xi)\}$, and thus, $\xi_\alpha\in\arg\min_{\xi\in\Xi}\{\lambda^*\ \|\xi-\hat{\xi}^i\| - \Psi(\xi)\}$. Therefore, we can assume without loss of generality that, if $\Psi(\xi^{i1}) = \Psi^{j_1}(\xi^{i1})$ and $\Psi(\xi^{i2}) = \Psi^{j_2}(\xi^{i2})$, then $j_1 \ne j_2$. Thus, we can consider distributions of the form

$$\frac{1}{N}\sum_{i=1}^N\sum_{j=1}^J p_{ij}\delta_{\xi^{ij}},\ \text{where}\ \sum_{j=1}^J p_{ij} = 1,\ \Psi(\xi^{ij}) = \Psi^j(\xi^{ij}),\ \forall i = 1,\dots,N.$$

If $\lambda^* = \kappa$, then by Lemma 7 and the concavity of Ψ^j, we can also consider distributions of the preceding form. Therefore, the original primal problem is equivalent to

$$\sup_{p_{ij}\ge 0,\ \xi^{ij}\in\Xi}\left\{\frac{1}{N}\sum_{i=1}^N\sum_{j=1}^J p_{ij}\Psi^j(\xi^{ij}) : \frac{1}{N}\sum_{i=1}^N\sum_{j=1}^J p_{ij}\|\xi^{ij}-\hat{\xi}^i\| \le \theta,\ \sum_{j=1}^J p_{ij} = 1,\ \forall i = 1,\dots,N\right\}.$$

Next, let $\tilde{\xi}^{ij} := \hat{\xi}^i + p_{ij}(\xi^{ij} - \hat{\xi}^i)$. Then, by the positive homogeneity of norms and the convexity-preserving property of perspective functions (cf. Boyd and Vandenberghe [15, section 2.3.3]), the primal problem can be reformulated as the following convex optimization problem:

$$\sup_{\substack{p_{ij}\ge 0,\ \sum_j p_{ij} = 1\\ \tilde{\xi}^{ij}\in\mathbb{R}^K}}\left\{\frac{1}{N}\sum_{i=1}^N\sum_{j=1}^J p_{ij}\Psi^j\left(\hat{\xi}^i + \frac{\tilde{\xi}^{ij}-\hat{\xi}^i}{p_{ij}}\right) : \frac{1}{N}\sum_{i=1}^N\sum_{j=1}^J\|\tilde{\xi}^{ij}-\hat{\xi}^i\| \le \theta,\ \hat{\xi}^i + \frac{\tilde{\xi}^{ij}-\hat{\xi}^i}{p_{ij}}\in\Xi,\ \forall i,j\right\}.$$

This establishes Esfahani and Kuhn [23, theorem 4.4], which was obtained therein by a separate procedure of dualizing twice the reformulation (28).

Example 7 (Uncertainty Quantification). Consider a metric space (Ξ,d) such that every bounded subset in (Ξ,d) is totally bounded, and any open proper Borel subset $C\subsetneq\Xi$. Let $\Psi = -\mathbb{1}_C$, and consider the uncertainty quantification problem

$$\sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[\Psi(\xi)] = \sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[-\mathbb{1}_C(\xi)] = -\min_{\mu\in\mathfrak{M}} \mu(C). \tag{31}$$

Next, we show that it follows from Corollary 1(i) that Problem (31) has an optimal distribution.

It follows from the definition of κ that $\kappa = 0$. Because C is open, $\Psi = -\mathbb{1}_C$ is upper semicontinuous. It follows from Lemma 5(v) that the dual objective function h has a minimizer $\lambda^* \in [0,\infty)$. Next, we consider two cases:

Case 1: There exists a dual minimizer $\lambda^* > 0$. Then, it follows from Corollary 1(i)(a) that a worst-case distribution exists.

Case 2: $\lambda^* = 0$ is the unique dual minimizer. Because C is a proper subset of Ξ, it follows that $\arg\max_{\xi\in\Xi}\Psi(\xi) = \arg\min_{\xi\in\Xi} \mathbb{1}_C(\xi)$ is nonempty and

$$\Phi(0,\zeta) = \inf_{\xi\in\Xi}\{-\Psi(\xi)\} = \inf_{\xi\in\Xi} \mathbb{1}_C(\xi) = 0.$$

Also, for any $\lambda > 0$, it holds that

$$\begin{aligned} 0 &= -\int_\Xi \Phi(0,\zeta)\nu(d\zeta) = h(0) < h(\lambda) \\ &= \lambda\theta^p - \int_\Xi \inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta) + \mathbb{1}_C(\xi)\}\nu(d\zeta) \\ &= \lambda\theta^p - \int_\Xi \min\left\{1,\, \lambda \inf_{\xi\in\Xi\setminus C} d^p(\xi,\zeta)\right\}\nu(d\zeta). \end{aligned}$$

Thus, for any $\lambda > 0$, it holds that

$$\int_\Xi \min\left\{\frac{1}{\lambda},\, \inf_{\xi\in\Xi\setminus C} d^p(\xi,\zeta)\right\}\nu(d\zeta) < \theta^p.$$

Next, let $\lambda \to 0$, and then, it follows from the monotone convergence theorem that

$$\int_\Xi \underline{D}_0\,(0,\zeta)\nu(d\zeta) = \int_\Xi \inf_{\xi\in\Xi\setminus C} d^p(\xi,\zeta) \le \theta^p.$$

Therefore, it follows from Corollary 1(i)(c) that a worst-case distribution exists.

Depending on the metric d, for $\zeta\in C$, $\arg\min_{\xi\in\Xi\setminus C} d^p(\xi,\zeta)$ may or may not be on the boundary of C. Next, suppose that d is an intrinsic metric (so that, for $\zeta \in C$, it holds that $\arg\min_{\xi\in\Xi\setminus C} d^p(\xi,\zeta) \subset \partial C$) and ν has finite support, say, $\nu = 1/N\sum_{i=1}^N \delta_{\hat\xi^i}$. Then, the worst-case distribution μ^* of Problem (31) has an intuitive form. The worst-case distribution perturbs ν such that the set C contains as little probability mass as possible, which can be achieved in a *greedy* fashion as follows. Suppose that $\{\hat\xi^i\}_{i=1}^N$ are sorted such that $\hat\xi^1,\dots,\hat\xi^I \in C$, $\hat\xi^{I+1},\dots,\hat\xi^N \notin C$, and $d^p(\hat\xi^1,\Xi\setminus C) \le \cdots \le d^p(\hat\xi^I,\Xi\setminus C)$. Then, to save the total budget of perturbation, $\hat\xi^{I+1},\dots,\hat\xi^N$ stay at the same place, and the points $\hat\xi^i$ with smaller indices i have priority to be transported to their closest points in ∂C. Proceeding in this greedy way, points $\hat\xi^1,\dots,\hat\xi^{i_0-1}$ ($i_0 \le I$) are transported in full (with full mass $1/N$) to their closest points in ∂C. The next point $\hat\xi^{i_0}$ is transported in part or in full to its closest point in ∂C; it may be the case that transporting $\hat\xi^{i_0}$ in full to its closest point in ∂C violates the Wasserstein distance constraint. Thus, only part p_0/N with $0 < p_0 \le 1$ of its mass is transported, and the remaining mass $(1-p_0)/N$ stays at point $\hat\xi^{i_0}$ (see Figure 3). Points $\hat\xi^{i_0+1},\dots,\hat\xi^N$ stay at the same place. Therefore, the worst-case distribution has the form

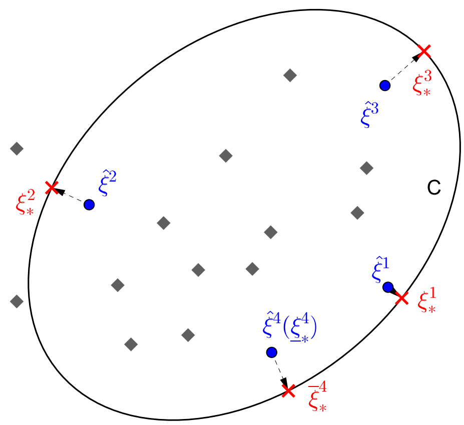
$$\mu^* = \frac{1}{N}\sum_{i=1}^{i_0-1}\delta_{\xi_*^i} + \frac{p_0}{N}\delta_{\xi_*^{i_0}} + \frac{1-p_0}{N}\delta_{\hat\xi^{i_0}} + \frac{1}{N}\sum_{i=i_0+1}^{N}\delta_{\hat\xi^i}.$$

In fact, if $d^p(\hat\xi^{i_0-1},\Xi\setminus C) < d^p(\hat\xi^{i_0},\Xi\setminus C)$, then the dual optimizer λ^* is such that

$$\xi_*^i \in \arg\min_{\xi\in\Xi}\left\{\lambda^* d^p(\xi,\hat\xi^i) + \mathbb{1}_C(\xi)\right\} = \arg\min_{\xi\in\partial C} d^p(\xi,\hat\xi^i)$$

for all $i = 1,\dots,i_0-1$, and

Figure 3. (Color online) When $\Psi = -\mathbb{1}_C$, then the worst-case distribution perturbs the nominal distribution in a greedy fashion.



Notes. The solid and diamond dots are the support of the nominal distribution ν. $\hat{\xi}^1, \hat{\xi}^2, \hat{\xi}^3$ are the three interior points closest to ∂C and, thus, are transported to $\xi_*^1, \xi_*^2, \xi_*^3$, respectively. $\hat{\xi}^4$ is the interior point fourth closest to ∂C, but its full mass cannot be transported to ∂C because of the Wasserstein distance constraint, so it is split and the parts are moved to $\overline{\xi}_*^4$ and $\underline{\xi}_*^4 = \hat{\xi}^4$.

$$\xi_*^{i_0} \in \arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi, \hat{\xi}^{i_0}) + \mathbb{1}_C(\xi)\} = \begin{cases} \{\hat{\xi}^{i_0}\} \cup \arg\min_{\xi\in\partial C} d^p(\xi, \hat{\xi}^{i_0}) & \text{if } p_0 < 1, \\ \arg\min_{\xi\in\partial C} d^p(\xi, \hat{\xi}^{i_0}) & \text{if } p_0 = 1. \end{cases}$$

In this discussion, we assume that C is open. The next proposition shows that, for the normed space $(\mathbb{R}^K, \|\cdot\|)$, uncertainty quantification of any Borel set can be reduced to that of an open set.

Proposition 3 (Continuity with Respect to the Boundary). *Consider a metric space (Ξ, d) such that every bounded subset in (Ξ, d) is totally bounded. Let $\nu \in \mathcal{P}(\Xi)$, $\theta > 0$, and $\mathfrak{M} = \{\mu \in \mathcal{P}(\Xi) : W_p(\mu, \nu) \le \theta\}$. Then, for any proper Borel subset $C \subsetneq \Xi$, it holds that*

$$\inf_{\mu\in\mathfrak{M}} \mu(C) = \min_{\mu\in\mathfrak{M}} \mu(\mathrm{int}(C)).$$

Example 8 (Finite Domain). Next consider the special case when Ξ is finite, say, $\Xi = \{\xi^1, \ldots, \xi^B\}$ for some positive integer B. The nominal distribution ν is given by $\nu = \sum_{i=1}^B \nu_i \delta_{\xi^i}$. Then, the DRSO becomes

$$\min_{x\in X} \max_{\mu:=(\mu_1,\ldots,\mu_B)\in\Delta_B} \left\{ \sum_{i=1}^B \mu_i \Psi(x, \xi^i) : W_p(\mu, \nu) \le \theta \right\}. \tag{32}$$

Corollary 3. *Problem (32) has a strong dual*

$$\min_{x\in X,\ \lambda\ge 0,\ y:=(y_1,\ldots,y_B)} \left\{ \lambda\theta^p + \sum_{i=0}^B \nu_i y_i : y_i \ge \Psi(x, \xi^j) - \lambda d^p(\xi^i, \xi^j),\ \forall i, j = 1, \ldots, B \right\}. \tag{33}$$

For any ν and x, the worst-case distribution can be computed by solving

$$\max_{\mu\in\Delta_B,\ \gamma\in\mathbb{R}_+^{B\times B}} \left\{ \sum_{i=1}^B \mu_i \Psi(x, \xi^i) : \sum_{i,j=1}^B d^p(\xi^i, \xi^j)\gamma_{ij} \le \theta^p,\ \sum_{j=1}^B \gamma_{ij} = \mu_i\ \forall i,\ \sum_{i=1}^B \gamma_{ij} = \nu_j\ \forall j \right\}. \tag{34}$$

Proof. Reformulation (33) follows from Theorem 1, and (34) follows from the equivalent definition of Wasserstein distance in Example 2. □

4. Applications

In this section, we apply our results to various application problems, including on/off system control and intensity estimation for point processes. In these problems, the space Ξ is a space of counting measures (sample paths of a point process), which is nonconvex and infinite dimensional, and the nominal distribution is a point process. Hence, the results in Esfahani and Kuhn [23] and Zhao and Guan [57] cannot be applied to these problems.

4.1. On/Off System Control

In this problem, the decision maker faces a point process and controls a two-state (on/off) system. The point process is assumed to be exogenous, that is, the arrival times are not affected by the on/off state of the system. When the system is switched on, a cost of c per unit time is incurred, and each arrival while the system is on contributes one unit of revenue. When the system is off, no cost is incurred, and no revenue is earned. The decision maker wants to choose a control to maximize the total profit during a finite time horizon. This problem is a prototype for problems in sensor networks and revenue management.

Let the finite time horizon be denoted with $[0,1]$, and let

$$\Xi := \left\{ \xi := \sum_{m=1}^{M} \delta_{\eta_m} : M \in \mathbb{Z}_+,\ \eta_m \in [0,1],\ m = 1,\dots,M \right\}$$

be the set of sample paths of arrival times, which can also be described as the space of finite counting measures on $[0,1]$. In many practical settings, the decision maker does not have a probability distribution for the point process. Instead, the decision maker has observations of sample paths of the point process, which constitute an empirical point process as nominal distribution. Specifically, suppose that we have data of N sample paths $\hat{\xi}^i = \sum_{m=1}^{M_i} \delta_{\hat{\eta}^i_m}$, $i = 1,\dots,N$, where M_i is a nonnegative integer and $\hat{\eta}^i_m \in [0,1]$ for all i and m. Then, the nominal (empirical) distribution is $\nu = 1/N \sum_{i=1}^N \delta_{\hat{\xi}^i}$.

Note that, if one maximizes the expected profit with respect to the nominal distribution, then it yields a degenerate control in which the system is switched on only momentarily at the arrival time points of the observed sample paths. Consequently, if future arrival times differ from the historically observed arrival times by even a small amount, then the system is switched off at future arrival times and no revenue is earned. Because of such degeneracy and instability of optimization with respect to the nominal distribution, we resort to the distributionally robust approach.

As pointed out before, the choice of ambiguity set $\mathfrak{M}$ matters a great deal. For example, suppose that one chose $\mathfrak{M}$ to be the KL-divergence ball $\mathfrak{M}_{\phi_{KL}}$, that is, the set of all distributions $\mu \in \mathcal{P}(\Xi)$ such that $I_{\phi_{KL}}(\mu,\nu) \le \theta$ for some $\theta > 0$. Then, for $I_{\phi_{KL}}(\mu,\nu) \le \theta$ to hold, μ can put positive probability on observed sample paths $\hat{\xi}^i$ only. Thus, even if one solves a DRSO problem with KL-divergence ball $\mathfrak{M}_{\phi_{KL}}$, the resulting solution is a degenerate control in which the system is switched on only momentarily at the arrival time points of the historically observed sample paths, the same solution as when maximizing the expected profit with respect to the nominal distribution. In this section, we show that solving a DRSO problem with a Wasserstein ball results in sensible solutions.

We assume that the metric d on Ξ satisfies the following conditions. (Note that, in this section, when we write the W_1 distance between two measures, we use the extended definition mentioned in Section 2).

i. The metric space (Ξ, d) is a Polish space.

ii. For any nonnegative integer M and any $\xi = \sum_{m=1}^M \delta_{\eta_m}$ and $\hat{\xi} = \sum_{m=1}^M \delta_{\hat{\eta}_m}$, where $\{\eta_m\}_{m=1}^M, \{\hat{\eta}_m\}_{m=1}^M \subset [0,1]$, it holds that

$$d(\xi,\hat{\xi}) = W_1(\xi,\hat{\xi}) = \sum_{m=1}^{M} \left| \eta_{(m)} - \hat{\eta}_{(m)} \right|,$$

where $\{\eta_{(m)}\}$ and $\{\hat{\eta}_{(m)}\}$ are the order statistics of $\{\eta_m\}$ and $\{\hat{\eta}_m\}$, respectively.

iii. For any Borel set $C \subset [0,1]$, $\theta \ge 0$, positive integer M, and $\hat{\xi} = \sum_{m=1}^M \delta_{\hat{\eta}_m}$, where $\{\hat{\eta}_m\}_{m=1}^M \subset [0,1]$, it holds that

$$\inf_{\xi \in \Xi} \{\xi(C) : d(\xi,\hat{\xi}) \le \theta\} = \inf_{\xi \in \Xi} \{\xi(C) : W_1(\xi,\hat{\xi}) \le \theta\}.$$

Note that condition (ii) is imposed only on $\xi, \hat{\xi} \in \Xi$ such that $\xi([0,1]) = \hat{\xi}([0,1])$ and that conditions (ii) and (iii) do not imply that $d = W_1$. Examples of metrics d that satisfy these conditions are

$$d\left(\sum_{m=1}^{M}\delta_{\eta_m}, \sum_{l=1}^{L}\delta_{\hat{\eta}_l}\right) = \sum_{m=1}^{\min\{M,L\}}\left|\eta_{(m)} - \hat{\eta}_{(m)}\right| + |M-L|,$$

$$d\left(\sum_{m=1}^{M}\delta_{\eta_m}, \sum_{l=1}^{L}\delta_{\hat{\eta}_l}\right) = \begin{cases}\max\{M,L\}, & M \neq L,\\ \sum_{m=1}^{M}\left|\eta_{(m)} - \hat{\eta}_{(m)}\right|, & M = L,\end{cases}$$

and

$$d\left(\sum_{m=1}^{M}\delta_{\eta_m}, \sum_{l=1}^{L}\delta_{\hat{\eta}_l}\right) = \begin{cases}+\infty, & M \neq L,\\ \sum_{m=1}^{M}\left|\eta_{(m)} - \hat{\eta}_{(m)}\right|, & M = L.\end{cases} \tag{35}$$

These metrics are similar to the ones in Barbour and Brown [5] and Chen and Xia [17].

The set of point processes on $[0,1]$ is defined by the set $\mathcal{P}(\Xi)$ of Borel probability measures on Ξ. Given the metric d, we choose the distance between two point processes $\mu, \nu \in \mathcal{P}(\Xi)$ to be $W_1(\mu,\nu)$ as defined in (1). Then, the ambiguity set $\mathfrak{M} = \{\mu \in \mathcal{P}(\Xi) : W_1(\mu,\nu) \leq \theta\}$. Let X denote the set of all functions $x : [0,1] \longmapsto \{0,1\}$ such that $x^{-1}(1) := \{t \in [0,1] : x(t) = 1\}$ is a Borel set. The decision maker is looking for a control $x \in X$ that maximizes the total profit by solving the problem

$$v^* := \sup_{x \in X}\left\{v(x) := -c\int_0^1 x(t)dt + \inf_{\mu \in \mathfrak{M}} \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))]\right\}. \tag{36}$$

Next, we investigate the structure of the optimal control. Let $\operatorname{int}(x^{-1}(1))$ denote the interior of the set $x^{-1}(1)$ on the space $[0,1]$ with canonical topology (and, thus, $0, 1 \in \operatorname{int}([0,1])$).

Proposition 4. *Suppose that $\nu = 1/N\sum_{i=1}^{N}\delta_{\hat{\xi}^i}$ with $\hat{\xi}^i = \sum_{m=1}^{M_i}\delta_{\hat{\eta}^i_m}$. For any control x, it holds that*

$$\inf_{\mu \in \mathfrak{M}} \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] = \inf_{\rho \in \mathcal{P}(\Xi^2)}\left\{\mathbb{E}_{(\xi,\hat{\xi})\sim\rho}[\xi(x^{-1}(1))] : \mathbb{E}_{(\xi,\hat{\xi})\sim\rho}[W_1(\xi,\hat{\xi})] \leq \theta,\ \pi^2_{\#}\rho = \nu\right\} \tag{37}$$

$$= \sup_{\lambda \geq 0}\left\{-\lambda\theta + \frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{M_i}\min_{\eta\in[0,1]}\left\{\mathbb{1}_{\{\eta \in \operatorname{int}(x^{-1}(1))\}} + \lambda\left|\eta - \hat{\eta}^i_m\right|\right\}\right\}. \tag{38}$$

Moreover, there exists a nonnegative integer J such that

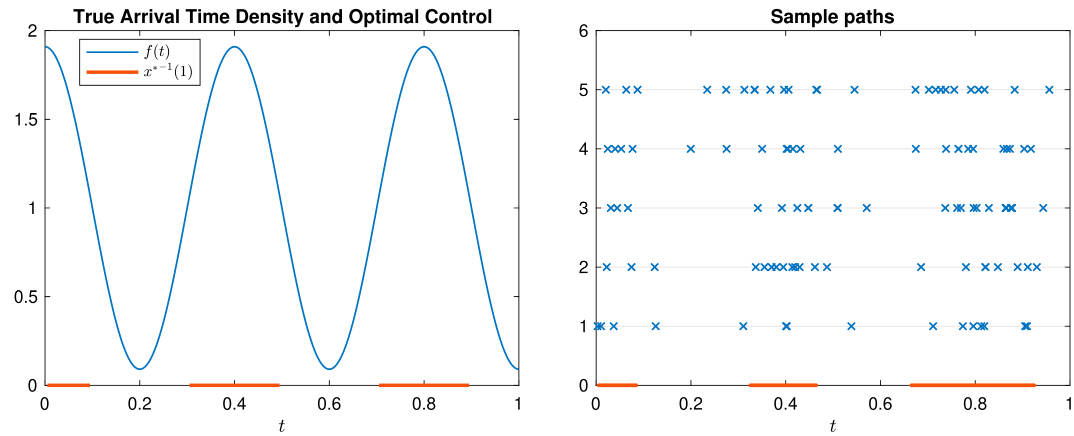
$$v^* = \sup_{\substack{\underline{x}_j, \overline{x}_j \in [0,1],\\ \underline{x}_j < \overline{x}_j < \underline{x}_{j'} < \overline{x}_{j'}\ \forall 1\leq j<j'\leq J}}\left\{v\left(\sum_{j=1}^{J}\mathbb{1}_{[\underline{x}_j,\overline{x}_j]}\right) := -c\sum_{j=1}^{J}(\overline{x}_j - \underline{x}_j) + \inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}\left[\xi\left(\cup_{j=1}^{J}[\underline{x}_j, \overline{x}_j]\right)\right]\right\}. \tag{39}$$

Note that

$$\inf_{\mu \in \mathfrak{M}} \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] = \inf_{\gamma \in \mathcal{P}(\Xi^2)}\left\{\mathbb{E}_{(\xi,\hat{\xi})\sim\gamma}[\xi(x^{-1}(1))] : \mathbb{E}_{\gamma}[d(\xi,\hat{\xi})] \leq \theta,\ \pi^2_{\#}\gamma = \nu\right\}.$$

Hence, (37) shows that, without changing the optimal value, we can replace d by W_1 in the constraint, and (38) provides an equivalent dual reformulation that can be interpreted as the dual problem of the uncertainty quantification problem in which the nominal distribution is a Borel measure $1/n\sum_{i=1}^{n}\sum_{m=1}^{M_i}\hat{\eta}^i_m$ on $[0,1]$. Then, by Example 7, it can be solved by a greedy algorithm. Moreover, (39) shows that, if ν is an empirical point process, then it suffices to consider the set of controls such that the system is on during a finite disjoint union of intervals with positive length.

Figure 4. (Color online) Optimal on/off system control for the true process and the DRSO.



Example 9. Figure 4 shows results for an instance of the on/off system control problem. Suppose that the number of arrivals is Poisson distributed with mean λ, and given the number of arrivals, the arrival points are independent and identically distributed (i.i.d.) with density $f(t)$, $t \in [0,1]$. For example, $f \equiv 1$ corresponds to the Poisson process with rate λ. The optimization problem based on knowing the process is $\max_x \int_{x^{-1}(1)} [-c + \lambda f(t)]dt$ with optimal control $x^*(t) = \mathbb{1}_{\{\lambda f(t) > c\}}$.

Let $f(t) = k[a + \sin(wt + s)]$ with $a > 1$ and $k = 1/[a + (\cos(s) - \cos(w+s))/w]$. Particularly, let $w = 5\pi$, $s = \pi/2$, $a = 1.1$, and $c = \lambda = 20$. Thus, $x^{*-1}(1) = [0, 0.1] \cup [0.3, 0.5] \cup [0.7, 0.9]$. The left of Figure 4 shows $f(t)$ and x^*. Suppose we have $n = 5$ sample paths $\hat{\xi}^1, \ldots, \hat{\xi}^5$, each of which contains $M_i \sim Poisson(\lambda)$ i.i.d. arrival points. The right of Figure 4 shows five sample paths and the resulting DRSO solution. Even with a relatively small number of sample paths, the two controls are close to each other, and the DRSO provides not only a sensible solution (unlike the approach of optimizing the expected profit with respect to the empirical distribution), but even a good solution for the true process control problem.

4.2. Intensity Estimation for Nonhomogeneous Poisson Processes

Consider the problem of estimating the intensity function $a(t)$ of a nonhomogeneous Poisson process $A(t)$ on $[0, T]$ using the maximum likelihood method. Given N i.i.d. sample paths $\hat{\xi}^i = \sum_{m=1}^{M_i} \delta_{\hat{\eta}^i_m}$, $i = 1, \ldots, N$, the log-likelihood function (see, e.g., Daley and Vere-Jones [18]) is written as

$$\mathscr{L}(a) = -\int_0^T a(t)\,dt + \frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{M_i} \ln(a(\hat{\eta}^i_m)).$$

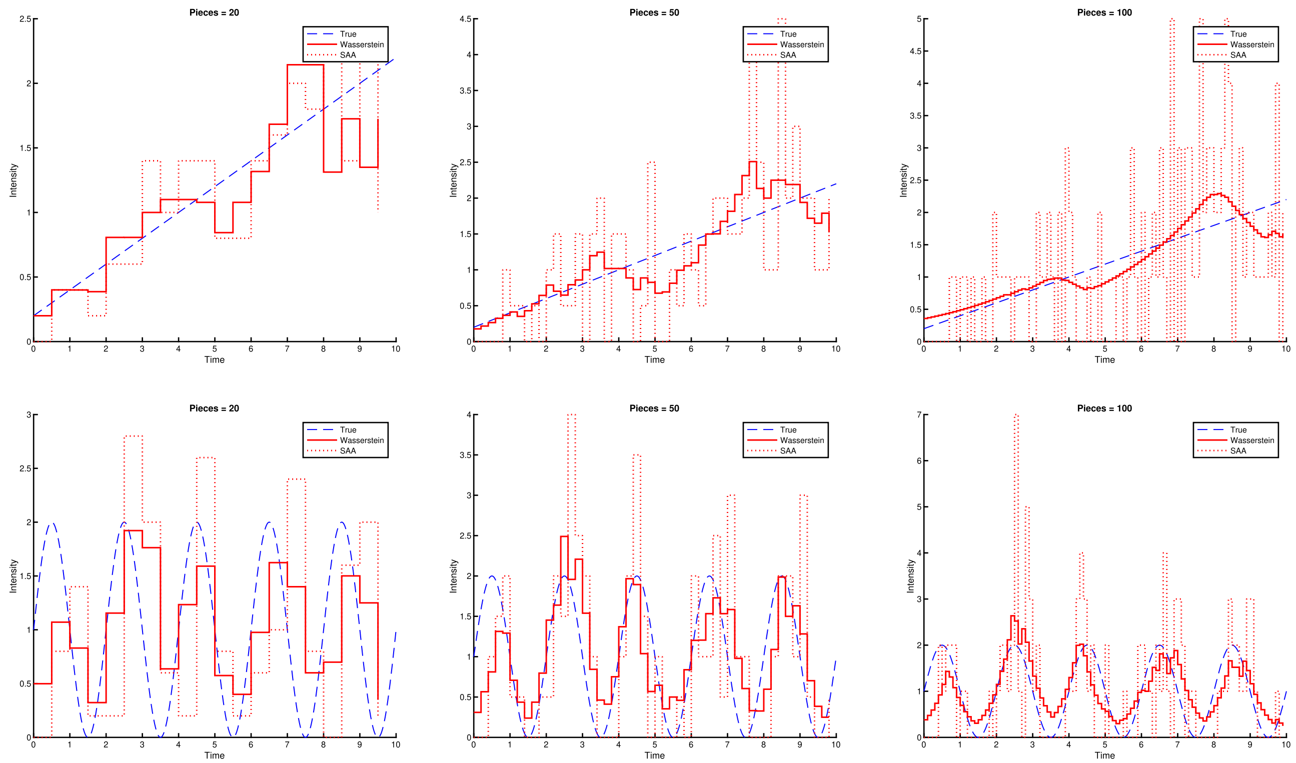
A common practice is to partition the time horizon $[0, T]$ into several intervals and assume that $a(t)$ is piecewise constant with constant value on each of the chosen intervals. Then, the maximum likelihood estimator reduces to the average arrival rate on each interval. Such an approach suffers from the drawback that the estimator is sensitive to the partition of the time horizon into intervals. If the partition is very coarse, then the estimator remains constant during long intervals and fails to capture time-varying arrival rates over shorter time periods. On the other hand, if the partition is very fine, then many intervals have zero observations, and the estimator varies too erratically over short intervals of time. A DRSO formulation with $\mathfrak{M}$ chosen to be the KL-divergence ball $\mathfrak{M}_{\phi_{KL}}$ has a similar problem because the resulting estimator vanishes on intervals with no observations.

Consider the DRSO formulation with Wasserstein distance

$$\min_{a(t),\ 0 \le t \le T} \left\{ v_\theta(a) := \int_0^T a(t)\,dt + \max_{\mu \in \mathfrak{M}} \mathbb{E}_{\eta \sim \mu}\left[-\int_0^T \ln(a(t))\,\eta(dt) \right] \right\}, \tag{40}$$

where $\mathfrak{M}$ is the Wasserstein ball centered at the empirical process. To facilitate further analysis, we choose d defined in (35) as the distance between two counting measures. Our strong duality results imply that the dual

Figure 5. (Color online) Estimated intensity functions using wasserstein DRSO and MLE.



reformulation of (40) is given by

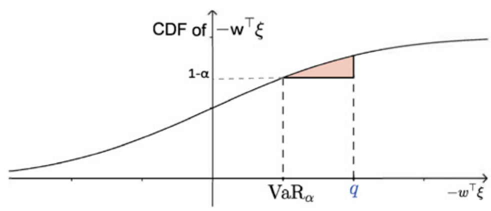
$$\min_{\substack{a(t)\\ \lambda\geq 0}} \left\{ \int_0^T a(t)dt + \lambda\theta + \frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{M_i} \sup_{\eta\in[0,T]} \left\{ -\ln\left(a(\xi)\right) - \lambda\left|\eta - \hat{\eta}_m^i\right| \right\} \right\}.$$

Next, we present numerical results for underlying true intensity functions given by $a(t) = 0.2 + 0.2t$ and $a(t) = 1 + \sin(\pi t)$, $t \in [0, 10]$. The sample size (number of sample paths) is $n = 20$. For both the maximum likelihood estimator and the DRSO estimator, we optimized over piecewise constant $a(t)$ with number of pieces in {20, 50, 100}. The Wasserstein radius θ was chosen via a cross-validation method, in which half of the sample paths were used for estimating $a_\theta \in \arg\min_a v_\theta(a)$ for each value of θ, and the other half of the sample paths were used for selecting the value of $\theta \in \arg\max_\theta \mathcal{Q}(a_\theta)$ with the largest log-likelihood. The estimation results for both the maximum likelihood estimator (MLE) and the Wasserstein DRSO estimator are shown in Figure 5. Table 1 shows the L^2 distances between the estimated intensity functions and the true intensity functions. The Wasserstein DRSO estimator has superior performance in all cases. Also, the performance of the DRSO estimator is less sensitive to the fineness of the partition for the piecewise constant function than the performance of the maximum likelihood estimator; in these results, the performance of the maximum likelihood estimator behaves poorly when the number of pieces is large.

Table 1. L2 distances between the true intensity functions and the estimated intensity functions of Wasserstein DRSO and MLE.

	$0.2 + 0.2t$			$1 + \sin(\pi t)$		
Pieces	20	50	100	20	50	100
Wasserstein	0.854	0.893	0.544	2.008	2.157	2.087
MLE	1.510	6.536	11.906	6.160	6.591	11.766

Figure 6. (Color online) Worst case VaR.



Notes. When $-w^\top\xi$ has a continuous cumulative distribution function, and $p=1$, then $\mathrm{VaR}_\alpha^{\mathfrak{M}}$ is equal to the value of q such that the area of the shaded region is equal to $\theta\|w\|_*$.

4.3. Worst-Case Value at Risk

Value at risk is a popular risk measure in finance. Given a real-valued random variable Z that represents losses with measure ν that has a positive density on an interval, and $\alpha\in(0,1)$, the value at risk $\mathrm{VaR}_\alpha^\nu[Z]$ of Z is defined by

$$\mathrm{VaR}_\alpha^\nu[Z] := \inf\{t : \mathbb{P}_\nu\{Z<t\}>1-\alpha\}.$$

In the spirit of El Ghaoui et al. [21], we consider the following worst-case VaR problem. Consider a portfolio consisting of K assets with allocation weights $w:=(w_1,\ldots,w_K)\ge 0$ such that $\sum_{k=1}^K w_k=1$. Let ξ_k denote the (random) return of asset k, $k=1,\ldots,K$. Assume that $(\Xi,d)=(\mathbb{R}^K,\|\cdot\|)$. The worst-case VaR with respect to the set of probability distributions $\mathfrak{M}$ is defined as

$$\mathrm{VaR}_\alpha^{\mathfrak{M}}(w) := \sup_{\mu\in\mathfrak{M}} \mathrm{VaR}_\alpha^\mu[-w^\top\xi] = \min\left\{q : \inf_{\mu\in\mathfrak{M}} \mathbb{P}_\mu\{-w^\top\xi<q\}>1-\alpha\right\}.$$

Given w and q, let $C:=\{\xi : -w^\top\xi<q\}$. Then, $\inf_{\mu\in\mathfrak{M}}\mathbb{P}_\mu\{-w^\top\xi<q\}=\inf_{\mu\in\mathfrak{M}}\mathbb{E}_\mu[\mathbb{1}_C(\xi)]=\min_{\mu\in\mathfrak{M}}\mu(C)$, similar to the uncertainty quantification Problem (31) considered in Example 7. Using the rationale developed there, we obtain the following result.

Proposition 5. *Suppose that ν has a positive density on $(\Xi,d)=(\mathbb{R}^K,\|\cdot\|)$, $\alpha\in(0,1)$, and $\theta>0$. Let w be given, and let $\nu_w\{(-\infty,s)\}:=\nu\{\xi : -w^\top\xi<s\}$ for all $s\in\mathbb{R}$. Then, $\mathrm{VaR}_\alpha^{\mathfrak{M}}(w)$ equals the unique solution q of the following equation:*

$$\int_{\mathrm{VaR}_\alpha^\nu[-w^\top\xi]}^q (q-s)^p\,\nu_w(ds)=\theta^p\|w\|_*^p.$$

Example 10 (Worst-Case VaR with Gaussian Nominal Distribution). Suppose that $\nu=N(\overline{\mu},\Sigma)$ and $p=1$. It follows that $-w^\top\xi\sim N(-w^\top\overline{\mu},w^\top\Sigma w)$ and $\mathrm{VaR}_\alpha^\nu[-w^\top\xi]=-w^\top\overline{\mu}+\sqrt{w^\top\Sigma w}\,\Phi^{-1}(1-\alpha)$, where $\Phi=N(0,1)$. By Proposition 5, $\mathrm{VaR}_\alpha^{\mathfrak{M}}(-w^\top\xi)$ is the value of q such that (see Figure 6)

$$f(q):=\frac{1}{\sqrt{2\pi w^\top\Sigma w}}\int_{\mathrm{VaR}_\alpha^\nu[-w^\top\xi]}^q (q-y)\,e^{-\frac{(y+w^\top\overline{\mu})^2}{2w^\top\Sigma w}}\,dy=\theta\|w\|_*. \tag{41}$$

Because $f(q)$ is increasing and continuous, (41) can be solved efficiently via a one-dimensional search algorithm.

Proposition 5 indicates that finding the worst-case VaR is tractable. It should also be noted that finding the best allocation weight, that is, optimizing over w, still may be hard because the VaR constraint is essentially a chance constraint.

5. Discussions

In this section, we discuss some of the advantages of using the Wasserstein ambiguity set. In Section 5.1, we compare DRSO with the Wasserstein ambiguity set to DRSO with ϕ-divergence ambiguity sets for the newsvendor problem. In Section 5.2, we illustrate how the close connection between DRSO and robust optimization (Corollary 3(iii)) can be used to solve DRSO problems.

5.1. Newsvendor Problem: Comparison with ϕ-Divergence

In this section, we consider distributionally robust newsvendor problems and compare results obtained with the Wasserstein ambiguity set and results obtained with ϕ-divergence ambiguity sets. In the newsvendor problem,

Table 2. Examples of ϕ-divergence.

Divergence	Kullback–Leibler	Burg entropy	χ^2-distance	Modified χ^2	Hellinger	Total variation
ϕ	ϕ_{kl}	ϕ_b	ϕ_{χ^2}	$\phi_{m\chi^2}$	ϕ_h	ϕ_{tv}
$\phi(t), t \geq 0$	$t\log t$	$-\log t$	$\frac{1}{t}(t-1)^2$	$(t-1)^2$	$(\sqrt{t}-1)^2$	$\lvert t-1\rvert$
$I_\phi(\mu,\nu)$	$\sum p_j \log\left(\frac{p_j}{q_j}\right)$	$\sum q_j \log\left(\frac{q_j}{p_j}\right)$	$\sum \frac{(p_j-q_j)^2}{p_j}$	$\sum \frac{(p_j-q_j)^2}{q_j}$	$\sum(\sqrt{p_j}-\sqrt{q_j})^2$	$\sum \lvert p_j - q_j\rvert$

the decision maker chooses the initial inventory level before the unknown demand is observed, facing both overage and underage costs. If the demand distribution μ is known, the problem can be formulated as

$$\min_{x\geq 0} \mathbb{E}_\mu[h[x-\xi]^+ + b[\xi-x]^+],$$

where x denotes the chosen initial inventory level; ξ denotes the random demand; and h, b represent, respectively, the overage and underage costs per unit. Assume that $h, b > 0$, and μ is supported on $\{0,1,\ldots,B\}$ for some positive integer B. Consider the usual metric $d(i,j) = |i-j|$ on $\{0,1,\ldots,B\}$.

Usually, the demand distribution μ is not known. Then, the decision maker may want to consider the DRSO problem

$$\min_{x\geq 0} \sup_{\mu\in\Delta_B} \left\{\mathbb{E}_\mu[h[x-\xi]^+ + b[\xi-x]^+] : W_p(\mu,\nu) \leq \theta\right\}.$$

Using Corollary 3, we obtain a convex optimization reformulation

$$\min_{x\geq 0,\, \lambda\geq 0,\, y:=(y_0,\ldots,y_B)} \left\{\lambda\theta^p + \sum_{i=0}^{B} \nu_i y_i : y_i \geq \max\{h(x-j), b(j-x)\} - \lambda|i-j|^p,\ \forall 0\leq i,j\leq B\right\}.$$

One may also consider ϕ-divergence ambiguity sets (Table 2 shows some common ϕ-divergences). As mentioned in Section 1, the worst-case distribution in a ϕ-divergence ambiguity set may be problematic. For example, when $\lim_{t\to\infty}\phi(t)/t = \infty$, such as ϕ_{kl} and $\phi_{m\chi^2}$, the ϕ-divergence ambiguity set fails to include many relevant distributions. Specifically, because $0\phi(p_j/0) := p_j \lim_{t\to\infty}\phi(t)/t = \infty$ for all $p_j > 0$, the ϕ_{kl}- and $\phi_{m\chi^2}$-divergence ambiguity sets do not include any distribution that is not absolutely continuous with respect to the nominal distribution ν.

When $\lim_{t\to\infty}\phi(t)/t < \infty$, such as ϕ_b, ϕ_{χ^2}, ϕ_h, and ϕ_{tv}, the situation is also bad. Let $I_0 := \{j\in\{0,\ldots,B\} : q_j > 0\}$. Assume that $\Psi(\xi^j)$ are different from each other so that $j_M \in \arg\max_{j\in\{0,\ldots,B\}}\{\Psi(\xi^j) : q_j = 0\}$ is unique. Then according to Ben-Tal et al. [8] and Bayraksan and Love [6], the worst-case distribution p^* satisfies

$$\frac{p_j^*}{q_j} \in \partial\phi^*\left(\frac{\Psi(\xi^j)-\beta^*}{\lambda^*}\right), \qquad \forall i\in I_0, \tag{42a}$$

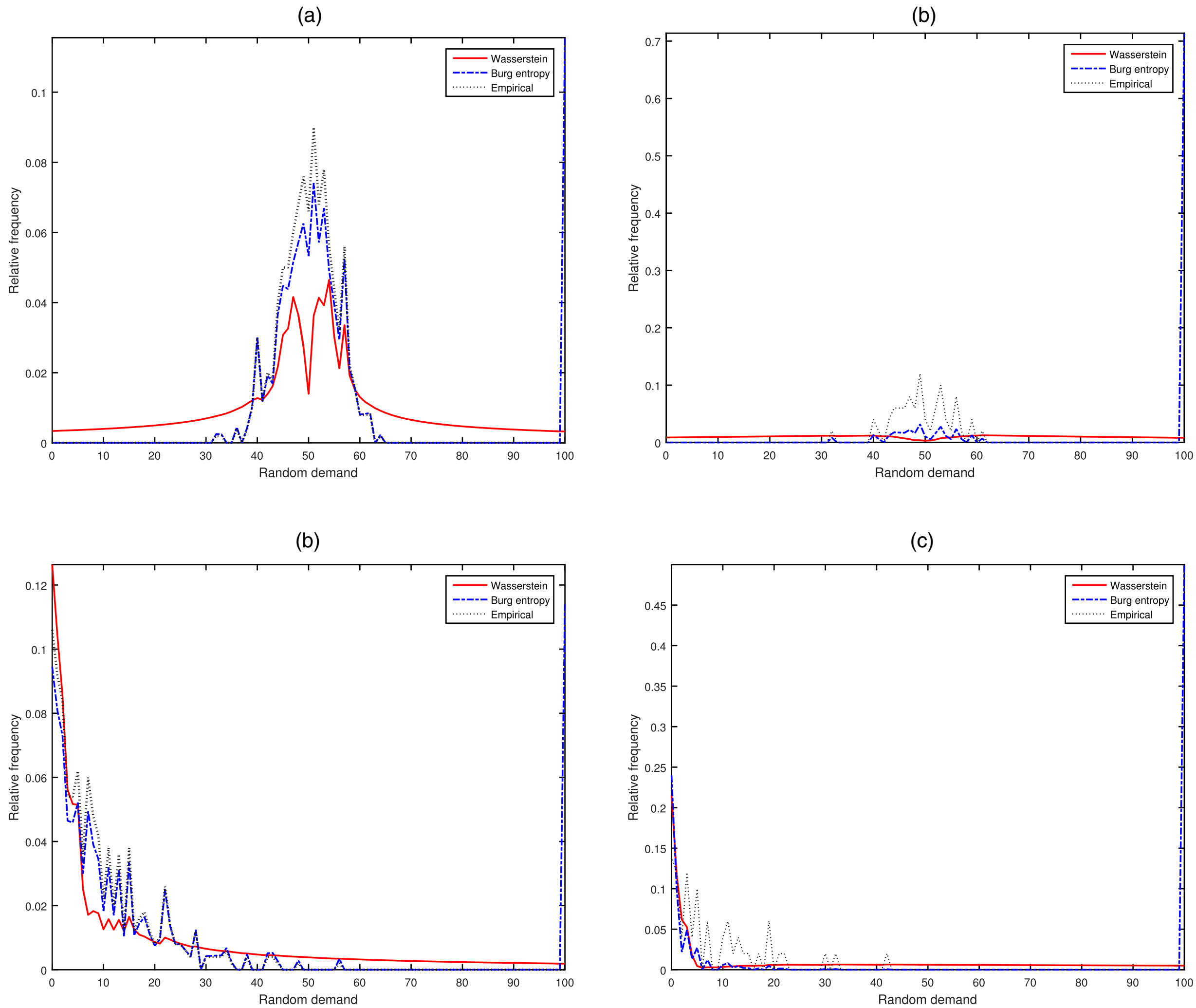
$$p_j^* = 0, \qquad \forall j\notin I_0\cup\{j_M\}, \tag{42b}$$

$$p_{j_M}^* = \begin{cases} 1 - \sum_{i\in I_0} p_j^* & \text{if } \beta^* = \Psi(\xi^{j_M}) - \lambda^* \lim_{t\to\infty}\phi(t)/t, \\ 0 & \text{if } \beta^* > \Psi(\xi^{j_M}) - \lambda^* \lim_{t\to\infty}\phi(t)/t, \end{cases} \tag{42c}$$

for some $\lambda^* \geq 0$ and $\beta^* \geq \Psi(\xi^{j_M}) - \lambda^* \lim_{t\to\infty}\phi(t)/t$, where ϕ^* denotes the convex conjugate function of ϕ. According to (42b), the support of the worst-case distribution and that of the nominal distribution can differ by at most one point ξ^{j_M}. According to (42c), the probability mass $p_{j_M}^*$ is moved away from scenarios in I_0 to the worst scenario ξ^{j_M}. Note that, in many applications in which the support of ξ is unknown, the choice of the underlying space Ξ (such as $\{0,1,\ldots,B\}$) may be somewhat arbitrary, and therefore, the worst scenario ξ^{j_M} (such as 0 or B) may be arbitrary. Hence, the worst-case behavior is too sensitive to arbitrary choices in the specification of Ξ.

We present numerical examples of which the setup is similar to that in Wang et al. [52] and Ben-Tal et al. [8]. Let $b = h = 1$, $B = 100$, and $N \in \{50, 500\}$ representing small and large data sets. Random data are generated from binomial(100, 0.5) and geometric(0.1) truncated on $[0, 100]$. We estimate the radius of the ambiguity set such that it covers the underlying distribution with probability greater than 0.95 (see Appendix C). For Burg entropy, we

Figure 7. (Color online) Histograms of worst-case distributions resulting from Wasserstein distance and Burg entropy.



Notes. (a) Binomial$(100, 0.5)$, $n = 500$. (b) Binomial$(100, 0.5)$, $n = 50$. (c) Truncated geometric(0.1), $n = 500$. (d) Truncated geometric(0.1), $n = 50$.

use the estimation developed in [52], while for Wasserstein distance, the radius is determined by a classical concentration inequality for Wasserstein distance developed in [14] (see Appendix D for more details). Figure 7 shows the histograms of the worst-case distributions under the optimal initial inventory levels determined by solving the DRSO problems with Wasserstein and Burg entropy ambiguity sets.

When the underlying distribution is binomial, the symmetry of the binomial distribution and $b = h = 1$ implies that the optimal initial inventory level is close to 50. Intuitively, the corresponding worst-case distribution should make provision for the possibility of both high and low demand. This intuition is consistent with the worst-case distributions in the Wasserstein ambiguity set shown by the solid curves in Figure 7, (a) and (b). The tails of these worst-case distributions are heavy on both the high and low sides and are quite smooth and reasonable for both small and large data sets. In contrast, if Burg entropy is used to define the ambiguity set, then the worst-case distributions have disconnected support as shown by the dashed curves in Figure 7, (a) and (b). There is a large spike on the boundary $B = 100$, displaying the "popping" behavior mentioned in Bayraksan and Love [6]. Especially when the data set is small, the worst-case spike is huge, which makes the solution too conservative.

When the underlying distribution is geometric, the worst-case distributions in the Wasserstein ambiguity set make provision for the possibility of both high and low demand with fairly smooth variation of probability in between as shown by solid curves in Figure 7, (c) and (d). As before, if Burg entropy is used, then the tail has unrealistic spikes on the boundary, and thus, the worst-case distribution is very sensitive to the somewhat arbitrary choice of truncation value B. In these examples, the Wasserstein ambiguity set seems to yield a more reasonable worst-case distribution.

5.2. Two-stage DRSO: Connection with Robust Optimization

In Corollary 2(iii), we establish the close connection between the DRSO problem and robust optimization. More specifically, we show that every DRSO problem can be approximated with high accuracy by robust optimization problems, which facilitates practical application of DRSO problems. To illustrate this point, in this section, we show the tractability of two-stage linear DRSO problems.

Consider the two-stage distributionally robust stochastic optimization problem

$$\min_{x\in X} c^\top x + \sup_{\mu\in\mathfrak{M}} \mathbb{E}_\mu[\Psi(x,\xi)], \tag{43}$$

where $\Psi(x,\xi)$ is the optimal objective value of the second stage problem

$$\Psi(x,\xi) := \min_{y\in\mathbb{R}^m} \{q(\xi)^\top y : T(\xi)x + W(\xi)y \le h(\xi)\}$$

and

$$q(\xi) = q^0 + \sum_{l=1}^{s} \xi_l q^l,\ T(\xi) = T^0 + \sum_{l=1}^{s} \xi_l T^l,\ W(\xi) = W^0 + \sum_{l=1}^{s} \xi_l W^l,\ h(\xi) = h^0 + \sum_{l=1}^{s} \xi_l h^l.$$

Assume that $p = 2$ and $\Xi = \mathbb{R}^s$ with Euclidean distance d. In general, the two-stage problem (43) is NP-hard (see Guslitser [31, section 3.3.2]). However, with tools from robust optimization, we are able to obtain a tractable approximation of (43). Let $\mathfrak{M}_1 := \{(\xi^1,\dots,\xi^N) \in \Xi^N : 1/N\sum_{i=1}^N \|\xi^i - \hat{\xi}^i\|_2^2 \le \theta^2\}$. Using Corollary 2(iii) with $K = 1$, we obtain an adjustable robust optimization approximation

$$\begin{aligned}
&\min_{x\in X}\left\{c^\top x + \sup_{(\xi^i)_{i=1}^N\in\mathfrak{M}_1} \frac{1}{N}\sum_{i=1}^N \Psi(x,\xi^i)\right\}\\
&= \min_{\substack{x\in X, t\in\mathbb{R},\\ y:\Xi\mapsto\mathbb{R}^m}} \left\{ t : \begin{array}{l} t \ge c^\top x + \dfrac{1}{N}\displaystyle\sum_{i=1}^N q(\xi^i)^\top y(\xi^i),\ \forall (\xi^i)_{i=1}^N \in \mathfrak{M}_1,\\[2ex] T(\xi)x + W(\xi)y(\xi) \le h(\xi),\ \forall \xi \in \displaystyle\bigcup_{i=1}^N \{\xi' \in \Xi : \|\xi' - \hat{\xi}^i\|_2 \le \theta\sqrt{N}\} \end{array}\right\},
\end{aligned} \tag{44}$$

where the second set of inequalities follows from the fact that $T(\xi)x + W(\xi)y(\xi) \le h(\xi)$ should hold for any realization ξ with positive probability for some distribution in $\mathfrak{M}_1$. Although Problem (44) in general is still intractable, there is a substantial literature on different approximations for Problem (44). One approach is to consider the affinely adjustable robust counterpart (AARC) as follows. Consider y that is an affine function of ξ:

$$y(\xi) = y^0 + \sum_{l=1}^{s} \xi_l y^l,\ \forall \xi \in \bigcup_{i=1}^N B^i,$$

for some chosen $y^0, y^l \in \mathbb{R}^m$, where $B^i := \{\xi' \in \Xi : \|\xi' - \hat{\xi}^i\|_2 \le \theta\sqrt{N}\}$. Then, the AARC of (44) is

$$\begin{aligned}
\min_{\substack{x\in X, t\in\mathbb{R},\\ y^l\in\mathbb{R}^m, l=0,\dots,s}} \Bigg\{ t : {} & c^\top x + \frac{1}{N}\sum_{i=1}^N \left(q^0 + \sum_{l=1}^s \xi_l^i q^l\right)^\top \left(y^0 + \sum_{l=1}^s \xi_l^i y^l\right) - t \le 0,\ \forall (\xi^i)_{i=1}^N \in \mathfrak{M}_1,\\
& \left(T^0 + \sum_{l=1}^s \xi_l T^l\right)x + \left(W^0 + \sum_{l=1}^s \xi_l W^l\right)\left(y^0 + \sum_{l=1}^s \xi_l y^l\right) - \left(h^0 + \sum_{l=1}^s \xi_l h^l\right) \le 0,\ \forall \xi \in \bigcup_{i=1}^N B^i \Bigg\}.
\end{aligned} \tag{45}$$

Set $\zeta_{il} := \xi_l^i - \hat{\xi}_l^i$ for $i = 1,\dots,N$ and $l = 1,\dots,s$. Note that $(\xi^1,\dots,\xi^N) \in \mathfrak{M}_1$ if and only if

$$\zeta := (\zeta_{il})_{i,l} \in \mathcal{U} := \left\{(\zeta'_{il})_{i,l} : \sum_{i=1}^N\sum_{l=1}^s \zeta'^2_{il} \le N\theta^2\right\}.$$

Set $z := (x, t, \{y^l\}_{l=0}^s)$, and let

$$\alpha_0(z) := -\left[c^\top x + \frac{1}{N}\sum_{i=1}^N\left(q^0 + \sum_{l=1}^s \hat{\xi}_l^i q^l\right)^\top\left(y^0 + \sum_{l=1}^s \hat{\xi}_l^i y^l\right) - t\right],$$

$$\beta_0^{il}(z) := -\frac{\left[\left(q^0 + \sum_{l'=1}^s \hat{\xi}_{l'}^i q^{l'}\right)^\top y^l + q^{l\top}\left(y^0 + \sum_{l'=1}^s \hat{\xi}_{l'}^i y^{l'}\right)\right]}{2N}, \qquad \forall i = 1, \ldots, N,\ l = 1, \ldots, s.$$

$$\Gamma_0^{(l,l')}(z) := -\frac{q^{l\top}y^{l'} + q^{l'\top}y^l}{2N}, \qquad \forall l, l' = 1, \ldots, s.$$

Then, the first set of constraints in (45) is equivalent to

$$\alpha_0(z) + 2\sum_{i=1}^N\sum_{l=1}^s \beta_0^{il}(z)\zeta_{il} + \sum_{i=1}^N\sum_{l=1}^s\sum_{l'=1}^s \Gamma_0^{(l,l')}(z)\zeta_{il}\zeta_{il'} \geq 0, \qquad \forall \zeta \in \mathcal{U}. \tag{46}$$

It follows from Ben-Tal et al. [7, theorem 4.2] that (46) holds if and only if there exists $\lambda_0 \geq 0$ such that

$$(\alpha_0(z) - \lambda_0)v^2 + 2v\sum_{i=1}^N\sum_{l=1}^s \beta_0^{il}(z)w_{il} + \sum_{i=1}^N\sum_{l=1}^s\sum_{l'=1}^s \Gamma_0^{(l,l')}(z)w_{il}w_{il'} + \frac{\lambda_0}{N\theta^2}\sum_{i=1}^N\sum_{l=1}^s w_{il}^2 \geq 0,$$
$$\forall v \in \mathbb{R},\ w_{il} \in \mathbb{R},\quad i = 1, \ldots, N,\ l = 1, \ldots, s.$$

In matrix form,

$$\exists \lambda_0 \geq 0 : \begin{pmatrix} \Gamma_0(z) \otimes I_N + \frac{\lambda_0}{N\theta^2} I_{sN} & \operatorname{vec}(\beta_0(z)) \\ \operatorname{vec}(\beta_0(z))^\top & \alpha_0(z) - \lambda_0 \end{pmatrix} \succeq 0, \tag{47}$$

where I_N (I_{sN}) is the $N \times N$ ($sN \times sN$) identity matrix, $\otimes$ is the Kronecker product of matrices, $\operatorname{vec}(\beta_0)$ is the vectorization of matrix $\beta_0(z)$, and $\Gamma_0(z)$ is a matrix whose (l, l')-element equal to $\Gamma_0^{(l,l')}(z)$.

Next, we reformulate the second set of constraints in (45). Let T_j^l and W_j^l denote row j of T^l and W^l, respectively, $j = 1, \ldots, J$. For all $i = 1, \ldots, N$, $j = 1, \ldots, J$, and $l, l' = 1, \ldots, s$, set

$$\alpha_{ij}(z) := -\left[\left(T_j^0 + \sum_{l=1}^s \hat{\xi}_l^i T_j^l\right)x + \left(W_j^0 + \sum_{l=1}^s \hat{\xi}_l^i W_j^l\right)\left(y^0 + \sum_{l=1}^s \hat{\xi}_l^i y^l\right) - \left(h_j^0 + \sum_{l=1}^s \hat{\xi}_l^i h_j^l\right)\right],$$

$$\beta_{ij}^l(z) := -\frac{\left[T_j^l x + \left(W_j^0 + \sum_{l=1}^s \hat{\xi}_l^i W_j^l\right)y^l + W_j^l\left(y^0 + \sum_{l=1}^s \hat{\xi}_l^i y^l\right) - h_j^l\right]}{2},$$

$$\Gamma_j^{(l,l')}(z) := -\frac{W_j^l y^{l'} + W_j^{l'} y^l}{2}.$$

Let $\eta^i := \xi - \hat{\xi}^i$, $\beta_{ij}(z) := (\beta_{ij}^l(z))_l$, and $\Gamma_j(z) := (\Gamma_j^{(l,l')}(z))_{l,l'}$. Then, the second set of constraints in (45) is equivalent to

$$\alpha_{ij}(z) + 2\beta_{ij}(z)^\top \eta^i + \eta^{i\top}\Gamma_j(z)\eta^i \geq 0, \qquad \forall \eta^i \in \{\eta' \in \mathbb{R}^s : \|\eta'\|_2 \leq \theta\sqrt{N}\},\ i = 1, \ldots, N,\ j = 1, \ldots, J.$$

Again, by Ben-Tal et al. [7, theorem 4.2], the second set of constraints in (45) is equivalent to

$$\exists \lambda_{ij} \geq 0 : \begin{pmatrix} \Gamma_j(z) + \frac{\lambda_{ij}}{N\theta^2} I_s & \beta_{ij}(z) \\ \beta_{ij}(z)^\top & \alpha_{ij}(z) - \lambda_{ij} \end{pmatrix} \succeq 0, \qquad \forall i = 1, \ldots, N,\ j = 1, \ldots, J. \tag{48}$$

Combining (47) and (48), we obtain the following result.

Proposition 6. *An exact semidefinite program reformulation of the AARC of* (44) *is given by*

$$\min_{\substack{x\in X,\, t\in\mathbb{R},\, y^l\in\mathbb{R}^m,\, l=0,\dots,s,\\ \lambda_0,\lambda_{ij}\ge 0,\, i=1,\dots,N,\, j=1,\dots,J}} \{t : (47), (48)\ holds\}. \tag{49}$$

By Corollary 2(iii), (44) is a fairly good approximation of the original two-stage DRSO problem (43). Hence, as long as the AARC of (44) is reasonably good, its semidefinite program reformulation (49) provides a good tractable approximation of the two-stage linear DRSO (43).

6. Conclusions

In this paper, we developed a constructive proof method to derive the dual reformulation of distributionally robust stochastic optimization with Wasserstein distance for a general setting. This approach allows us to obtain a precise structural description of the worst-case distribution. It also facilitates a connection between distributionally robust stochastic optimization and robust optimization. We showed how the results can be used to obtain theoretical and computational conclusions for a variety of problems.

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Appendix A. Proofs for Section 2

Proof of Lemma 1. Let (u_0, v_0) be any feasible solution for the maximization problem in (2). For any $t\in\mathbb{R}$ and any $\xi,\zeta\in\Xi$, let $u_t(\xi) := u_0(\xi)+t$ and $v_t(\zeta) := v_0(\zeta)-t$. Then, it follows that $u_t(\xi)+v_t(\zeta)\le d^p(\xi,\zeta)$ for all $\xi,\zeta\in\Xi$, and

$$\int_\Xi u_t(\xi)\mu(d\xi) + \int_\Xi v_t(\zeta)\nu(d\zeta) = \int_\Xi u_0(\xi)\mu(d\xi) + \int_\Xi v_0(\zeta)\nu(d\zeta) + t[\mu(\Xi)-\nu(\Xi)].$$

Because $\mu(\Xi)\neq\nu(\Xi)$,

$$\sup_{t\in\mathbb{R}}\left\{\int_\Xi u_t(\xi)\mu(d\xi) + \int_\Xi v_t(\zeta)\nu(d\zeta)\right\} = \infty,$$

and thus, $W_p^p(\mu,\nu)=\infty$. □

Appendix B. Proofs for Section 3

B.1. Proofs for Section 3.1

B.1.1. Auxiliary Results.

Lemma B.1. *Consider any $p\ge 1$ and any $\varepsilon>0$. Then, there exists $C_p(\varepsilon)\ge 1$ such that*

$$(x+y)^p \le (1+\varepsilon)x^p + C_p(\varepsilon)y^p$$

for all $x,y\ge 0$.

Proof of Lemma B.1. Note that, if $x = 0$, then the inequality holds for any $C_p(\varepsilon)\ge 1$. Next, we consider the case with $x > 0$, and we let $t := y/x$. Let

$$t_0(\varepsilon) := \sup\{t>0 : 1+\varepsilon \ge (1+t)^p\}.$$

Note that $t_0(\varepsilon)>0$. Next, let

$$C_p(\varepsilon) := \max\left\{1, \sup_{t\ge t_0(\varepsilon)} \frac{(1+t)^{p-1}}{t^{p-1}}\right\}.$$

Note that $C_p(\varepsilon)<\infty$ because $\lim_{t\to\infty}(1+t)^{p-1}/t^{p-1} = 1$. Next, consider

$$f(t) := 1+\varepsilon + C_p(\varepsilon)t^p - (1+t)^p.$$

Note that $f(t)\ge 0$ for all $t\in[0,t_0(\varepsilon)]$. Also, $f'(t) = C_p(\varepsilon)pt^{p-1} - p(1+t)^{p-1}\ge 0$ for all $t\in[t_0(\varepsilon),\infty)$. Therefore, $f(t)\ge 0$ for all $t\ge 0$, which establishes the inequality for $x > 0$. □

Lemma B.2. *Consider any $\zeta^0 \in \Xi$. Then, for any $\lambda > \lambda_1 > \kappa$, there exists a constant $C > 0$ such that*

$$\frac{\lambda - \lambda_1}{2}\overline{D}(\lambda, \zeta) \le \Phi(\lambda, \zeta) - \Phi(\lambda_1, \zeta^0) + \lambda_1 C d^p(\zeta, \zeta^0)$$

for all $\zeta \in \Xi$.

Proof of Lemma B.2. It follows from Lemma B.1 with $\varepsilon := \frac{\lambda - \lambda_1}{2\lambda_1}$ that

$$\lambda_1 d^p(\xi, \zeta^0) \;\le\; \frac{\lambda + \lambda_1}{2} d^p(\xi, \zeta) + \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0)$$

for all $\xi, \zeta, \zeta^0 \in \Xi$. Thus,

$$\begin{aligned}
\lambda d^p(\xi, \zeta) - \Psi(\xi) &= \frac{\lambda - \lambda_1}{2} d^p(\xi, \zeta) - \Psi(\xi) + \frac{\lambda + \lambda_1}{2} d^p(\xi, \zeta) \\
&\ge \frac{\lambda - \lambda_1}{2} d^p(\xi, \zeta) - \Psi(\xi) + \lambda_1 d^p(\xi, \zeta^0) - \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0) \\
&\ge \frac{\lambda - \lambda_1}{2} d^p(\xi, \zeta) + \Phi(\lambda_1, \zeta^0) - \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0).
\end{aligned}$$

Hence, for every $\xi \in \Xi$ that satisfies $\lambda d^p(\xi, \zeta) - \Psi(\xi) < \Phi(\lambda, \zeta) + \delta$ for some $\delta \ge 0$, it holds that

$$\begin{aligned}
&\frac{\lambda - \lambda_1}{2} d^p(\xi, \zeta) < \Phi(\lambda, \zeta) - \Phi(\lambda_1, \zeta^0) + \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0) + \delta \\
\Rightarrow &\frac{\lambda - \lambda_1}{2} \limsup_{\delta \downarrow 0} \left\{ \sup_{\xi \in \Xi} \{ d^p(\xi, \zeta) : \lambda d^p(\xi, \zeta) - \Psi(\xi) < \Phi(\lambda, \zeta) + \delta \} \right\} \\
&\qquad \le \limsup_{\delta \downarrow 0} \{ \Phi(\lambda, \zeta) - \Phi(\lambda_1, \zeta^0) + \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0) + \delta \} \\
\Rightarrow &\frac{\lambda - \lambda_1}{2} \overline{D}(\lambda, \zeta) \le \Phi(\lambda, \zeta) - \Phi(\lambda_1, \zeta^0) + \lambda_1 C_p(\varepsilon) d^p(\zeta, \zeta^0). \quad \square
\end{aligned}$$

B.1.2. Proof of Proposition 1.

Note that

$$\begin{aligned}
v_P &= \sup_{\mu \in \mathcal{P}(\Xi)} \left\{ \int_\Xi \Psi(\xi) \mu(d\xi) : W_p^p(\mu, \nu) \le \theta^p \right\} \\
&= \sup_{\gamma \in \mathcal{P}(\Xi^2)} \left\{ \int_{\Xi^2} \Psi(\xi) \gamma(d\xi, d\zeta) : \int_{\Xi^2} d^p(\xi, \zeta) \gamma(d\xi, d\zeta) \le \theta^p,\ \pi^2_{\#}\gamma = \nu \right\}.
\end{aligned}$$

Thus, for any $\lambda \ge 0$, it holds that

$$v_P \le \sup_{\gamma \in \mathcal{P}(\Xi^2)} \left\{ \int_{\Xi^2} \Psi(\xi) \gamma(d\xi, d\zeta) + \lambda \left[\theta^p - \int_{\Xi^2} d^p(\xi, \zeta) \gamma(d\xi, d\zeta) \right] : \pi^2_{\#}\gamma = \nu \right\}.$$

Hence,

$$\begin{aligned}
v_P &\le \inf_{\lambda \ge 0} \left\{ \lambda \theta^p + \sup_{\gamma \in \mathcal{P}(\Xi^2)} \int_{\Xi^2} [\Psi(\xi) - \lambda d^p(\xi, \zeta)] \gamma(d\xi, d\zeta) : \pi^2_{\#}\gamma = \nu \right\} \\
&\le \inf_{\lambda \ge 0} \left\{ \lambda \theta^p + \sup_{\gamma \in \mathcal{P}(\Xi^2)} \int_{\Xi^2} \sup_{\xi' \in \Xi} [\Psi(\xi') - \lambda d^p(\xi', \zeta)] \gamma(d\xi, d\zeta) : \pi^2_{\#}\gamma = \nu \right\} \\
&= \inf_{\lambda \ge 0} \left\{ \lambda \theta^p + \int_\Xi \sup_{\xi \in \Xi} [\Psi(\xi) - \lambda d^p(\xi, \zeta)] \nu(d\zeta) \right\} \\
&= v_D. \quad \square
\end{aligned}$$

B.1.3. Proof of Lemma 2.

i. We prove the result by contradiction. Suppose that, for some $\zeta^0, \zeta^1 \in \Xi$, it holds that

$$\kappa^0 := \limsup_{\xi \in \Xi : d(\xi, \zeta^0) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta^0)\}}{d^p(\xi, \zeta^0)} < \kappa^1 := \limsup_{\xi \in \Xi : d(\xi, \zeta^1) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta^1)\}}{d^p(\xi, \zeta^1)}$$

($\kappa^1 = \infty$ is allowed). Choose any $\varepsilon \in (0, \kappa^1 - \kappa^0)$. Then, there exists an R such that, for all ξ with $d(\xi, \zeta^0) > R$, it holds that

$$\begin{aligned}\Psi(\xi) - \Psi(\zeta^1) &= \Psi(\xi) - \Psi(\zeta^0) + \Psi(\zeta^0) - \Psi(\zeta^1) \\ &\le \max\{0, \Psi(\xi) - \Psi(\zeta^0)\} + \Psi(\zeta^0) - \Psi(\zeta^1) \\ &< (\kappa^0 + \varepsilon) d^p(\xi, \zeta^0) + [\Psi(\zeta^0) - \Psi(\zeta^1)] \\ &\le (\kappa^0 + \varepsilon)\big[d(\xi, \zeta^1) + d(\zeta^1, \zeta^0)\big]^p + [\Psi(\zeta^0) - \Psi(\zeta^1)].\end{aligned}$$

Because $\kappa^1 > 0$, it follows that

$$\begin{aligned}\kappa^1 &= \limsup_{\xi \in \Xi : d(\xi, \zeta^1) \to \infty} \frac{\Psi(\xi) - \Psi(\zeta^1)}{d^p(\xi, \zeta^1)} \\ &\le \limsup_{\xi \in \Xi : d(\xi, \zeta^1) \to \infty} \frac{(\kappa^0 + \varepsilon)\big[d(\xi, \zeta^1) + d(\zeta^1, \zeta^0)\big]^p + [\Psi(\zeta^0) - \Psi(\zeta^1)]}{d^p(\xi, \zeta^1)} \\ &= \kappa^0 + \varepsilon < \kappa^1,\end{aligned}$$

which is a contradiction.

ii. First, we show that, if there exists $\zeta^0 \in \Xi$ and $L, M > 0$ such that $\Psi(\xi) - \Psi(\zeta^0) \le L d^p(\xi, \zeta^0) + M$ for all $\xi \in \Xi$, then $\kappa < \infty$. Let $\kappa^0 := 0$ if Ξ is bounded, and let

$$\kappa^0 := \limsup_{\xi \in \Xi : d(\xi, \zeta^0) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta^0)\}}{d^p(\xi, \zeta^0)} \le L < \infty$$

if Ξ is unbounded. If Ξ is unbounded, then it follows from (i) that

$$\kappa^0 = \limsup_{\xi \in \Xi : d(\xi, \zeta) \to \infty} \frac{\max\{0, \Psi(\xi) - \Psi(\zeta)\}}{d^p(\xi, \zeta)} \quad \forall \zeta \in \Xi. \tag{B.1}$$

We show that $\int_\Xi \Phi(\lambda, \zeta) \nu(d\zeta) > -\infty$ for all $\lambda > \kappa^0$, and therefore, $\kappa \le \kappa^0 < \infty$.

First, we show that $\Phi(\lambda, \zeta) > -\infty$ for any $\lambda > \kappa^0$ and $\zeta \in \Xi$. If Ξ is bounded, then choose any $R(\zeta) > 0$ such that $d^p(\xi, \zeta) \le R(\zeta)$ for all $\xi \in \Xi$. If Ξ is unbounded, then it follows from (B.1) that, for any $\zeta \in \Xi$, there is a $R(\zeta) > 0$ such that, for all $\xi \in \Xi$ with $d^p(\xi, \zeta) > R(\zeta)$, it holds that

$$\frac{\Psi(\xi) - \Psi(\zeta)}{d^p(\xi, \zeta)} < \frac{\lambda + \kappa^0}{2},$$

that is, $\left(\frac{\lambda + \kappa^0}{2}\right) d^p(\xi, \zeta) - \Psi(\xi) > -\Psi(\zeta)$. Thus, for all $\xi \in \Xi$ with $d^p(\xi, \zeta) > R(\zeta)$, it holds that

$$\begin{aligned}\lambda d^p(\xi, \zeta) - \Psi(\xi) &= \frac{\lambda + \kappa^0}{2} d^p(\xi, \zeta) - \Psi(\xi) + \frac{\lambda - \kappa^0}{2} d^p(\xi, \zeta) \\ &> -\Psi(\zeta) + \frac{\lambda - \kappa^0}{2} R(\zeta),\end{aligned}$$

and hence,

$$\inf_{\xi \in \Xi} \{\lambda d^p(\xi, \zeta) - \Psi(\xi) : d^p(\xi, \zeta) > R(\zeta)\} \ge -\Psi(\zeta) + \frac{\lambda - \kappa^0}{2} R(\zeta) > -\infty.$$

Also, by assumption, for any $\xi \in \Xi$, it holds that

$$\begin{aligned}\Psi(\xi) - \Psi(\zeta^0) &\le L d^p(\xi, \zeta^0) + M \\ &\le L[d(\xi, \zeta) + d(\zeta, \zeta^0)]^p + M \\ &\le 2^{p-1} L[d^p(\xi, \zeta) + d^p(\zeta, \zeta^0)] + M,\end{aligned}$$

where the second inequality follows from the elementary inequality $(a+b)^p \le 2^{p-1}(a^p+b^p)$ for any $a,b \ge 0$, and $p \ge 1$. Thus,

$$\inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi) : d^p(\xi,\zeta)\le R(\zeta)\} \ge \inf_{\xi\in\Xi}\{-\Psi(\xi) : d^p(\xi,\zeta)\le R(\zeta)\}$$
$$\ge -\Psi(\zeta^0) - 2^{p-1}LR(\zeta) - 2^{p-1}Ld^p(\zeta,\zeta^0) - M > -\infty.$$

Therefore, $\Phi(\lambda,\zeta) > -\infty$ for all $\zeta\in\Xi$ and $\lambda>\kappa^0$.

Next, we show that $\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) > -\infty$ for any $\lambda>\kappa^0$. Consider any $\lambda_1\in(\kappa^0,\lambda)$ and any $\zeta^0\in\Xi$. It follows from Lemma B.2 that there is a constant C such that

$$\Phi(\lambda,\zeta) \ge \frac{\lambda-\lambda_1}{2}\overline{D}(\lambda,\zeta) + \Phi(\lambda_1,\zeta^0) - Cd^p(\zeta,\zeta^0) \ge \Phi(\lambda_1,\zeta^0) - Cd^p(\zeta,\zeta^0).$$

Thus,

$$\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) \ge \Phi(\lambda_1,\zeta^0) - C\int_\Xi d^p(\zeta,\zeta^0)\nu(d\zeta) > -\infty.$$

Therefore, $\kappa \le \kappa^0 < \infty$.

Next, we show that, if there does not exist $\zeta^0\in\Xi$ and $L,M>0$ such that $\Psi(\xi)-\Psi(\zeta^0)\le Ld^p(\xi,\zeta^0)+M$ for all $\xi\in\Xi$, then $\kappa=\infty$. First, observe that, if there exists $\zeta^0\in\Xi$ and $L,M>0$ such that $\Psi(\xi)-\Psi(\zeta^0)\le Ld^p(\xi,\zeta^0)+M$ for all $\xi\in\Xi$, then for any $\xi,\zeta\in\Xi$, it holds that

$$\begin{aligned}\Psi(\xi)-\Psi(\zeta) &= \Psi(\xi)-\Psi(\zeta^0)+\Psi(\zeta^0)-\Psi(\zeta)\\ &\le Ld^p(\xi,\zeta^0)+M+\Psi(\zeta^0)-\Psi(\zeta)\\ &\le L[d(\xi,\zeta)+d(\zeta,\zeta^0)]^p+M+\Psi(\zeta^0)-\Psi(\zeta)\\ &\le 2^{p-1}L[d^p(\xi,\zeta)+d^p(\zeta,\zeta^0)]+M+\Psi(\zeta^0)-\Psi(\zeta).\end{aligned}$$

It follows that there exists $\zeta^0\in\Xi$ and $L,M>0$ such that $\Psi(\xi)-\Psi(\zeta^0)\le Ld^p(\xi,\zeta^0)+M$ for all $\xi\in\Xi$ if and only if there exists $L' := 2^{p-1}L \ge 0$ and $M(\zeta) := 2^{p-1}Ld^p(\zeta,\zeta^0)+M+\Psi(\zeta^0)-\Psi(\zeta)\in L^1(\nu)$ such that $\Psi(\xi)-\Psi(\zeta)\le L'd^p(\xi,\zeta)+M(\zeta)$ for all $\xi,\zeta\in\Xi$, that is, there exists $L'\ge 0$ and $M(\zeta)\in L^1(\nu)$ such that $-\Psi(\zeta)-M(\zeta)\le \inf_{\xi\in\Xi}\{L'd^p(\xi,\zeta)-\Psi(\xi)\}$ for all $\zeta\in\Xi$. Therefore, if there does not exist $\zeta^0\in\Xi$ and $L,M>0$ such that $\Psi(\xi)-\Psi(\zeta^0)\le Ld^p(\xi,\zeta^0)+M$ for all $\xi\in\Xi$, then for any $\lambda\ge 0$, it holds that

$$\inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi)\}\notin L^1(\nu),$$

which implies that $\kappa=\infty$.

iii. It was established in the proof of (ii) that, if $\kappa<\infty$, then there exists $\zeta^0\in\Xi$ and $L,M>0$ such that $\Psi(\xi)-\Psi(\zeta^0)\le Ld^p(\xi,\zeta^0)+M$ for all $\xi\in\Xi$, and then,

$$\kappa \le \kappa^0 := \limsup_{\xi\in\Xi : d(\xi,\zeta^0)\to\infty} \frac{\max\{0,\Psi(\xi)-\Psi(\zeta^0)\}}{d^p(\xi,\zeta^0)}.$$

Next, we show that $\kappa\ge\kappa^0$. If $\kappa^0=0$, then it follows from the definition of κ that $\kappa\ge\kappa^0$. Next, suppose that $\kappa^0>0$, and consider any $\lambda\in[0,\kappa^0)$. We show that $\inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi)\}=-\infty$ for all $\zeta\in\Xi$. If $\lambda=0$, then it follows from $\kappa^0>0$ that $\inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi)\}=-\infty$ for all ζ. Next, consider any $\lambda\in(0,\kappa^0)$, any $\zeta\in\Xi$, any $M>0$, any $\lambda_2\in(\lambda,\kappa^0)$, and any $\varepsilon\in(0,(\lambda_2-\lambda)/\lambda)$. Because $[d(\xi,\zeta^0)+d(\zeta^0,\zeta)]^p/d^p(\xi,\zeta^0)\to 1$ as $d^p(\xi,\zeta^0)\to\infty$, it follows that there exists $R_1>0$ such that $d^p(\xi,\zeta)/d^p(\xi,\zeta^0)\le [d(\xi,\zeta^0)+d(\zeta^0,\zeta)]^p/d^p(\xi,\zeta^0)\le 1+\varepsilon$ for all $\xi\in\Xi$ such that $d^p(\xi,\zeta^0)>R_1$. Choose any $R>\max\{R_1,[M-\Psi(\zeta^0)]/(\lambda_2-\lambda-\lambda\varepsilon)\}$. It follows from the definition of κ^0 that there exists $\xi\in\Xi$ such that $d^p(\xi,\zeta^0)>R$ and

$$\begin{aligned}\Psi(\xi)-\Psi(\zeta^0) &> \lambda_2 d^p(\xi,\zeta^0)\\ &= \lambda d^p(\xi,\zeta^0)+(\lambda_2-\lambda)d^p(\xi,\zeta^0)\\ &\ge \lambda d^p(\xi,\zeta)+(\lambda_2-\lambda-\lambda\varepsilon)d^p(\xi,\zeta^0)\\ \Rightarrow \lambda d^p(\xi,\zeta)-\Psi(\xi) &< -\Psi(\zeta^0)-(\lambda_2-\lambda-\lambda\varepsilon)R < -M.\end{aligned}$$

Thus, $\inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi)\}=-\infty$ for all ζ, and hence, $\int_\Xi \inf_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta)-\Psi(\xi)\}\nu(d\zeta)=-\infty$ for all $\lambda\in[0,\kappa^0)$. Therefore, $\kappa\ge\kappa^0$. Next, recall that (i) established that κ^0 does not depend on the choice of ζ^0, and therefore, the result follows. □

B.1.4. Proof of Lemma 3.

i. By Ambrosio et al. [3, definition 1.11], ν has an extension, still denoted by ν, such that the measure space $(\Xi,\mathscr{B}_\nu,\nu)$ is complete. Note that, for any $b\in\mathbb{R}$, it holds that

$$\begin{aligned}\{\zeta\in\Xi:\Phi(\lambda,\zeta)<b\}&=\{\zeta\in\Xi:\exists\xi\in\Xi \text{ such that } \lambda d^p(\xi,\zeta)-\Psi(\xi)<b\}\\&=\pi^2(\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)<b\}).\end{aligned}$$

Note that the set $\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)<b\}$ on the right side is measurable. Because (Ξ,d) is Polish, it follows from the measurable projection theorem (cf. Aubin and Frankowska [4, theorem 8.3.2]) that $\Phi(\lambda,\cdot)$ is $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable.
Define functions $\overline{C},\underline{C}$ by

$$\begin{aligned}\overline{C}(\lambda,\zeta,\delta)&:=\sup_{\xi\in\Xi}\{d^p(\xi,\zeta):\lambda d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta\}\\\underline{C}(\lambda,\zeta,\delta)&:=\inf_{\xi\in\Xi}\{d^p(\xi,\zeta):\lambda d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta\}.\end{aligned}$$

For any $b\in\mathbb{R}$, it holds that

$$\begin{aligned}\{\zeta\in\Xi:\overline{C}(\lambda,\zeta,\delta)>b\}&=\{\zeta\in\Xi:\exists\xi\in\Xi\ \text{ such that } \lambda\ d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta,\ d^p(\xi,\zeta)>b\}\\&=\pi^2(\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta,\ d^p(\xi,\zeta)>b\}),\end{aligned}$$

and thus, it follows from the measurable projection theorem that $\overline{C}(\lambda,\cdot,\delta)$ is $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable. Similarly,

$$\begin{aligned}\{\zeta\in\Xi:\underline{C}(\lambda,\zeta,\delta)<b\}&=\{\zeta\in\Xi:\exists\xi\in\Xi\ \text{ such that } \lambda d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta,\ d^p(\xi,\zeta)<b\}\\&=\pi^2(\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)<\Phi(\lambda,\zeta)+\delta,\ d^p(\xi,\zeta)<b\}),\end{aligned}$$

and thus, $\underline{C}(\lambda,\cdot,\delta)$ is $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable.

Next, note that $\overline{D}(\lambda,\cdot)=\limsup_{\delta\downarrow 0}\overline{C}(\lambda,\cdot,\delta)$ and $\underline{D}(\lambda,\cdot)=\liminf_{\delta\downarrow 0}\underline{C}(\lambda,\cdot,\delta)$ are also $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable because measurability is preserved under lim sup and lim inf.
For any $b\in\mathbb{R}$, it holds that

$$\begin{aligned}\{\zeta\in\Xi:\overline{D}_0(\lambda,\zeta)>b\}&=\{\zeta\in\Xi:\exists\xi\in\Xi\ \text{ such that } \lambda d^p(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda,\zeta),\ d^p(\xi,\zeta)>b\}\\&=\pi^2(\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda,\zeta),\ d^p(\xi,\zeta)>b\}),\end{aligned}$$

and thus, it follows from the measurable projection theorem that $\overline{D}_0(\lambda,\cdot)$ is $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable. Similarly,

$$\begin{aligned}\{\zeta\in\Xi:\underline{D}_0(\lambda,\zeta)<b\}&=\{\zeta\in\Xi:\exists\xi\in\Xi \text{ such that } \lambda \mathrm{d}^{\mathrm{p}}(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda,\zeta),\ d^p(\xi,\zeta)<b\}\\&=\pi^2(\{(\xi,\zeta)\in\Xi\times\Xi:\lambda d^p(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda,\zeta),\ d^p(\xi,\zeta)<b\}),\end{aligned}$$

and thus, $\underline{D}_0(\lambda,\cdot)$ is $(\mathscr{B}_\nu,\mathscr{B}(\mathbb{R}))$-measurable.

ii. For each $\zeta\in\Xi$, it follows from the measurability of Ψ and $d^p(\cdot,\zeta)$ that $\overline{F}^\varepsilon_\delta(\lambda,\zeta)$ and $\underline{F}^\varepsilon_\delta(\lambda,\zeta)$ are in $\mathscr{B}(\Xi)$. Because (Ξ,d) is Polish and ν is a complete finite measure, it follows from Aumann's measurable selection theorem (see, e.g., Aliprantis and Border [2, theorem 18.26]) that ν-measurable selections $\overline{T}^\varepsilon_\delta(\lambda,\cdot),\underline{T}^\varepsilon_\delta(\lambda,\cdot):\Xi\longmapsto\Xi$ exist such that $\overline{T}^\varepsilon_\delta(\lambda,\zeta)\in\overline{F}^\varepsilon_\delta(\lambda,\zeta)$ and $\underline{T}^\varepsilon_\delta(\lambda,\zeta)\in\underline{F}^\varepsilon_\delta(\lambda,\zeta)$ for ν-almost all $\zeta\in\Xi$.

iii. The proof is the same as the proof of (ii).

iv. For each $\zeta\in E$, it follows from the measurability of Ψ and $d^p(\cdot,\zeta)$ that $F(\zeta)\in\mathscr{B}(\Xi)$. Then, using the same argument as in (ii), there exists a ν-measurable selection $T:E\longmapsto\Xi$ such that $T(\zeta)\in F(\zeta)$ for ν-almost all $\zeta\in E$.

v. For each $\zeta\in\Xi$, it follows from the measurability of Ψ, M, and $d^p(\cdot,\zeta)$ that $F(\zeta)\in\mathscr{B}(\Xi)$. Then, using the same argument as in (ii), there exists a ν-measurable selection $T:\Xi\longmapsto\Xi$ such that $T(\zeta)\in F(\zeta)$ for ν-almost all $\zeta\in\Xi$. □

B.1.5. Proof of Proposition 2.

If $\kappa=\infty$, then for any $n>0$, it holds that

$$\phi^n(\zeta):=\inf_{\xi\in\Xi}\{nd^p(\xi,\zeta)-\Psi(\xi)+\Psi(\zeta)\}\notin L^1(\nu).$$

Observe that $\phi^n(\zeta) = \Phi(n,\zeta) + \Psi(\zeta)$. Hence, for any $n > 0$, there exists $E^n \in \mathscr{B}_\nu$ with $\nu(E^n) > 0$ such that $\phi^n(\zeta) < -n$ for ν-almost all $\zeta \in E^n$. By Lemma 3(iv), there exists a ν-measurable mapping $T^n : E^n \mapsto \Xi$ such that

$$T^n(\zeta) \in F^n(\zeta) := \{\xi \in \Xi : \Psi(\xi) - \Psi(\zeta) > nd^p(\xi,\zeta) + n\}$$

for ν-almost all $\zeta \in E^n$. For $m = 1, 2, \ldots,$ consider the set

$$E^n_m := \{\zeta \in E^n : d^p(T^n(\zeta),\zeta) \le m\}.$$

Note that $E^n_m \in \mathscr{B}_\nu$ for all m. Then, $\lim_{m\to\infty} E^n_m = E^n$, and thus, $\lim_{m\to\infty}\nu(E^n_m) = \nu(E^n) > 0$. Hence, for each n, there exists m^n such that $\nu(E^n_{m^n}) > 0$, and $\int_{E^n_{m^n}} d^p(T^n(\zeta),\zeta)nu(d\zeta) \le m^n < \infty$. Let $\overline{T}^n$ be the restriction of T^n on $E^n_{m^n}$. Note that $\overline{T}^n$ is also ν-measurable.

For each n, let

$$p^n := \min\left\{1,\ \frac{\theta^p}{\int_{E^n_{m^n}} d^p(\overline{T}^n(\zeta),\zeta)\nu(d\zeta)}\right\},$$

and define the distribution

$$\mu^n := p^n \overline{T}^n_{\#}\nu + (1 - p^n)\nu.$$

Then, μ^n is a primal feasible solution, and

$$\int_\Xi \Psi d\mu^n - \int_\Xi \Psi d\nu \ge p^n\left(n\int_{E^n_{m^n}} d^p(\overline{T}^n(\zeta),\zeta)\nu(d\zeta) + n\right) \ge \min\{n\theta^p, n\}.$$

Because n can be chosen arbitrarily large, it follows that $v_P = \infty = v_D$. □

B.1.6. Proof of Lemma 4.

i. For any $\zeta \in \Xi$, $\Phi(\cdot,\zeta)$ is the infimum of nondecreasing functions. Thus, $\Phi(\cdot,\zeta)$ is nondecreasing for all $\zeta \in \Xi$. Also, for any $\zeta \in \Xi$, $\Phi(\cdot,\zeta)$ is the infimum of continuous functions. Thus, $\Phi(\cdot,\zeta)$ is upper semicontinuous for all $\zeta \in \Xi$. Consider any sequence $\{\lambda_n\}_n$ such that $\lambda_n \downarrow \kappa$ as $n \to \infty$. Because $\int_\Xi \Phi(\lambda_n,\zeta)\nu(d\zeta) > -\infty$, it holds that there is a set $B_n \in \mathscr{B}_\nu(\Xi)$ such that $\nu(B_n) = 1$ and $\Phi(\lambda_n,\zeta) > -\infty$ for all $\zeta \in B_n$. Then, it follows from $\Phi(\cdot,\zeta)$ being nondecreasing that $\Phi(\lambda,\zeta) > -\infty$ for all $\lambda \ge \lambda_n$ and all $\zeta \in B_n$. Let $B := \cap_n B_n$. Then, $B \in \mathscr{B}_\nu(\Xi)$ and $\nu(B) = 1$, and $\Phi(\lambda,\zeta) > -\infty$ for all $\lambda > \kappa$ and all $\zeta \in B$. Because $\Phi(\cdot,\zeta)$ is the infimum of affine functions of λ, and $\Phi(\lambda,\zeta) < \infty$ for all $\lambda \ge 0$ and all $\zeta \in \Xi$, and $\Phi(\lambda,\zeta) > -\infty$ for all $\lambda > \kappa$ and all $\zeta \in B$, it follows that $\Phi(\cdot,\zeta)$ is concave on $[0,\infty)$ for all $\zeta \in B$.

For the second part, consider any $\varepsilon \ge 0$, any $\lambda_2 > \lambda_1$, and any $\zeta \in \Xi$ such that $\Phi(\lambda_1,\zeta) > -\infty$. Consider any $\xi' \in \Xi$ such that $d^p(\xi',\zeta) > \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}$. If no such $\xi' \in \Xi$ exists, then it follows immediately that $\sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta) + \varepsilon\} \le \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}$.

Otherwise, consider any ξ'' such that $\lambda_1 d^p(\xi'',\zeta) - \Psi(\xi'') < \Phi(\lambda_1,\zeta) + \min\{\varepsilon, (\lambda_2 - \lambda_1)[d^p(\xi',\zeta) - \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}]\}$. Because $\lambda_1 d^p(\xi'',\zeta) - \Psi(\xi'') < \Phi(\lambda_1,\zeta) + \varepsilon$, it follows that $d^p(\xi'',\zeta) \le \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\} < d^p(\xi',\zeta)$. Thus, $\lambda_1 d^p(\xi',\zeta) - \Psi(\xi') > \Phi(\lambda_1,\zeta) + \varepsilon$ and

$$\begin{aligned}
&\lambda_1 d^p(\xi'',\zeta) - \Psi(\xi'') < \Phi(\lambda_1,\zeta)\\
&\qquad\qquad + (\lambda_2 - \lambda_1)\left[d^p(\xi',\zeta) - \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}\right]\\
&\qquad < \lambda_1 d^p(\xi',\zeta) - \Psi(\xi') - \varepsilon\\
&\qquad\qquad + (\lambda_2 - \lambda_1)\left[d^p(\xi',\zeta) - \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}\right]\\
\Rightarrow\ &\lambda_1[d^p(\xi',\zeta) - d^p(\xi'',\zeta)] > \Psi(\xi') - \Psi(\xi'') + \varepsilon\\
&\qquad\qquad - (\lambda_2 - \lambda_1)\left[d^p(\xi',\zeta) - \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}\right]\\
\Rightarrow\ &\lambda_2[d^p(\xi',\zeta) - d^p(\xi'',\zeta)] > \Psi(\xi') - \Psi(\xi'') + \varepsilon + (\lambda_2 - \lambda_1)[d^p(\xi',\zeta) - d^p(\xi'',\zeta)]\\
&\qquad\qquad - (\lambda_2 - \lambda_1)\left[d^p(\xi',\zeta) - \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta) + \varepsilon\}\right]\\
&\qquad\qquad \ge \Psi(\xi') - \Psi(\xi'') + \varepsilon\\
\Rightarrow\ &\lambda_2 d^p(\xi',\zeta) - \Psi(\xi') > \lambda_2 d^p(\xi'',\zeta) - \Psi(\xi'') + \varepsilon \ge \Phi(\lambda_2,\zeta) + \varepsilon.
\end{aligned}$$

That is, every $\xi' \in \Xi$ such that $d^p(\xi',\zeta) > \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta)+\varepsilon\}$ satisfies $\lambda_2 d^p(\xi',\zeta) - \Psi(\xi') > \Phi(\lambda_2,\zeta)+\varepsilon$, and thus, $\sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta)+\varepsilon\} \le \sup_{\xi\in\Xi}\{d^p(\xi,\zeta) : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta)+\varepsilon\}$.

For the third part, consider any $\lambda_2 > \lambda_1$ and any $\zeta \in \Xi$ such that $\Phi(\lambda_1,\zeta) > -\infty$. For any $\delta_i > 0$, consider any $\xi_i^{\delta_i} \in \Xi$ such that $\lambda_i d^p(\xi_i^{\delta_i},\zeta) - \Psi(\xi_i^{\delta_i}) \le \Phi(\lambda_i,\zeta)+\delta_i$ for $i = 1, 2$. It follows that

$$\begin{aligned}
\lambda_2 d^p(\xi_2^{\delta_2},\zeta) - \Psi(\xi_2^{\delta_2}) &\le \Phi(\lambda_2,\zeta)+\delta_2\\
&\le \lambda_2 d^p(\xi_1^{\delta_1},\zeta) - \Psi(\xi_1^{\delta_1}) + \delta_2\\
&= (\lambda_2-\lambda_1) d^p(\xi_1^{\delta_1},\zeta) + \lambda_1 d^p(\xi_1^{\delta_1},\zeta) - \Psi(\xi_1^{\delta_1}) + \delta_2\\
&\le (\lambda_2-\lambda_1) d^p(\xi_1^{\delta_1},\zeta) + \Phi(\lambda_1,\zeta) + \delta_1 + \delta_2\\
&\le (\lambda_2-\lambda_1) d^p(\xi_1^{\delta_1},\zeta) + \lambda_1 d^p(\xi_2^{\delta_2},\zeta) - \Psi(\xi_2^{\delta_2}) + \delta_1 + \delta_2\\
\Rightarrow d^p(\xi_2^{\delta_2},\zeta) - \frac{\delta_2}{\lambda_2-\lambda_1} &\le d^p(\xi_1^{\delta_1},\zeta) + \frac{\delta_1}{\lambda_2-\lambda_1}\\
\Rightarrow \sup_{\xi\in\Xi}&\left\{d^p(\xi,\zeta) - \frac{\delta_2}{\lambda_2-\lambda_1} : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta)+\delta_2\right\}\\
\le \inf_{\xi\in\Xi}&\left\{d^p(\xi,\zeta) + \frac{\delta_1}{\lambda_2-\lambda_1} : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta)+\delta_1\right\}\\
\Rightarrow \limsup_{\delta_2\downarrow 0}&\left\{\sup_{\xi\in\Xi}\left\{d^p(\xi,\zeta) - \frac{\delta_2}{\lambda_2-\lambda_1} : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta)+\delta_2\right\}\right\}\\
\le \liminf_{\delta_1\downarrow 0}&\left\{\inf_{\xi\in\Xi}\left\{d^p(\xi,\zeta) + \frac{\delta_1}{\lambda_2-\lambda_1} : \lambda_1 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_1,\zeta)+\delta_1\right\}\right\}\\
\Rightarrow \overline{D}(\lambda_2,\zeta) &\le \underline{D}(\lambda_1,\zeta).
\end{aligned}$$

Also, it follows from the definition of $\overline{D}$ and $\underline{D}$ that $\underline{D}(\lambda_1,\zeta) \le \overline{D}(\lambda_1,\zeta)$.

ii. It follows from the definition of Φ that, for all $\xi,\zeta \in \Xi$, it holds that

$$\Psi(\xi) \le \lambda_1 d^p(\xi,\zeta) - \Phi(\lambda_1,\zeta).$$

Also, for every $\xi \in \Xi$ that satisfies $\lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta)+\delta$ for some $\delta \ge 0$, it holds that

$$\lambda_2 d^p(\xi,\zeta) - \Psi(\xi) - \delta \le -\Psi(\zeta).$$

Combining these two inequalities yields that

$$\begin{aligned}
&\lambda_2 d^p(\xi,\zeta) + \Psi(\zeta) - \delta \le \lambda_1 d^p(\xi,\zeta) - \Phi(\lambda_1,\zeta)\\
&\Rightarrow (\lambda_2-\lambda_1) d^p(\xi,\zeta) - \delta \le -\Psi(\zeta) - \Phi(\lambda_1,\zeta)\\
&\Rightarrow \limsup_{\delta\downarrow 0}\left\{\sup_{\xi\in\Xi}\{(\lambda_2-\lambda_1) d^p(\xi,\zeta) - \delta : \lambda_2 d^p(\xi,\zeta) - \Psi(\xi) \le \Phi(\lambda_2,\zeta)+\delta\}\right\} \le -\Psi(\zeta) - \Phi(\lambda_1,\zeta)\\
&\Rightarrow (\lambda_2-\lambda_1)\overline{D}(\lambda_2,\zeta) \le -\Psi(\zeta) - \Phi(\lambda_1,\zeta).
\end{aligned}$$

iii. Consider any $\zeta \in B$ and any $\lambda_2 > \lambda_1 > \kappa$. For any $\delta > 0$, choose any $\xi_i^\delta \in \Xi$ such that $\lambda_i d^p(\xi_i^\delta,\zeta) - \Psi(\xi_i^\delta) \le \Phi(\lambda_i,\zeta)+\delta$ for $i = 1, 2$. Then,

$$\Phi(\lambda_1,\zeta) - \Phi(\lambda_2,\zeta) \le \lambda_1 d^p(\xi_2^\delta,\zeta) - \Psi(\xi_2^\delta) - [\lambda_2 d^p(\xi_2^\delta,\zeta) - \Psi(\xi_2^\delta)] + \delta = (\lambda_1-\lambda_2) d^p(\xi_2^\delta,\zeta) + \delta.$$

Similarly, $\Phi(\lambda_2,\zeta) - \Phi(\lambda_1,\zeta) \le (\lambda_2-\lambda_1) d^p(\xi_1^\delta,\zeta) + \delta$. It follows that

$$d^p(\xi_2^\delta,\zeta) - \frac{\delta}{\lambda_2-\lambda_1} \le \frac{\Phi(\lambda_2,\zeta)-\Phi(\lambda_1,\zeta)}{\lambda_2-\lambda_1} \le d^p(\xi_1^\delta,\zeta) + \frac{\delta}{\lambda_2-\lambda_1}.$$

Then, it follows from the definitions of $\overline{D}$ and $\underline{D}$ that

$$\overline{D}(\lambda_2,\zeta) \le \frac{\Phi(\lambda_2,\zeta)-\Phi(\lambda_1,\zeta)}{\lambda_2-\lambda_1} \le \underline{D}(\lambda_1,\zeta).$$

Because $\lambda_1 \in \operatorname{int}(\operatorname{dom}(\Phi(\cdot,\zeta)))$, there is a $\lambda_0 < \lambda_1$ such that $\Phi(\cdot,\zeta)$ is finite-valued and concave on (λ_0,∞), and the left and right derivatives $\partial\Phi(\lambda,\zeta)/\partial_{\lambda\pm}$ exist for all $\lambda \in (\lambda_0,\infty)$. Setting $\lambda_2 = \lambda$ and letting $\lambda_1 \uparrow \lambda$, it follows that

$$\overline{D}(\lambda,\zeta) \le \frac{\partial\Phi(\lambda,\zeta)}{\partial\lambda-} \le \lim_{\lambda_1\uparrow\lambda} \underline{D}(\lambda_1,\zeta).$$

Similarly, setting $\lambda_1 = \lambda$ and letting $\lambda_2 \downarrow \lambda$ in the preceding inequality, it follows that

$$\lim_{\lambda_2\downarrow\lambda} \overline{D}(\lambda_2,\zeta) \le \frac{\partial\Phi(\lambda,\zeta)}{\partial\lambda+} \le \underline{D}(\lambda,\zeta). \quad \square$$

B.1.7. Proof of Lemma 5.

i. It follows from Definition 4 of κ that $\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) = -\infty$ for all $\lambda < \kappa$ and $\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta)$ is finite for all $\lambda > \kappa$, and thus, $h(\lambda) = \infty$ for all $\lambda < \kappa$ and $h(\lambda)$ is finite for all $\lambda > \kappa$.

ii. It follows from Lemma 4(i) that $h(\lambda)$ is the sum of a linear function $\lambda\theta^p$ and an (extended real-valued) convex function $-\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta)$ on $[0,\infty)$. Thus, h is convex.

iii. Note that (i) and (ii) imply that h is continuous everywhere except possibly at κ. We show that h is lower semicontinuous at κ. Consider any sequence $\{\lambda_n\}_n$ such that $\lambda_n \downarrow \kappa$. Because $\Phi(\cdot,\zeta)$ is upper semicontinuous for all $\zeta \in \Xi$, it follows that $\Phi(\kappa,\zeta) \ge \limsup_{n\to\infty}\Phi(\lambda_n,\zeta)$. Also, $\Phi(\lambda,\zeta) \le -\Psi(\zeta)$ for all λ and ζ. Thus, $\liminf_{n\to\infty}[-\Phi(\lambda_n,\zeta)] - \Psi(\zeta) \ge -\Phi(\kappa,\zeta) - \Psi(\zeta) \ge 0$ for all ζ. Hence, it follows from Fatou's lemma that

$$\begin{aligned}
\liminf_{n\to\infty} h(\lambda_n) - \int_\Xi \Psi(\zeta)\nu(d\zeta) &= \liminf_{n\to\infty}\left\{\lambda_n\theta^p + \int_\Xi [-\Phi(\lambda_n,\zeta) - \Psi(\zeta)]\nu(d\zeta)\right\} \\
&\ge \kappa\theta^p + \int_\Xi \liminf_{n\to\infty}[-\Phi(\lambda_n,\zeta) - \Psi(\zeta)]\nu(d\zeta) \\
&\ge \kappa\theta^p + \int_\Xi [-\Phi(\kappa,\zeta) - \Psi(\zeta)]\nu(d\zeta) \\
&= h(\kappa) - \int_\Xi \Psi(\zeta)\nu(d\zeta).
\end{aligned}$$

Because $\left|\int_\Xi \Psi(\zeta)\nu(d\zeta)\right| < \infty$, it follows that $\liminf_{n\to\infty} h(\lambda_n) \ge h(\kappa)$, and thus, h is lower semicontinuous.

iv. Because $\Phi(\lambda,\zeta) \le -\Psi(\zeta)$, it follows that $h(\lambda) \ge \lambda\theta^p + \int_\Xi \Psi(\zeta)\nu(d\zeta) \to \infty$ as $\lambda \to \infty$.

v. The result follows from (i)–(iv). $\square$

B.1.8. Proof of Corollary 1.

i. Recall from Lemma 4(i) that there is a set $B \in \mathscr{B}_\nu(\Xi)$ such that $\nu(B) = 1$, and $\Phi(\lambda,\zeta) > -\infty$ for all $\lambda > \kappa$ and all $\zeta \in B$. Note that, if Ψ is upper semicontinuous and bounded subsets of (Ξ,d) are totally bounded, then for $\delta = 0$, it holds that $\underline{F}(\lambda,\zeta)$ and $\overline{F}(\lambda,\zeta)$ in Lemma 3(iii) are nonempty for all $\lambda > \kappa$ and all $\zeta \in B$. Next, we show that $\overline{D}_0(\cdot,\zeta)$ and $\underline{D}_0(\cdot,\zeta)$ are nonincreasing. Consider any $\lambda_2 > \lambda_1$ and any $\zeta \in \Xi$ such that $\Phi(\lambda_1,\zeta) > -\infty$. Consider any $\xi_i \in \Xi$ such that $\lambda_i d^p(\xi_i,\zeta) - \Psi(\xi_i) = \Phi(\lambda_i,\zeta)$ for $i = 1, 2$. Then, it follows as in the proof of Lemma 4(i) that $d^p(\xi_2,\zeta) \le d^p(\xi_1,\zeta)$. Therefore, $\overline{D}_0(\lambda_2,\zeta) \le \underline{D}_0(\lambda_1,\zeta) \le \overline{D}_0(\lambda_1,\zeta)$.

Next, we show that, for all $\zeta \in B$, it holds that $\overline{D}_0(\cdot,\zeta)$ is upper semicontinuous and $\underline{D}_0(\cdot,\zeta)$ is lower semicontinuous at all $\lambda > \kappa$. Consider any $\lambda > \kappa$ and any sequence $\{\lambda_n\}_n$ such that $\lambda_n \to \lambda$ as $n \to \infty$ and $\lambda_n \in ((\lambda+\kappa)/2, \lambda+\delta)$ for all n for some $\delta > 0$. For each n and each $\zeta \in B$, consider any $\xi^n \in \arg\min_{\xi\in\Xi}\{\lambda_n d^p(\xi,\zeta) - \Psi(\xi)\}$. Note that $d^p(\xi^n,\zeta) \in [\overline{D}_0(\lambda+\delta,\zeta), \underline{D}_0((\lambda+\kappa)/2,\zeta)]$ for all n. Because bounded subsets of (Ξ,d) are totally bounded, it is sufficient to consider subsequences of $\{\xi^n\}_n$ that converge to some $\xi^* \in \Xi$. It follows from the upper semicontinuity of Ψ and the continuity of $\Phi(\cdot,\zeta)$ at all $\lambda > \kappa$ that

$$\lambda d^p(\xi^*,\zeta) - \Psi(\xi^*) \le \liminf_{n\to\infty}\{\lambda_n d^p(\xi^n,\zeta) - \Psi(\xi^n)\} = \liminf_{n\to\infty}\Phi(\lambda_n,\zeta) = \Phi(\lambda,\zeta),$$

and thus, $\xi^* \in \arg\min_{\xi\in\Xi}\{\lambda d^p(\xi,\zeta) - \Psi(\xi)\}$. Because $\underline{D}_0(\lambda,\zeta) \le d^p(\xi^*,\zeta) = \lim_{n\to\infty} d^p(\xi^n,\zeta) \le \overline{D}_0(\lambda,\zeta)$, it follows that $\overline{D}_0(\cdot,\zeta)$ is upper semicontinuous and $\underline{D}_0(\cdot,\zeta)$ is lower semicontinuous at all $\lambda > \kappa$ for all $\zeta \in B$.

Next, we show that, for all $\lambda > \kappa$ and all $\zeta \in B$, it holds that $\partial\Phi(\lambda,\zeta)/\partial\lambda- = \overline{D}_0(\lambda,\zeta)$ and $\partial\Phi(\lambda,\zeta)/\partial\lambda+ = \underline{D}_0(\lambda,\zeta)$. Consider any $\zeta \in B$ and any $\lambda_2 > \lambda_1 > \kappa$. Consider any $\xi^i \in \arg\min_{\xi\in\Xi}\{\lambda_i d^p(\xi,\zeta) - \Psi(\xi)\}$ for $i = 1, 2$. Then, it follows as in the proof of Lemma 4(iii) that

$$d^p(\xi^2,\zeta) \le \frac{\Phi(\lambda_2,\zeta) - \Phi(\lambda_1,\zeta)}{\lambda_2 - \lambda_1} \le d^p(\xi^1,\zeta).$$

Then, it follows from the definitions of $\overline{D}_0$ and $\underline{D}_0$ that

$$\overline{D}_0(\lambda_2,\zeta) \le \frac{\Phi(\lambda_2,\zeta)-\Phi(\lambda_1,\zeta)}{\lambda_2-\lambda_1} \le \underline{D}_0(\lambda_1,\zeta).$$

Setting $\lambda_2=\lambda$ and letting $\lambda_1\uparrow\lambda$, it follows from the upper semicontinuity of $\overline{D}_0(\cdot,\zeta)$ that

$$\overline{D}_0(\lambda,\zeta) \le \frac{\partial}{\partial\lambda-}\Phi(\lambda,\zeta) \le \lim_{\lambda_1\uparrow\lambda}\underline{D}_0(\lambda_1,\zeta) \le \lim_{\lambda_1\uparrow\lambda}\overline{D}_0(\lambda_1,\zeta) \le \overline{D}_0(\lambda,\zeta),$$

and hence,

$$\frac{\partial}{\partial\lambda-}\Phi(\lambda,\zeta) = \overline{D}_0(\lambda,\zeta).$$

Similarly, setting $\lambda_1=\lambda$ and letting $\lambda_2\downarrow\lambda$, it follows from the lower semicontinuity of $\underline{D}_0(\cdot,\zeta)$ that

$$\underline{D}_0(\lambda,\zeta) \le \lim_{\lambda_2\downarrow\lambda}\underline{D}_0(\lambda_2,\zeta) \le \lim_{\lambda_2\downarrow\lambda}\overline{D}_0(\lambda_2,\zeta) \le \frac{\partial}{\partial\lambda+}\Phi(\lambda,\zeta) \le \underline{D}_0(\lambda,\zeta),$$

and hence,

$$\frac{\partial}{\partial\lambda+}\Phi(\lambda,\zeta) = \underline{D}_0(\lambda,\zeta).$$

Next, we show that, if condition (a) or (b) or (c) holds, then there exists a primal optimal distribution. First, suppose that condition (a) holds: there exists a dual minimizer $\lambda^*>\kappa$. Because, for $\delta=0$, it holds that $\underline{F}(\lambda^*,\zeta)$ and $\overline{F}(\lambda^*,\zeta)$ in Lemma 3(iii) are nonempty for all $\zeta\in B$, it follows that there exists ν-measurable mappings $\overline{T},\underline{T}:\Xi\longmapsto\Xi$ such that

$$\begin{aligned}\overline{T}(\zeta) &\in \{\xi\in\Xi:\lambda^*d^p(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda^*,\zeta),\ d^p(\xi,\zeta)=\overline{D}_0(\lambda^*,\zeta)\},\\ \underline{T}(\zeta) &\in \{\xi\in\Xi:\lambda^*d^p(\xi,\zeta)-\Psi(\xi)=\Phi(\lambda^*,\zeta),\ d^p(\xi,\zeta)=\underline{D}_0(\lambda^*,\zeta)\}\end{aligned}$$

for ν-almost all $\zeta\in\Xi$. As in the proof of Lemma 6, it follows from the first order optimality conditions $\frac{\partial}{\partial\lambda-}h(\lambda^*)\le 0$ and $\frac{\partial}{\partial\lambda+}h(\lambda^*)\ge 0$ that

$$\begin{aligned}\theta^p &\ge \frac{\partial}{\partial\lambda+}\left(\int_\Xi\Phi(\lambda^*,\zeta)\nu(d\zeta)\right) = \int_\Xi\frac{\partial}{\partial\lambda+}\Phi(\lambda^*,\zeta)\nu(d\zeta) = \int_\Xi\underline{D}_0(\lambda^*,\zeta)\nu(d\zeta) = \int_\Xi d^p(\underline{T}(\zeta),\zeta)\nu(d\zeta)\\ \theta^p &\le \frac{\partial}{\partial\lambda-}\left(\int_\Xi\Phi(\lambda^*,\zeta)\nu(d\zeta)\right) = \int_\Xi\frac{\partial}{\partial\lambda-}\Phi(\lambda^*,\zeta)\nu(d\zeta) = \int_\Xi\overline{D}_0(\lambda^*,\zeta)\nu(d\zeta) = \int_\Xi d^p(\overline{T}(\zeta),\zeta)\nu(d\zeta).\end{aligned} \tag{B.2}$$

Let $q\in[0,1]$ be such that

$$q\int_\Xi d^p(\underline{T}(\zeta),\zeta)\nu(d\zeta)+(1-q)\int_\Xi d^p(\overline{T}(\zeta),\zeta)\nu(d\zeta)=\theta^p.$$

Let

$$\mu^* := q\underline{T}_\#\nu+(1-q)\overline{T}_\#\nu. \tag{B.3}$$

Then,

$$W_p^p(\mu^*,\nu) \le q\int_\Xi d^p(\underline{T}(\zeta),\zeta)\nu(d\zeta)+(1-q)\int_\Xi d^p(\overline{T}(\zeta),\zeta)\nu(d\zeta)=\theta^p,$$

and thus, μ^* is primal feasible. Also,

$$\begin{aligned}\int_\Xi\Psi(\xi)\mu^*(d\xi) &= q\int_\Xi\Psi(\underline{T}(\zeta))\nu(d\zeta)+(1-q)\int_\Xi\Psi(\overline{T}(\zeta))\nu(d\zeta)\\ &= q\int_\Xi[\lambda^*d^p(\underline{T}(\zeta),\zeta)-\Phi(\lambda^*,\zeta)]\nu(d\zeta)+(1-q)\int_\Xi[\lambda^*d^p(\overline{T}(\zeta),\zeta)-\Phi(\lambda^*,\zeta)]\nu(d\zeta)\\ &= \lambda^*\theta^p-\int_\Xi\Phi(\lambda^*,\zeta)\nu(d\zeta)=v_D.\end{aligned}$$

Therefore, μ^* is primal optimal.

Suppose that condition (b) holds: $\lambda^* = \kappa > 0$ is the unique dual minimizer, $\nu(\{\zeta \in \Xi : \arg\min_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta) - \Psi(\xi)\} = \emptyset\}) = 0$, and

$$\int_\Xi \underline{D}_0(\kappa,\zeta)\nu(d\zeta) \le \theta^p \le \int_\Xi \overline{D}_0(\kappa,\zeta)\nu(d\zeta).$$

Then, it follows in the same way as in the proof for condition (a) that there exists a primal optimal distribution.

Suppose that condition (c) holds: $\lambda^* = \kappa = 0$ is the unique dual minimizer, $\arg\max_{\xi\in\Xi}\{\Psi(\xi)\}$ is nonempty, and

$$\int_\Xi \underline{D}_0(0,\zeta)\nu(d\zeta) \le \theta^p.$$

Then, for $\delta = 0$, the sets $\underline{F}(\lambda^*,\zeta)$ in Lemma 3(iii) are given by

$$\begin{aligned}\underline{F}(\lambda^*,\zeta) &= \{\xi \in \Xi : -\Psi(\xi) = \Phi(\lambda^*,\zeta),\, d^p(\xi,\zeta) \le \underline{D}_0(\lambda^*,\zeta)\}\\ &= \left\{\xi \in \arg\max_{\xi\in\Xi}\{\Psi(\xi)\} : d^p(\xi,\zeta) = \underline{D}_0(0,\zeta)\right\}\end{aligned}$$

and are nonempty for ν-almost all $\zeta \in \Xi$. Thus, there exists a ν-measurable mapping $\underline{T} : \Xi \longmapsto \Xi$ such that $\underline{T}(\zeta) \in \underline{F}(\lambda^*,\zeta)$ for ν-almost all $\zeta \in \Xi$. Let $\mu^* := \underline{T}_\#\nu$. Then,

$$W_p^p(\mu^*,\nu) \le \int_\Xi d^p(\underline{T}(\zeta),\zeta)\nu(d\zeta) = \int_\Xi \underline{D}_0(0,\zeta)\nu(d\zeta) \le \theta^p,$$

and thus, μ^* is primal feasible. Furthermore,

$$\int_\Xi \Psi(\xi)\mu^*(d\xi) = \int_\Xi \Psi(\underline{T}(\zeta))\nu(d\zeta) = \max_{\xi\in\Xi}\Psi(\xi) = v_D,$$

and thus, μ^* is primal optimal. Therefore, we showns that, if condition (a) or (b) or (c) holds, then there exists a primal optimal distribution.

Next, we show that, if there exists a primal optimal distribution, then condition (a) or (b) or (c) holds. Consider any primal feasible distribution μ. Let $\gamma \in \mathcal{P}(\Xi\times\Xi)$ denote the corresponding optimal solution in Definition (1) of Wasserstein distance $W_p(\mu,\nu)$, and let γ_ζ denote the corresponding conditional distribution of ξ given ζ. Because μ is feasible, it holds that $\int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta) \le \theta^p$. Lemma 5(v) establishes existence of a dual minimizer $\lambda^* \in [\kappa,\infty)$. Note that

$$\begin{aligned} v_D - \int_\Xi \Psi(\xi)\mu(d\xi) &= \lambda^*\theta^p - \int_\Xi \Phi(\lambda^*,\zeta)\nu(d\zeta)\\ &\quad - \left[\int_{\Xi^2}[\Psi(\xi) - \lambda^* d^p(\xi,\zeta)]\gamma_\zeta(d\xi)\nu(d\zeta) + \int_{\Xi^2}\lambda^* d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta)\right]\\ &= \lambda^*\left[\theta^p - \int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta)\right]\\ &\quad + \int_\Xi\int_\Xi[\lambda^* d^p(\xi,\zeta) - \Psi(\xi) - \Phi(\lambda^*,\zeta)]\gamma_\zeta(d\xi)\nu(d\zeta). \end{aligned}$$

For μ to be primal optimal, it must hold that $v_D - \int_\Xi \Psi(\xi)\mu(d\xi) = 0$. Because $\lambda^* \ge 0$, $\theta^p - \int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta) \ge 0$, and $\lambda^* d^p(\xi,\zeta) - \Psi(\xi) - \Phi(\lambda^*,\zeta) \ge 0$ for all (ξ,ζ), it follows that all of the following must hold for μ to be primal optimal: $\lambda^*\left[\theta^p - \int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta)\right] = 0$.

$\int_\Xi[\lambda^* d^p(\xi,\zeta) - \Psi(\xi) - \Phi(\lambda^*,\zeta)]\gamma_\zeta(d\xi) = 0$ for ν-almost all ζ, which, in turn, implies that $\arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\zeta) - \Psi(\xi)\} \ne \emptyset$ for ν-almost all ζ, and the conditional distribution γ_ζ should be supported on $\arg\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\zeta) - \Psi(\xi)\}$ for ν-almost all ζ.

Next, we show that these conditions imply that condition (a) or (b) or (c) holds. Because there is a dual minimizer $\lambda^* \in [\kappa,\infty)$, one of the following conditions must hold:

1° There is a dual minimizer $\lambda^* > \kappa$.
2° The unique dual minimizer satisfies $\lambda^* = \kappa > 0$.
3° The unique dual minimizer satisfies $\lambda^* = \kappa = 0$.

If 1° holds, then condition (a) holds, and the proof is complete.

Next, suppose that 2° holds and that μ is a primal optimal solution. Condition (B) implies that $\nu(\{\zeta\in\Xi : \arg\min_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta)-\Psi(\xi)\}=\emptyset\})=0$. Next, we show that

$$\int_\Xi \underline{D}_0(\kappa,\zeta)\nu(d\zeta)\le\theta^p\le\int_\Xi\overline{D}_0(\kappa,\zeta)\nu(d\zeta).$$

It follows from (A) that

$$\theta^p=\int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta)\le\int_\Xi\overline{D}_0(\kappa,\zeta)\nu(d\zeta).$$

If $\theta^p<\int_\Xi\underline{D}_0(\kappa,\zeta)\nu(d\zeta)$, then it follows as in (B.2) that

$$\frac{\partial}{\partial\lambda+}\int_\Xi\Phi(\kappa,\zeta)\nu(d\zeta)=\int_\Xi\underline{D}_0(\kappa,\zeta)\nu(d\zeta)>\theta^p.$$

Then, there exists a $\lambda>\kappa$ such that $\lambda\theta^p-\int_\Xi\Phi(\lambda,\zeta)\nu(d\zeta)<\kappa\theta^p-\int_\Xi\Phi(\kappa,\zeta)\nu(d\zeta)$, contradicting $\lambda^*=\kappa$ being a dual minimizer. Therefore, if 2° holds, then condition (b) holds.

Next suppose that 3° holds. Condition (B) implies that $\arg\max_{\xi\in\Xi}\{\Psi(\xi)\}\neq\emptyset$. Suppose that μ is a primal optimal solution. Then,

$$\int_\Xi\underline{D}_0(0,\zeta)\nu(d\zeta)\le\int_\Xi\int_\Xi d^p(\xi,\zeta)\gamma_\zeta(d\xi)\nu(d\zeta)\le\theta^p.$$

Therefore, if 3° holds, then condition (c) holds.

If $-\Psi(\zeta)\le\inf_{\xi\in\Xi}\{\kappa d^p(\xi,\zeta)-\Psi(\xi)\}$ for ν-almost ζ, then $-\Psi(\zeta)\le\kappa d^p(\xi,\zeta)-\Psi(\xi)\le\lambda d^p(\xi,\zeta)-\Psi(\xi)$ for ν-almost ζ, for all $\xi\in\Xi$, and for all $\lambda\ge\kappa$. Then, $\Phi(\lambda,\zeta)=-\Psi(\zeta)$ and $h(\lambda)=\lambda\theta^p+\int_\Xi\Psi(\zeta)\nu(d\zeta)$ for all $\lambda\ge\kappa$, and hence, the dual optimal solution $\lambda^*=\kappa$ for all $\theta>0$.

Otherwise, there exists a set $E\subset\Xi$ such that $\nu(E)>0$ and $-\Psi(\zeta)>\Phi(\kappa,\zeta)$ for all $\zeta\in E$, and thus, $-\int_\Xi\Psi(\zeta)\nu(d\zeta)>\int_\Xi\Phi(\kappa,\zeta)\nu(d\zeta)$. Then, by continuity (follows from concavity) of $\int_\Xi\Phi(\cdot,\zeta)\nu(d\zeta)$ on $[\kappa,\infty)$, there exists a $\lambda_0>\kappa$ such that $-\int_\Xi\Psi(\zeta)\nu(d\zeta)>\int_\Xi\Phi(\lambda_0,\zeta)\nu(d\zeta)$. Using the assumptions that Ψ is upper semicontinuous and that bounded subsets of (Ξ,d) are totally bounded as well as Lemma 3, it follows that there exists a ν-measurable map $T(\lambda_0,\cdot):\Xi\mapsto\Xi$ such that $\lambda_0 d^p(T(\lambda_0,\zeta),\zeta)-\Psi(T(\lambda_0,\zeta))=\Phi(\lambda_0,\zeta)$, and

$$\varepsilon:=\int_\Xi d^p(T(\lambda_0,\zeta),\zeta)\nu(d\zeta)>0$$

because, otherwise, $-\int_\Xi\Psi(\zeta)\nu(d\zeta)=\int_\Xi\Phi(\lambda_0,\zeta)\nu(d\zeta)$. Then, because $\Phi(\kappa,\zeta)\le\kappa d^p(T(\lambda_0,\zeta),\zeta)-\Psi(T(\lambda_0,\zeta))$, it follows that, for any $\theta<\varepsilon^{1/p}$, it holds that

$$\begin{aligned}
h(\kappa)&=\kappa\theta^p-\int_\Xi\Phi(\kappa,\zeta)\nu(d\zeta)\\
&\ge\kappa\theta^p-\int_\Xi\kappa d^p(T(\lambda_0,\zeta),\zeta)\nu(d\zeta)+\int_X\Psi(T(\lambda_0,\zeta))\nu(d\zeta)\\
&=\kappa\theta^p-\kappa\varepsilon+\int_\Xi\lambda_0 d^p(T(\lambda_0,\zeta),\zeta)\nu(d\zeta)-\int_\Xi\Phi(\lambda_0,\zeta)\nu(d\zeta)\\
&=\kappa\theta^p+(\lambda_0-\kappa)\varepsilon-\int_\Xi\Phi(\lambda_0,\zeta)\nu(d\zeta)\\
&>\kappa\theta^p+(\lambda_0-\kappa)\theta^p-\int_\Xi\Phi(\lambda_0,\zeta)\nu(d\zeta)=h(\lambda_0),
\end{aligned}$$

and therefore, $\lambda=\kappa$ cannot be dual optimal for $\theta<\varepsilon^{1/p}$.

ii. It follows from the proof of (i) that, whenever there is a worst-case distribution, then condition (a) or (b) or (c) holds. As shown in the proof of (i), in each case, there are maps $\overline{T}^*,\underline{T}^*:\Xi\mapsto\Xi$ such that (27) holds, and there is a $p^*\in[0,1]$ such that $\mu^*:=p^*\overline{T}^*_\#\nu+(1-p^*)\underline{T}^*_\#\nu$ is a worst-case distribution.

Next, consider $\gamma^T \in \mathcal{P}(\Xi \times \Xi)$. Note that, for any measurable set $B \subset \Xi$, it holds that

$$\pi^1_{\#}\gamma^T(B) = \gamma^T(B \times \Xi) = p^*\nu(B) + (1 - p^*)\nu(B) = \nu(B)$$

and

$$\begin{aligned}\pi^2_{\#}\gamma^T(B) &= \gamma^T(\Xi \times B) = p^*\nu(\{\zeta : \overline{T}^*(\zeta) \in B\}) + (1 - p^*)\nu(\{\zeta : \underline{T}^*(\zeta) \in B\}) \\ &= p^*\overline{T}^*_{\#}\nu(B) + (1 - p^*)\underline{T}^*_{\#}\nu(B) = \mu^*(B).\end{aligned}$$

Thus, γ^{T} is a feasible joint distribution in the definition (1) of $W^p_p(\mu^*, \nu)$. Next, consider any $\gamma \in \mathcal{P}(\Xi \times \Xi)$ such that $\pi^1_{\#}\gamma = \nu$ and $\pi^2_{\#}\gamma = \mu^*$, and let γ_ζ denote the corresponding conditional distribution of ξ given ζ. Then,

$$\begin{aligned}&\int_{\Xi\times\Xi}(\lambda^* d^p(\xi,\zeta) - \Psi(\xi))\gamma(d\xi, d\zeta) = \int_{\Xi}\int_{\Xi}(\lambda^* d^p(\xi,\zeta) - \Psi(\xi))\gamma_\zeta(d\xi)\nu(d\zeta) \\ &\geq \int_{\Xi}\min_{\xi\in\Xi}\{\lambda^* d^p(\xi,\zeta) - \Psi(\xi)\}\nu(d\zeta) \\ &= p^*\int_{\Xi}(\lambda^* d^p(\overline{T}^*(\zeta),\zeta) - \Psi(\overline{T}^*(\zeta))\}\nu(d\zeta) + (1-p^*)\int_{\Xi}(\lambda^* d^p(\underline{T}^*(\zeta),\zeta) - \Psi(\underline{T}^*(\zeta))\}\nu(d\zeta) \\ &= \int_{\Xi\times\Xi}(\lambda^* d^p(\xi,\zeta) - \Psi(\xi))\gamma^T(d\xi, d\zeta) \\ \Rightarrow &\int_{\Xi\times\Xi}\lambda^* d^p(\xi,\zeta)\gamma(d\xi,d\zeta) - \int_{\Xi\times\Xi}\Psi(\xi)\gamma(d\xi,d\zeta) \geq \int_{\Xi\times\Xi}\lambda^* d^p(\xi,\zeta)\gamma^T(d\xi,d\zeta) - \int_{\Xi\times\Xi}\Psi(\xi)\gamma^T(d\xi,d\zeta) \\ \Rightarrow &\int_{\Xi\times\Xi}\lambda^* d^p(\xi,\zeta)\gamma(d\xi,d\zeta) - \int_{\Xi}\Psi(\xi)\mu^*(d\xi) \geq \int_{\Xi\times\Xi}\lambda^* d^p(\xi,\zeta)\gamma^T(d\xi,d\zeta) - \int_{\Xi}\Psi(\xi)\mu^*(d\xi) \\ \Rightarrow &\int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma(d\xi,d\zeta) \geq \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^T(d\xi,d\zeta).\end{aligned}$$

Therefore, γ^{T} is an optimal joint distribution in the definition (1) of $W^p_p(\mu^*, \nu)$.

iii. Observe that

$$\{T_{\#}\nu : W_p(T_{\#}\nu, \nu) \leq \theta,\ T : \Xi \longmapsto \Xi \text{ is } \nu\text{-measurable}\} \subset \{\mu \in \mathcal{P}(\Xi) : W_p(\mu, \nu) \leq \theta\},$$

and thus,

$$v_P \geq \sup\left\{\mathbb{E}_{T_{\#}\nu}[\Psi(\xi)] : W_p(T_{\#}\nu, \nu) \leq \theta,\ T : \Xi \longmapsto \Xi \text{ is } \nu\text{-measurable}\right\}.$$

To show that equality holds, we consider the two cases in Lemmas 6 and 7:

Case 1: h has a minimizer $\lambda^* > \kappa$.

Consider any $\delta, \varepsilon > 0$ and any λ_1, λ_2 such that $\kappa < \lambda_1 < \lambda^* < \lambda_2$. Let $q^\varepsilon_\delta(\lambda_1, \lambda_2) \in [0,1]$, $q^\delta = \theta^p/(\theta^p + \delta)$, $\overline{T}^\varepsilon_\delta(\lambda_1, \cdot), \underline{T}^\varepsilon_\delta(\lambda_2, \cdot) : \Xi \longmapsto \Xi$ be as in the proof of Lemma 6. Let $T^\varepsilon_\delta(\lambda_1, \lambda_2, \cdot) : \Xi \longmapsto \Xi$ be given by

$$T^\varepsilon_\delta(\lambda_1, \lambda_2, \zeta) := q^\delta q^\varepsilon_\delta(\lambda_1, \lambda_2)\overline{T}^\varepsilon_\delta(\lambda_1, \zeta) + q^\delta(1 - q^\varepsilon_\delta(\lambda_1, \lambda_2))\underline{T}^\varepsilon_\delta(\lambda_2, \zeta) + (1 - q^\delta)\zeta.$$

Note that $T^\varepsilon_\delta(\lambda_1, \lambda_2, \zeta) \in \Xi$ for all $\zeta \in \Xi$ and that $T^\varepsilon_\delta(\lambda_1, \lambda_2, \cdot)$ is ν-measurable. It follows from $d^p(\cdot, \zeta)$ being convex that

$$\begin{aligned}W^p_p(T^\varepsilon_\delta(\lambda_1, \lambda_2, \cdot)_{\#}\nu, \nu) &\leq \int_{\Xi} d^p\Big(q^\delta q^\varepsilon_\delta(\lambda_1, \lambda_2)\overline{T}^\varepsilon_\delta(\lambda_1, \zeta) + q^\delta(1 - q^\varepsilon_\delta(\lambda_1, \lambda_2))\underline{T}^\varepsilon_\delta(\lambda_2, \zeta) + (1 - q^\delta)\zeta,\ \zeta\Big)\nu(d\zeta) \\ &\leq q^\delta q^\varepsilon_\delta(\lambda_1, \lambda_2)\int_{\Xi} d^p(\overline{T}^\varepsilon_\delta(\lambda_1, \zeta), \zeta)\nu(d\zeta) + q^\delta(1 - q^\varepsilon_\delta(\lambda_1, \lambda_2))\int_{\Xi} d^p(\underline{T}^\varepsilon_\delta(\lambda_2, \zeta), \zeta)\nu(d\zeta) \\ &= q^\delta(\theta^p + [1 - 2q^\varepsilon_\delta(\lambda_1, \lambda_2)]\delta) \leq \theta^p.\end{aligned}$$

Also, it follows from Ψ being concave that

$$\begin{aligned}\mathbb{E}_{T^{\varepsilon}_{\delta}(\lambda_1,\lambda_2,\cdot)_{\#}\nu}[\Psi(\xi)] &= \int_{\Xi}\Psi\Big(q^{\delta}q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2)\overline{T}^{\varepsilon}_{\delta}(\lambda_1,\zeta)+q^{\delta}(1-q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2))\underline{T}^{\varepsilon}_{\delta}(\lambda_2,\zeta)+(1-q^{\delta})\zeta\Big)\nu(d\zeta)\\ &\geq q^{\delta}q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2)\int_{\Xi}\Psi(\overline{T}^{\varepsilon}_{\delta}(\lambda_1,\zeta))\nu(d\zeta)+q^{\delta}(1-q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2))\int_{\Xi}\Psi(\underline{T}^{\varepsilon}_{\delta}(\lambda_2,\zeta))\nu(d\zeta)\\ &\quad+(1-q^{\delta})\int_{\Xi}\Psi(\zeta)\nu(d\zeta)\\ &\geq q^{\delta}\lambda_1[\theta^p+(1-2q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2))\delta]-q^{\delta}q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2)\int_{\Xi}\Phi(\lambda_1,\zeta)\nu(d\zeta)\\ &\quad-q^{\delta}(1-q^{\varepsilon}_{\delta}(\lambda_1,\lambda_2))\int_{\Xi}\Phi(\lambda_2,\zeta)\nu(d\zeta)-q^{\delta}\varepsilon+(1-q^{\delta})\int_{\Xi}\Psi(\zeta)\nu(d\zeta).\end{aligned}\tag{B.4}$$

Thus, given any $\varepsilon>0$, choose $\lambda^{\varepsilon}_1\in(\max\{\kappa,\lambda^*-\varepsilon\},\lambda^*)$ such that $\int_{\Xi}\Phi(\lambda^{\varepsilon}_1,\zeta)\nu(d\zeta)\leq\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)+\varepsilon$ and $(\lambda^*-\lambda^{\varepsilon}_1)\theta^p\leq\varepsilon$, choose $\lambda^{\varepsilon}_2\in(\lambda^*,\lambda^*+\varepsilon)$ such that $\int_{\Xi}\Phi(\lambda^{\varepsilon}_2,\zeta)\nu(d\zeta)\leq\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)+\varepsilon$, and choose $\delta^{\varepsilon}\in(0,\varepsilon)$ such that $2\lambda^{\varepsilon}_1\delta^{\varepsilon}\leq\varepsilon$, $-\delta^{\varepsilon}\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)\leq\theta^p\varepsilon$, and $-\delta^{\varepsilon}\int_{\Xi}\Psi(\zeta)\nu(d\zeta)\leq\theta^p\varepsilon$. Set $T^{\varepsilon}:=T^{\varepsilon}_{\delta^{\varepsilon}}(\lambda^{\varepsilon}_1,\lambda^{\varepsilon}_2,\cdot)$. Then, it follows from (B.4) that

$$\mathbb{E}_{T^{\varepsilon}_{\#}\nu}[\Psi(\xi)]\geq\lambda^*\theta^p-\int_{\Xi}\Phi(\lambda^*,\zeta)\nu(d\zeta)-6\varepsilon=v_D-6\varepsilon.$$

Because $\varepsilon>0$ can be arbitrarily small, it follows that

$$\sup\Big\{\mathbb{E}_{T_{\#}\nu}[\Psi(\xi)]:W_p(T_{\#}\nu,\nu)\leq\theta,\ T:\Xi\longmapsto\Xi\text{ is }\ \nu\text{-measurable}\Big\}=v_P.$$

Case 2: κ is the unique minimizer of h.

Consider any $\lambda>\kappa$. Consider any $\varepsilon\in\Big(0,(\lambda-\kappa)\theta^p-\int_{\Xi}[\Phi(\lambda,\zeta)-\Phi(\kappa,\zeta)]\nu(d\zeta)\Big)$. Let $\underline{T}_{\varepsilon}(\lambda,\cdot):\Xi\to\Xi$ be as in the proof of Lemma 7. Then,

$$W^p_p(\underline{T}_{\varepsilon}\,(\lambda,\cdot)_{\#}\nu,\nu)\leq\int_{\Xi}d^p(\underline{T}_{\varepsilon}(\lambda,\zeta),\zeta)\nu(d\zeta)<\theta^p.$$

Case 2.1: $\kappa=0$.

Then,

$$\begin{aligned}&\mathbb{E}_{\underline{T}_{\varepsilon}(\lambda,\cdot)_{\#}\nu}[\Psi(\xi)]=\int_{\Xi}\Psi(\xi)\underline{T}_{\varepsilon}(\lambda,\cdot)_{\#}\nu(d\xi)\geq-\int_{\Xi}\Phi(\lambda,\zeta)\nu(d\zeta)-\varepsilon\\ &\Rightarrow\lim_{\lambda\downarrow0}\ \mathbb{E}_{\underline{T}_{\varepsilon}(\lambda,\cdot)_{\#}\nu}[\Psi(\xi)]\geq\lim_{\lambda\downarrow0}\left\{-\int_{\Xi}\Phi(\lambda,\zeta)\nu(d\zeta)-\varepsilon\right\}=v_D-\varepsilon\\ &\Rightarrow\sup\Big\{\mathbb{E}_{T_{\#}\nu}[\Psi(\xi)]:W_p(T_{\#}\nu,\nu)\leq\theta,\ T:\Xi\longmapsto\Xi\text{is}\nu\text{-measurable}\Big\}=v_P.\end{aligned}$$

Case 2.2: $\kappa>0$.

Consider any $\lambda>\kappa$, $\kappa'\in(0,\kappa)$, and $R>\theta^p$. Let $\overline{T}^R(\kappa',\cdot):\Xi\longmapsto\Xi$ and $q^R_{\varepsilon}(\kappa',\lambda)\in(0,1)$ be as in the proof of Lemma 7. Let $T^R_{\varepsilon}(\kappa',\lambda,\cdot):\Xi\longmapsto\Xi$ be given by

$$T^R_{\varepsilon}(\kappa',\lambda,\zeta):=q^R_{\varepsilon}(\kappa',\lambda)\underline{T}_{\varepsilon}\,(\lambda,\zeta)+\Big(1-q^R_{\varepsilon}(\kappa',\lambda)\Big)\overline{T}^R(\kappa',\zeta).$$

Note that $T^R_{\varepsilon}(\kappa',\lambda,\zeta)\in\Xi$ for all $\zeta\in\Xi$ and that $T^R_{\varepsilon}(\kappa',\lambda,\cdot)$ is ν-measurable. It follows from $d^p(\cdot,\zeta)$ being convex that

$$\begin{aligned}W^p_p(T^R_{\varepsilon}(\kappa',\lambda,\cdot)_{\#}\nu,\nu)&\leq\int_{\Xi}d^p\Big(q^R_{\varepsilon}(\kappa',\lambda)\underline{T}_{\varepsilon}(\lambda,\zeta)+\Big(1-q^R_{\varepsilon}(\kappa',\lambda)\Big)\overline{T}^R(\kappa',\zeta),\ \zeta\Big)\nu(d\zeta)\\ &\leq q^R_{\varepsilon}(\kappa',\lambda)\int_{\Xi}d^p(\underline{T}_{\varepsilon}(\lambda,\zeta),\zeta)\nu(d\zeta)+\Big(1-q^R_{\varepsilon}(\kappa',\lambda)\Big)\int_{\Xi}d^p\Big(\overline{T}^R(\kappa',\zeta),\zeta\Big)\nu(d\zeta)=\theta^p.\end{aligned}$$

Also, it follows from Ψ being concave that

$$\begin{aligned}\mathbb{E}_{T_\varepsilon^R(\kappa',\lambda,\cdot)_\# \nu}[\Psi(\xi)] &= \int_\Xi \Psi\Big(q_\varepsilon^R(\kappa',\lambda)\underline{T}_\varepsilon(\lambda,\zeta) + \Big(1-q_\varepsilon^R(\kappa',\lambda)\Big)\overline{T}^R(\kappa',\zeta)\Big)\nu(d\zeta)\\ &\geq q_\varepsilon^R(\kappa',\lambda)\int_\Xi \Psi(\underline{T}_\varepsilon(\lambda,\zeta))\nu(d\zeta) + \Big(1-q_\varepsilon^R(\kappa',\lambda)\Big)\int_\Xi \Psi\Big(\overline{T}^R(\kappa',\zeta)\Big)\nu(d\zeta)\\ &\geq \kappa'\theta^p - q_\varepsilon^R(\kappa',\lambda)\int_\Xi \Phi(\lambda,\zeta)\nu(d\zeta) - q_\varepsilon^R(\kappa',\lambda)\varepsilon + \Big(1-q_\varepsilon^R(\kappa',\lambda)\Big)\int_\Xi \Psi(\zeta)\nu(d\zeta). \end{aligned} \tag{B.5}$$

Thus, given any $\varepsilon > 0$, choose $\lambda_1^\varepsilon \in (\kappa-\varepsilon,\kappa)$ such that $(\kappa-\lambda_1^\varepsilon)\theta^p \leq \varepsilon$, choose $\lambda_2^\varepsilon \in (\kappa,\kappa+\varepsilon)$ such that $(\lambda_2^\varepsilon-\kappa)\theta^p \leq \varepsilon$, choose $\varepsilon' \in \Big(0,(\lambda_2^\varepsilon-\kappa)\theta^p - \int_\Xi[\Phi(\lambda_2^\varepsilon,\zeta)-\Phi(\kappa,\zeta)]\nu(d\zeta)\Big)$ such that $\varepsilon'\Big|\int_\Xi \Phi(\lambda_2^\varepsilon,\zeta)\nu(d\zeta)\Big| \leq \varepsilon$ and $\varepsilon'\Big|\int_\Xi \Psi(\zeta)\nu(d\zeta)\Big| \leq \varepsilon$ (note that $\varepsilon' \leq \varepsilon$), and choose $R \geq \theta^p + \theta^p/\varepsilon'$. Set $T^\varepsilon = T_{\varepsilon'}^R(\lambda_1^\varepsilon,\lambda_2^\varepsilon,\cdot)$. Then, it follows from (B.5) that

$$\mathbb{E}_{T_\#^\varepsilon \nu}[\Psi(\xi)] \geq \kappa\theta^p - \int_\Xi \Phi(\kappa,\zeta)\nu(d\zeta) - 5\varepsilon = v_D - 5\varepsilon.$$

Because $\varepsilon > 0$ can be arbitrarily small, it follows that

$$\sup\Big\{\mathbb{E}_{T_\#\nu}[\Psi(\xi)] : W_p(T_\#\nu,\nu) \leq \theta,\, T : \Xi \longmapsto \Xi \text{ is } \nu\text{-measurable}\Big\} = v_P.$$

Next, suppose that a worst-case distribution exists. Then, it follows from (i) that condition (a) or (b) or (c) holds.

Case a/b: Condition (a) or (b) holds.

Let $\underline{T},\overline{T} : \Xi \longmapsto \Xi$ and $q \in [0,1]$ be as in the proof of (i). Let $T^* : \Xi \longmapsto \Xi$ be given by $T^*(\zeta) := q\underline{T}(\zeta) + (1-q)\overline{T}(\zeta)$. Note that $T^*(\zeta) \in \Xi$ for all $\zeta \in \Xi$ and that T^* is ν-measurable. It follows from $d^p(\cdot,\zeta)$ being convex that

$$\begin{aligned}W_p^p(T_\#^*\nu,\nu) &\leq \int_\Xi d^p(q\underline{T}(\zeta)+(1-q)\overline{T}(\zeta),\zeta)\nu(d\zeta)\\ &\leq q\int_\Xi d^p(\underline{T}(\zeta),\zeta)\nu(d\zeta) + (1-q)\int_\Xi d^p(\overline{T}(\zeta),\zeta)\nu(d\zeta) = \theta^p.\end{aligned}$$

Also, it follows from Ψ being concave that

$$\begin{aligned}\mathbb{E}_{T_\#^*\nu}[\Psi(\xi)] &= \int_\Xi \Psi(q\underline{T}(\zeta)+(1-q)\overline{T}(\zeta))\nu(d\zeta)\\ &\geq q\int_\Xi \Psi(\underline{T}(\zeta))\nu(d\zeta) + (1-q)\int_\Xi \Psi(\overline{T}(\zeta))\nu(d\zeta) = v_D.\end{aligned}$$

This shows that $T_\#^*\nu$ is a primal optimal solution.

Case c: Condition (c) holds.

Let $\underline{T} : \Xi \longmapsto \Xi$ be as in the proof of (i). It follows from the proof of (i) that $\underline{T}_\#\nu$ is primal optimal. □

B.2. Proofs for Section 3.2

Proof of Proposition 3. Consider $\min_{\mu\in\mathfrak{M}}\mu(\text{int}(C))$. It was shown in Example 7 that Corollary 1 applies, and thus, a worst-case distribution $\mu^* \in \arg\min_{\mu\in\mathfrak{M}}\mu(\text{int}(C))$ exists.

First, suppose that $\mu^*(\text{int}(C)) = 1$. Because $1 = \mu^*(\text{int}(C)) = \min_{\mu\in\mathfrak{M}}\mu(\text{int}(C)) \leq \inf_{\mu\in\mathfrak{M}}\mu(C) \leq \mu^*(C) \leq 1$, it follows that $\inf_{\mu\in\mathfrak{M}}\mu(C) = \min_{\mu\in\mathfrak{M}}\mu(\text{int}(C))$ (and $\mu^* \in \arg\min_{\mu\in\mathfrak{M}}\mu(C)$).

Next, suppose that $\mu^*(\text{int}(C)) < 1$, and consider any $\varepsilon > 0$. Note that, for every $\xi \in (C\backslash\text{int}(C)) \subset \partial C$, it holds that $\{\zeta \in \Xi\backslash C : 0 < d(\xi,\zeta) < \varepsilon\} \neq \emptyset$, and that the graph

$$G_\varepsilon := \{(\xi,\zeta) \in (C\backslash\text{int}(C)) \times (\Xi\backslash C) : 0 < d(\xi,\zeta) < \varepsilon\}$$

is Borel in $\Xi\times\Xi$. It follows from the measurable selection theorem (see, e.g., Aliprantis and Border [2, theorem 18.26]) that there exists a Borel-measurable map $T^\varepsilon : \Xi \longmapsto \Xi$ such that $T^\varepsilon(\xi) = \xi$ for all $\xi \in \text{int}(C) \cup (\Xi\backslash C)$, and $T^\varepsilon(\xi) \in \Xi\backslash C$ such that $0 < d(\xi, T^\varepsilon(\xi)) < \varepsilon$ for all $\xi \in C \setminus \mathrm{int}(C)$. For any $q^\varepsilon \in (0,1)$, let

$$\mu^\varepsilon := (1-q^\varepsilon)T^\varepsilon_\# \mu^* + q^\varepsilon \nu.$$

Let $\gamma^{T\varepsilon} \in \arg\min_{\gamma \in \mathcal{P}(\Xi\times\Xi)} \left\{ \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\# \gamma = T^\varepsilon_\# \mu^*, \pi^2_\# \gamma = \nu \right\}$. Let $\nu^2 \in \mathcal{P}(\Xi\times\Xi)$ be given by $\nu^2(B) := \nu(\{\zeta : (\zeta,\zeta) \in B\})$. Note that $\pi^1_\# \nu^2 = \pi^2_\# \nu^2 = \nu$. Consider $\gamma^\varepsilon := (1-q^\varepsilon)\gamma^{T\varepsilon} + q^\varepsilon \nu^2$. Note that

$$\begin{aligned}
\pi^1_\# \gamma^\varepsilon &= (1-q^\varepsilon)\pi^1_\# \gamma^{T\varepsilon} + q^\varepsilon \pi^1_\# \nu^2 = (1-q^\varepsilon)T^\varepsilon_\# \mu^* + q^\varepsilon \nu = \mu^\varepsilon \\
\pi^2_\# \gamma^\varepsilon &= (1-q^\varepsilon)\pi^2_\# \gamma^{T\varepsilon} + q^\varepsilon \pi^2_\# \nu^2 = (1-q^\varepsilon)\nu + q^\varepsilon \nu = \nu.
\end{aligned}$$

Thus,

$$\begin{aligned}
W_p^p(\mu^\varepsilon, \nu) &= \min_{\gamma \in \mathcal{P}(\Xi\times\Xi)} \left\{ \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\# \gamma = \mu^\varepsilon, \pi^2_\# \gamma = \nu \right\} \\
&\le \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^\varepsilon(d\xi,d\zeta) \\
&= (1-q^\varepsilon)\int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^{T\varepsilon}(d\xi,d\zeta) + q^\varepsilon \int_{\Xi\times\Xi} d^p(\xi,\zeta)\nu^2(d\xi,d\zeta) \\
&= (1-q^\varepsilon)\int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^{T\varepsilon}(d\xi,d\zeta) \\
&= (1-q^\varepsilon) W_p^p(T^\varepsilon_\# \mu^*, \nu).
\end{aligned}$$

Next, let $\gamma^* \in \arg\min_{\gamma \in \mathcal{P}(\Xi\times\Xi)} \left\{ \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\# \gamma = \mu^*, \pi^2_\# \gamma = \nu \right\}$. Consider $\gamma^T \in \mathcal{P}(\Xi\times\Xi)$ given by $\gamma^T(B) := \gamma^*(\{(\xi,\zeta) : (T^\varepsilon(\xi),\zeta) \in B\})$. Note that

$$\begin{aligned}
\pi^1_\# \gamma^T(A) &= \gamma^T(A\times\Xi) = \gamma^*(\{(\xi,\zeta) : T^\varepsilon(\xi) \in A, \zeta \in \Xi\}) = \gamma^*((T^\varepsilon)^{-1}(A)\times\Xi) \\
&= \pi^1_\# \gamma^*((T^\varepsilon)^{-1}(A)) = \mu^*((T^\varepsilon)^{-1}(A)) = T^\varepsilon_\# \mu^*(A) \\
\pi^2_\# \gamma^T(A) &= \gamma^T(\Xi\times A) = \gamma^*(\{(\xi,\zeta) : T^\varepsilon(\xi) \in \Xi, \zeta \in A\}) = \gamma^*(\Xi\times A) \\
&= \pi^2_\# \gamma^*(A) = \nu(A).
\end{aligned}$$

Thus, $\pi^1_\# \gamma^T = T^\varepsilon_\# \mu^*$ and $\pi^2_\# \gamma^T = \nu$. Hence,

$$\begin{aligned}
W_p^p(T^\varepsilon_\# \mu^*, \nu) &= \min_{\gamma \in \mathcal{P}(\Xi\times\Xi)} \left\{ \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\# \gamma = T^\varepsilon_\# \mu^*, \pi^2_\# \gamma = \nu \right\} \\
&\le \int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^T(d\xi,d\zeta) \\
&= \int_{\Xi\times\Xi} d^p(T^\varepsilon(\xi),\zeta)\gamma^*(d\xi,d\zeta) \\
&\le \int_{\Xi\times\Xi} [d(T^\varepsilon(\xi),\xi) + d(\xi,\zeta)]^p \gamma^*(d\xi,d\zeta) \\
&\le \int_{\Xi\times\Xi} [\varepsilon + d(\xi,\zeta)]^p \gamma^*(d\xi,d\zeta) \\
&\le \left(\varepsilon + \left(\int_{\Xi\times\Xi} d^p(\xi,\zeta)\gamma^*(d\xi,d\zeta) \right)^{1/p} \right)^p \\
&= \left(\varepsilon + W_p(\mu^*,\nu) \right)^p,
\end{aligned}$$

where the last inequality follows from the triangle inequality for the Lp-norm. Therefore,

$$W_p^p(\mu^\varepsilon, \nu) \le (1-q^\varepsilon)(\varepsilon + W_p(\mu^*,\nu))^p \le (1-q^\varepsilon)(\varepsilon+\theta)^p.$$

Choose $q^\varepsilon = 1 - \theta^p/(\varepsilon+\theta)^p$. Then, $W_p(\mu^\varepsilon, \nu) \le \theta$, and thus, $\mu^\varepsilon \in \mathfrak{M}$ for all $\varepsilon > 0$. Also,

$$\begin{aligned}\min_{\mu\in\mathfrak{M}} \mu(\mathrm{int}(C)) \le \inf_{\mu\in\mathfrak{M}} \mu(C) \le \mu^\varepsilon(C) &= (1-q^\varepsilon) T^\varepsilon_\# \mu^*(C) + q^\varepsilon \nu(C) \\ &= (1-q^\varepsilon)\mu^*(\mathrm{int}(C)) + q^\varepsilon \nu(C) \\ &\le (1-q^\varepsilon) \min_{\mu\in\mathfrak{M}} \mu(\mathrm{int}(C)) + q^\varepsilon,\end{aligned}$$

and therefore, it follows that $\inf_{\mu\in\mathfrak{M}} \mu(C) = \min_{\mu\in\mathfrak{M}} \mu(\mathrm{int}(C))$. □

Appendix C. Proofs for Section 4

C.1. Proofs for Section 4.1

Proof of Proposition 4. It follows from Lemmas 6 and 7 that

$$\begin{aligned}&\inf_{\mu\in\mathfrak{M}} \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] \\ &= \inf_{\mu\in\mathcal{P}(\Xi)} \left\{ \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] : \min_{\gamma\in\mathcal{P}(\Xi^2)} \left\{ \int_{\Xi^2} d(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\#\gamma = \mu, \pi^2_\#\gamma = \nu \right\} \le \theta \right\} \\ &= \inf_{\substack{\{\xi^i_\pm\}_{i=1}^n \subset \Xi \\ q_1,q_2\in[0,1],\ \{\theta^i_\pm\}_{i=1}^n\subset\mathbb{R}_+}} \left\{ \frac{1}{n}\sum_{i=1}^n \left(q_1 \xi^i_+(x^{-1}(1)) + q_2 \xi^i_-(x^{-1}(1)) + (1-q_1-q_2)\hat{\xi}^i(x^{-1}(1)) \right) : q_1+q_2\le 1,\ d(\xi^i_\pm, \hat{\xi}^i) \le \theta^i_\pm\ \forall 1\le i\le n, \right. \\ &\qquad\qquad \left. \frac{1}{n}\sum_{i=1}^n \left(q_1\theta^i_+ + q_2\theta^i_- \right) \le \theta \right\} \\ &= \inf_{\substack{q_1,q_2\in[0,1],\\ \{\theta^i_\pm\}_{i=1}^n\subset\mathbb{R}_+}} \left\{ \frac{1}{n}\sum_{i=1}^n q_1 \left(\inf_{\xi^i_+\in\Xi} \left\{ \xi^i_+(x^{-1}(1)) : d(\xi^i_+,\hat{\xi}^i) \le \theta^i_+ \right\} + q_2 \inf_{\xi^i_-\in\Xi} \left\{ \xi^i_-(x^{-1}(1)) : d(\xi^i_-,\hat{\xi}^i) \le \theta^i_- \right\} \right.\right. \\ &\qquad\qquad \left.\left. + (1-q_1-q_2)\hat{\xi}^i(x^{-1}(1)) \right) : q_1+q_2\le 1,\ \frac{1}{n}\sum_{i=1}^n \left(q_1\theta^i_+ + q_2\theta^i_- \right) \le \theta \right\}.\end{aligned}$$

It follows from condition (iii) that

$$\inf_{\xi^i_\pm\in\Xi} \left\{ \xi^i_\pm(x^{-1}(1)) : d(\xi^i_\pm,\hat{\xi}^i) \le \theta^i_\pm \right\} = \inf_{\xi^i_\pm\in\Xi} \left\{ \xi^i_\pm(x^{-1}(1)) : W_1(\xi^i_\pm,\hat{\xi}^i) \le \theta^i_\pm \right\}.$$

Thus,

$$\begin{aligned}&\inf_{\mu\in\mathfrak{M}} \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] \\ &= \inf_{\substack{q_1,q_2\in[0,1],\\ \{\theta^i_\pm\}_{i=1}^n\subset\mathbb{R}_+}} \left\{ \frac{1}{n}\sum_{i=1}^n q_1 \left(\inf_{\xi^i_+\in\Xi} \left\{ \xi^i_+(x^{-1}(1)) : W_1(\xi^i_+,\hat{\xi}^i) \le \theta^i_+ \right\} + q_2 \inf_{\xi^i_-\in\Xi} \left\{ \xi^i_-(x^{-1}(1)) : W_1(\xi^i_-,\hat{\xi}^i) \le \theta^i_- \right\} \right.\right. \\ &\qquad \left.\left. +(1-q_1-q_2)\hat{\xi}^i(x^{-1}(1)) \right) : q_1+q_2\le 1,\ \frac{1}{n}\sum_{i=1}^n \left(q_1\theta^i_+ + q_2\theta^i_- \right) \le \theta \right\} \\ &= \inf_{\mu\in\mathcal{P}(\Xi)} \left\{ \mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))] : \min_{\gamma\in\mathcal{P}(\Xi^2)} \left\{ \int_{\Xi^2} W_1(\xi,\zeta)\gamma(d\xi,d\zeta) : \pi^1_\#\gamma = \mu, \pi^2_\#\gamma = \nu \right\} \le \theta \right\} \\ &= \sup_{\lambda\ge 0} \left\{ -\lambda\theta + \frac{1}{n}\sum_{i=1}^n \inf_{\xi\in\Xi} \left\{ \xi(x^{-1}(1)) + \lambda W_1(\xi,\hat{\xi}^i) \right\} \right\} \\ &\ge \sup_{\lambda\ge 0} \left\{ -\lambda\theta + \frac{1}{n}\sum_{i=1}^n \inf_{\mu\in\mathcal{B}([0,1])} \left\{ \mu(x^{-1}(1)) + \lambda W_1(\mu,\hat{\xi}^i) \right\} \right\},\end{aligned}$$

where the last equality follows from Theorem 1, and the inequality holds because $\Xi \subset \mathcal{B}([0,1])$. Note that the inner problem $\inf_{\mu\in\mathcal{B}([0,1])} \left\{ \mu(x^{-1}(1)) + \lambda W_1(\mu,\hat{\xi}^i) \right\}$ is similar to the problem considered in Example 7. It follows from Example 7 and

Proposition 3 that

$$\inf_{\mu\in\mathcal{B}([0,1])}\left\{\mu(x^{-1}(1))+\lambda W_1(\xi,\hat{\xi}^i)\right\}=\inf_{\mu\in\mathcal{B}([0,1])}\left\{\mu(\mathrm{int}(x^{-1}(1)))+\lambda W_1(\mu,\hat{\xi}^i)\right\}$$
$$=\sum_{m=1}^{M_i}\min_{\eta\in[0,1]}\left\{\mathbb{1}\{\eta\in\mathrm{int}(x^{-1}(1))\}+\lambda\left|\eta-\hat{\eta}^i_m\right|\right\}.$$

Hence,

$$\begin{aligned}
&\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))]\\
&\geq\sup_{\lambda\geq 0}\left\{-\lambda\theta+\frac{1}{n}\sum_{i=1}^{n}\sum_{m=1}^{M_i}\min_{\eta\in[0,1]}\left\{\mathbb{1}\{\eta\in\mathrm{int}(x^{-1}(1))\}+\lambda\left|\eta-\hat{\eta}^i_m\right|\right\}\right\}\\
&=\inf_{\mu\in\mathcal{B}([0,1])}\left\{\mu(\mathrm{int}(x^{-1}(1))):\min_{\gamma\in\mathcal{B}([0,1]^2)}\left\{\int_{[0,1]^2}|\eta-\hat{\eta}|\gamma(d\eta,d\hat{\eta}):\pi^1_{\#}\gamma=\mu,\pi^2_{\#}\gamma=\hat{\nu}\right\}\leq\theta\right\},
\end{aligned}\tag{C.1}$$

where $\hat{\nu}:=1/n\sum_{i=1}^{n}\sum_{m=1}^{M_i}\delta_{\hat{\eta}^i_m}\in\mathcal{B}([0,1])$, and the last equality follows from Theorem 1. This last problem is the same as the problem considered in Example 7, and it follows that it has an optimal solution $\hat{\mu}\in\mathcal{B}([0,1])$ of the form

$$\hat{\mu}=\frac{1}{n}\sum_{\substack{1\leq i\leq n,\,1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\delta_{\eta^i_m}+\frac{1-p_0}{n}\delta_{\eta^{i_0}_{m_0-}}+\frac{p_0}{n}\delta_{\eta^{i_0}_{m_0+}},$$

and that

$$W_1(\hat{\mu},\hat{\nu})=\frac{1}{n}\sum_{\substack{1\leq i\leq n,\,1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\left|\eta^i_m-\hat{\eta}^i_m\right|+\frac{1-p_0}{n}\left|\eta^{i_0}_{m_0-}-\hat{\eta}^{i_0}_{m_0}\right|+\frac{p_0}{n}\left|\eta^{i_0}_{m_0+}-\hat{\eta}^{i_0}_{m_0}\right|\leq\theta.$$

Consider the $2n$ sample paths $\{\xi^i_{\pm}\}_{i=1}^n\subset\Xi$ given by

$$\xi^i_{\pm}:=\sum_{\substack{1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\delta_{\eta^i_m}+\delta_{\eta^{i_0}_{m_0\pm}}\mathbb{1}\{i=i_0\},$$

and consider the point process $\mu^*\in\mathcal{P}(\Xi)$ with support on the $2n$ sample paths given by

$$\mu^*=\frac{1-p_0}{n}\sum_{i=1}^{n}\delta_{\xi^i_-}+\frac{p_0}{n}\sum_{i=1}^{n}\delta_{\xi^i_+}.$$

Note that

$$\begin{aligned}
&\mathbb{E}_{\xi\sim\mu^*}\left[\xi(\mathrm{int}(x^{-1}(1)))\right]\\
&=\frac{1-p_0}{n}\left(\sum_{i=1}^{n}\sum_{\substack{1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\mathbb{1}\{\eta^i_m\in\mathrm{int}(x^{-1}(1))\}+\mathbb{1}\{\eta^{i_0}_{m_0-}\in\mathrm{int}(x^{-1}(1))\}\right)\\
&\quad+\frac{p_0}{n}\left(\sum_{i=1}^{n}\sum_{\substack{1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\mathbb{1}\{\eta^i_m\in\mathrm{int}(x^{-1}(1))\}+\mathbb{1}\{\eta^{i_0}_{m_0+}\in\mathrm{int}(x^{-1}(1))\}\right)\\
&=\frac{1}{n}\sum_{\substack{1\leq i\leq n,1\leq m\leq M_i\\(i,m)\neq(i_0,m_0)}}\mathbb{1}\left\{\eta^i_m\in\mathrm{int}(x^{-1}(1))\right\}+\frac{1-p_0}{n}\,\mathbb{1}\left\{\eta^{i_0}_{m_0-}\in\mathrm{int}(x^{-1}(1))\right\}+\frac{p_0}{n}\,\mathbb{1}\left\{\eta^{i_0}_{m_0+}\in\mathrm{int}(x^{-1}(1))\right\}\\
&=\hat{\mu}(\mathrm{int}(x^{-1}(1))),
\end{aligned}$$

and that

$$\begin{aligned}
&W_1(\mu^*,\nu)\\
&\le \frac{1-p_0}{n}\sum_{i=1}^n d(\xi_-^i,\hat{\xi}^i)+\frac{p_0}{n}\sum_{i=1}^n d(\xi_+^i,\hat{\xi}^i)\\
&=\frac{1-p_0}{n}\left(\sum_{i=1}^n \sum_{\substack{1\le m\le M_i\\(i,m)\ne(i_0,m_0)}} \left|\eta_m^i-\hat{\eta}_m^i\right|+\left|\eta_{m_0-}^{i_0}-\hat{\eta}_{m_0}^{i_0}\right|\right)+\frac{p_0}{n}\left(\sum_{i=1}^n \sum_{\substack{1\le m\le M_i\\(i,m)\ne(i_0,m_0)}} \left|\eta_m^i-\hat{\eta}_m^i\right|+\left|\eta_{m_0+}^{i_0}-\hat{\eta}_{m_0}^{i_0}\right|\right)\\
&=\frac{1}{n}\sum_{\substack{1\le i\le n,1\le m\le M_i\\(i,m)\ne(i_0,m_0)}}\left|\eta_m^i-\hat{\eta}_m^i\right|+\frac{1-p_0}{n}\left|\eta_{m_0-}^{i_0}-\hat{\eta}_{m_0}^{i_0}\right|+\frac{p_0}{n}\left|\eta_{m_0+}^{i_0}-\hat{\eta}_{m_0}^{i_0}\right|\le\theta,
\end{aligned}$$

where the first equality follows from condition (ii). That is, $\mu^*\in\mathfrak{M}$, and μ^* is an optimal solution for $\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(\mathrm{int}(x^{-1}(1)))]$. It follows as in Proposition 3 that $\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(x^{-1}(1))]=\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(\mathrm{int}(x^{-1}(1)))]$, and hence, Inequality (C.1) holds as an equality.

Let $\hat{\Xi}:=\{\hat{\eta}_m^i: i=1,\dots,N, m=1,\dots,M_i\}$. Next, we show that it suffices to optimize over the set of functions $x:[0,1]\longmapsto\{0,1\}$ such that each connected component of $x^{-1}(1)$ contains at least one point in $\hat{\Xi}$. Consider any $x:[0,1]\longmapsto\{0,1\}$ and any connected component C0 of $x^{-1}(1)$ such that $C_0\cap\hat{\Xi}=\varnothing$. Then, for every $\hat{\eta}_m^i$ and every $\lambda\ge0$, it holds that

$$\min_{\eta\in[0,1]}\left\{\mathbb{1}\{\eta\in\mathrm{int}(x^{-1}(1))\}+\lambda\left|\eta-\hat{\eta}_m^i\right|\right\}=\min_{\eta\in[0,1]\setminus C_0}\left\{\mathbb{1}\{\eta\in\mathrm{int}(x^{-1}(1))\}+\lambda\left|\eta-\hat{\eta}_m^i\right|\right\},$$

and thus,

$$\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(\mathrm{int}(x^{-1}(1)))]=\inf_{\mu\in\mathfrak{M}}\mathbb{E}_{\xi\sim\mu}[\xi(\mathrm{int}(x^{-1}(1))\setminus C_0)].$$

Hence, $v(\mathbb{1}\{\mathrm{int}(x^{-1}(1))\setminus C_0\})\ge v(\mathbb{1}\{\mathrm{int}(x^{-1}(1))\})$. Similarly, $v(\mathbb{1}\{x^{-1}(1)\setminus C_0\})\ge v(x)$. Because $\hat{\Xi}$ is finite, there exists $\{\underline{x}_j,\overline{x}_j\}_{j=1}^J$, where $J\le\mathrm{card}(\hat{\Xi})$, such that (39) holds. □

C.2. Proofs for Section 4.3

Proof of Proposition 5. For each q, let $C_q:=\{\zeta:-w^\top\zeta<q\}$. For any $q>\mathrm{VaR}_\alpha^\nu[-w^\top\xi]$, a necessary condition for ν to be transported to $\mu\in\mathcal{P}(\Xi)$ such that $\mathrm{VaR}_\alpha^\mu[-w^\top\xi]=q$ is that $\nu_w\{(-\infty,q)\}-(1-\alpha)$ of ν-probability is transported from Cq to $\Xi\setminus C_q$. Note that the metric d given by $d(\zeta,\xi)=\|\xi-\zeta\|$ is an intrinsic metric, and thus, the least cost way to transport probability from Cq to $\Xi\setminus C_q$ is to transport it to ∂C_q. It follows from Example 7 that given θ and q, there exists a worst-case distribution μ_q^* such that $\mu_q^*(C_q)=\min_{\mu\in\mathfrak{M}}\mu(C_q)$ and that μ_q^* is obtained from ν by greedily transporting probability from points in Cq closest to ∂C_q to ∂C_q.

Specifically, there exists an ξ_w^* such that $\|\xi_w^*\|=1$ and $\|w\|_*:=\sup\{|w^\top\xi|:\|\xi\|=1\}=w^\top\xi_w^*$. Consider any $\zeta\in C_q$, and let $s=-w^\top\zeta<q$. Let $\xi_\zeta:=\zeta-(q-s)\xi_w^*/\|w\|_*$. Note that $-w^\top\xi_\zeta=-w^\top\zeta+(q-s)w^\top\xi_w^*/\|w\|_*=s+(q-s)\|w\|_*/\|w\|_*=q$, and thus, $\xi_\zeta\in\Xi\setminus C_q$. Consider any $\xi\in\Xi\setminus C_q$, that is, $-w^\top\xi\ge q$. It follows from the definition of $\|w\|_*$ that $\|\xi-\zeta\|\ge|w^\top(\xi-\zeta)|/\|w\|_*\ge-w^\top(\xi-\zeta)/\|w\|_*\ge(q-s)/\|w\|_*$. Note that $\|\xi_\zeta-\zeta\|=(q-s)\|\xi_w^*\|/\|w\|_*=(q-s)/\|w\|_*$. Thus, $\xi_\zeta\in\arg\min\{\|\xi-\zeta\|:\xi\in\Xi\setminus C_q\}$.

Note that, for $\zeta\in C_q$, it holds that $\min\{\|\xi-\zeta\|:\xi\in\Xi\setminus C_q\}=(q+w^\top\zeta)/\|w\|_*$ is increasing in $w^\top\zeta$, and thus, a worst-case distribution $\mu_q^*\in\arg\min_{\mu\in\mathfrak{M}}\mu(C_q)$ is obtained from ν by transporting probability from points $\zeta\in C_q$ with the smallest values of $w^\top\zeta$ to ∂C_q. For any $t<q$, let μ_t^q denote the distribution obtained from ν by transporting probability from points in $\{\zeta:t\le-w^\top\zeta<q\}$ to ∂C_q, that is, for any $A\in\mathscr{B}_\nu(\Xi)$, $\mu_t^q(A)=\nu([A\cap C_t]\cup\{\zeta:t\le-w^\top\zeta<q,\ \xi_\zeta\in A\}\cup[A\cap(\Xi\setminus C_q)])$. Note that

$$\mathrm{VaR}_\alpha^{\mu_t^q}[-w^\top\xi]=\begin{cases}q & \text{if } t\le\mathrm{VaR}_\alpha^\nu[-w^\top\xi]\\ \mathrm{VaR}_\alpha^\nu[-w^\top\xi] & \text{if } t>\mathrm{VaR}_\alpha^\nu[-w^\top\xi]\end{cases}.$$

Also note that $W_p^p(\mu_t^q,\nu)$ is nonincreasing in t. Therefore,

$$\mathrm{VaR}_\alpha^{\mathfrak{M}}[-w^\top\xi]=\sup\left\{q:t=\mathrm{VaR}_\alpha^\nu[-w^\top\xi],\ W_p^p(\mu_t^q,\nu)\le\theta^p\right\}.$$

Note that, with $t = \mathrm{VaR}_\alpha^\nu[-w^\top \xi]$, it holds that

$$W_p^p(\mu_t^q, \nu) = \int_{\mathrm{VaR}_\alpha^\nu[-w^\top\xi]}^q \frac{(q-s)^p}{\|w\|_*^p} \nu_w(ds),$$

and thus,

$$\mathrm{VaR}_\alpha^{\mathfrak{M}}[-w^\top \xi] = \sup\left\{q : \int_{\mathrm{VaR}_\alpha^\nu[-w^\top\xi]}^q \frac{(q-s)^p}{\|w\|_*^p} \nu_w(ds) \le \theta^p\right\}.$$

It follows from the definition of $\mathrm{VaR}_\alpha^\nu[-w^\top\xi]$ that $\nu_w\{(-\infty, q)\} > (1-\alpha)$ for all $q > \mathrm{VaR}_\alpha^\nu[-w^\top\xi]$, and thus, $W_p^p(\mu_t^q, \nu)$ is increasing in q on $[\mathrm{VaR}_\alpha^\nu[-w^\top\xi], \infty)$. In addition, $W_p^p(\mu_t^q, \nu)$ is continuous in q. Therefore, $\mathrm{VaR}_\alpha^{\mathfrak{M}}[-w^\top\xi]$ is equal to the unique solution q of

$$\int_{\mathrm{VaR}_\alpha^\nu[-w^\top\xi]}^q (q-s)^p\, \nu_w(ds) = \theta^p \|w\|_*^p,$$

and μ_t^q with $t = \mathrm{VaR}_\alpha^\nu[-w^\top\xi]$ and the resulting q is a worst-case distribution. □

Appendix D. Selecting Radius θ

The following approach for selecting the radius of the Wasserstein ball uses a result for Wasserstein distance from Bolley et al. [14]. Let ν_N denote the empirical distribution given by N i.i.d. observations from the underlying distribution ν_0. In Bolley et al. [14, theorem 1.1 (see also remark 1.4)], it is shown that $\mathbb{P}[W_1(\nu_N, \nu_0) > \theta] \le C(\theta) e^{-\lambda N \theta^2/2}$ for some constant λ dependent on ν_0, and C dependent on θ. Because their result holds for general distributions, here we simplify it for our purpose by explicitly computing the constants λ and C. For a more general analysis, we refer the reader to Bolley et al. [14, section 2.1].

By assumption supp $\nu_0 \subset [0, \overline{B}]$, and thus, the truncation step in Bolley et al. [14] is no longer needed. Hence, the probability bound in Bolley et al. [14, (2.12) (see also (2.15))] is reduced to

$$\mathbb{P}[W_1(\nu_N, \nu_0) > \theta] \le \max\left\{8e\frac{\overline{B}}{\delta}, 1\right\}^{\mathcal{N}(\delta/2)} e^{-\lambda N(\theta-\delta)^2/8},$$

for some constants $\lambda > 0$, any $\delta \in (0, \theta)$, where e is the natural logarithm, and $\mathcal{N}(\delta/2)$ is the minimal number of balls of radius $\delta/2$ needed to cover the support of ν_0. In this case, $\mathcal{N}(\delta/2) = \overline{B}/\delta$. Next ,we compute λ. By Bolley et al. [14, theorem 1.1], λ is the constant in the Talagrand inequality

$$W_1(\mu, \nu_0) \le \sqrt{\frac{2}{\lambda} I_{\phi_{kl}}(\mu, \nu_0)},$$

where the Kullback–Leibler divergence of μ with respect to ν is defined by $I_{\phi_{kl}}(\mu, \nu_0) = \infty$ if μ is not absolutely continuous with respect to ν_0; otherwise, $I_{\phi_{kl}}(\mu, \nu_0) = \int f \log f\, d\nu_0$, where f is the Radon–Nikodym derivative $d\mu/d\nu_0$. Bolley and Villani [13, corollary 4] shows that λ can be chosen as

$$\lambda = \left[\inf_{\zeta^0 \in \Xi, \alpha > 0} \frac{1}{\alpha}\left(1 + \log \int e^{\alpha d^2(\xi, \zeta^0)} \nu(d\xi)\right)\right]^{-1},$$

which can be estimated from data. Finally, we obtain a concentration inequality

$$\mathbb{P}[W_1(\nu_N, \nu_0) > \theta] \le \max\left\{8e\frac{\overline{B}}{\delta}, 1\right\}^{\overline{B}/\delta} e^{-\lambda N(\theta-\delta)^2/8}. \tag{D.1}$$

In the numerical experiment, we chose δ to make the right side of (D.1) as small as possible, and θ was chosen such that the right side of (D.1) was equal to 0.05.

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